MODELLING THE EPIDEMIOLOGY, ECOLOGY, AND CONTROL OF AEDES-BORNE DISEASES

SUN HAOYANG

BSc (*Hons.*), NUS

A THESIS SUBMITTED

FOR THE DEGREE OF DOCTOR OF PHILOSOPHY

SAW SWEE HOCK SCHOOL OF PUBLIC HEALTH

NATIONAL UNIVERSITY OF SINGAPORE

2021

Supervisor:

Associate Professor Alex R Cook

Examiners:

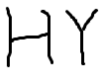
Professor Choi Kwok Pui

Dr Luo Wei

Declaration

I hereby declare that this thesis is my original work, and it has been written by me in its entirety. I have duly acknowledged all the sources of information which have been used in the thesis.

This thesis has also not been submitted for any degree in any university previously.



Sun Haoyang

October 27, 2021

Acknowledgements

First and foremost, I would like to thank my supervisor, Alex Cook, for his invaluable guidance and support during the past four years of my PhD journey.

I am also grateful to my thesis advisory committee, colleagues, and collaborators, Rick Ong, Roman Carrasco, Borame Dickens, Hannah Clapham, Mark Jit, Joel Koo, Lim Jue Tao, Ng Lee Ching, Joel Aik, Daniel Richards, and Muhammad Hassim for sharing their knowledge, ideas, and data.

Last but not least, I would like to express my deepest gratitude to my family and friends for their constant encouragement and support throughout the years.

Table of Contents

Summary ····· v

Published work in the thesis ····· vi

List of figures ····· vii

List of tables ····· viii

Chapter 1: Introduction ····· 1

1.1 Epidemiology of *Aedes*-borne diseases ····· 1

1.2 Ecology of *Aedes*-borne diseases ····· 4

1.3 Control of *Aedes*-borne diseases ····· 7

1.4 Aims of thesis ····· 11

Chapter 2: Mapping the cryptic spread of the 2015–2016 global Zika virus epidemic ····· 13

2.1 Abstract ····· 13

2.2 Introduction ····· 14

2.3 Methods ····· 15

2.4 Results ····· 29

2.5 Discussion ····· 36

2.6 Conclusion ····· 41

Chapter 3: Spatio-temporal analysis of the main dengue vector populations in Singapore ····· 42

3.1 Abstract ····· 42

3.2 Introduction ····· 43

3.3 Methods ····· 44

3.4 Results ····· 47

3.5 Discussion ····· 56

3.6 Conclusion ····· 59

Chapter 4: Short-term and long-term epidemiological impacts of sustained vector control in various dengue endemic settings: A modelling study ····· 60

4.1 Abstract ····· 60

4.2 Introduction ····· 61

4.3 Methods ····· 62

4.4 Results ····· 71

4.5 Discussion ····· 84

4.6 Conclusion ····· 88

Chapter 5: Discussion and conclusion ····· 89

5.1 Main findings and their public health implications ····89
5.2 Future work ···· 91
5.3 Final statement ····92
Bibliography ···· 93
Appendix A: Mapping the cryptic spread of the 2015–2016 global Zika virus epidemic ····108
A1. Additional information on the collection and pre-processing of data ····108
A2. Additional information on the global risk model validation ····110
A3. Data and tables ····111
Appendix B: Spatio-temporal analysis of the main dengue vector populations in Singapore ····112

Summary

Aedes-borne diseases are a major contributor to human mortality and morbidity worldwide. The past 50 years have witnessed an unprecedented expansion of *Aedes*-borne diseases, most notably dengue, Zika, yellow fever, and chikungunya. Due to the ongoing climate change and urban growth, it is anticipated that both *Aedes aegypti* and *Ae. albopictus* will continue to expand their geographical range, thereby further increasing the global health burden. In this thesis, I aim to use mathematical and statistical modelling to generate new insights and knowledge about the epidemiology and ecology of *Aedes*-borne diseases as well as the impact of *Aedes* control measures on disease transmission at global and metropolitan scales.

In my first data chapter, I developed a mathematical modelling framework integrating travel, demographic, climate, entomological, and human case data and identified a total of 24 locations at a high risk of cryptic Zika virus spread during the 2015—2016 global epidemic. Notably, some of these locations have recently reported their first Zika outbreaks in history, which add to the credibility of our model results and highlight the importance of continued investments in Zika virus research and surveillance systems to prepare for future epidemics.

Next, I examined how different environmental and anthropogenic variables influence the spatio-temporal variation of *Aedes* abundance in Singapore's high-rise public residential zones. My results showed building age to be the strongest predictor, and contrary to prior expectations, none of the weather variables were found to have a substantial impact on *Aedes* abundance. The evidence generated from this study can be used to guide the refinement of *Aedes* control strategies in Singapore.

In my final data chapter, I developed an agent-based model to study the short-term and long-term impacts of sustained *Aedes* control in a wide range of dengue endemic settings. I showed here that in the absence of a highly effective intervention technology that is able to reduce the basic reproduction number to below one, a fixed level of reduction in the *Aedes* abundance would only cause temporary reduction in dengue transmission. Furthermore, the time point at which interventions cease to be effective is strongly influenced not only by the intensity of vector control and the pre-intervention transmission intensity, but also by the individual-level heterogeneity in biting risk. The insights obtained will be useful to inform future model-based studies that seek to predict the impact of dengue vector control in their local contexts.

Published work in the thesis

1. Sun, H., Dickens, B.L., Jit, M. *et al.* Mapping the cryptic spread of the 2015–2016 global Zika virus epidemic. *BMC Med* 18, 399 (2020).

2. Sun, H., Dickens, B.L., Richards, D. *et al.* Spatio-temporal analysis of the main dengue vector populations in Singapore. *Parasites Vectors* 14, 41 (2021).

List of figures

Figure 2.1: Schematic overview of the methods ····18

Figure 2.2: Model validation results ····32

Figure 2.3: Median estimate of ZIKV R_0 for each spatial unit obtained from the global risk model ····33

Figure 2.4: Estimated ZIKV R_0 presented as a function of temperature and vector-to-host ratio ····34

Figure 2.5: ZIKV importation in each susceptible spatial unit during 2015–2016 ····35

Figure 2.6: Susceptible spatial units with a high chance of cryptic ZIKV spread during 2015–2016 ····36

Figure 3.1: Observed *Aedes* abundance (mean catch per trap per week) of each site during 2017–2018 ····49

Figure 3.2: Estimated percentage change in the expected value of *Aedes* abundance (weekly mean catch per trap) due to a 1-standard deviation increase in each covariate ····51

Figure 3.3: Predicted values of *Aedes* abundance (weekly mean catch per trap) at different values of lagged weekly maximum temperature with all the other variables held fixed at their average values ····54

Figure 3.4: Boxplots of the observed site-level *Aedes* abundance (mean catch per trap per week) during 2017–2018 within each quintile based on the out-of-sample model predictions ····55

Figure 4.1: Schematic overview of the individual-based model ····63

Figure 4.2: Simulated monthly number of dengue infections during the 30-year intervention period in different dengue endemic settings ····73

Figure 4.3: Impact of different predictor variables on the critical time point ····74

Figure 4.4: Impact of different predictor variables on the vector control effectiveness across the entire 30-year intervention period ····75

Figure 4.5: Boxplots of the intervention effectiveness under different levels of vector control intensity ····76

Figure 4.6: Impact of dengue importation on the predicted vector control effectiveness ····77

Figure 4.7: Impact of baseline PE9 on the predicted vector control effectiveness ···· 78

Figure 4.8: Impact of individual-level heterogeneity in the biting risk score on the predicted vector control effectiveness ····79

Figure 4.9: Impact of population ageing on the predicted vector control effectiveness ····80

Figure 4.10: Impact of host mobility on the predicted vector control effectiveness ····81

Figure 4.11: Impact of the strengths of immune interactions assumed in the simulation model on the critical time point and vector control effectiveness across the entire 30-year intervention period ···· 82

Figure 4.12: Impact of antibody-dependent enhancement on the predicted vector control effectiveness ···· 83

Figure 4.13: Impact of cross-protection on the predicted vector control effectiveness ····84

List of tables

Table 2.1: Parameters for the global risk model ····· 25

Table 2.2: Susceptible spatial units with a high chance of cryptic ZIKV spread during 2015–2016 ····· 31

Table 3.1: Summary statistics of all the variables included in the study ····· 48

Table 3.2: Posterior estimates of the Bayesian spatio-temporal model parameters ····· 52

Table 3.3: Percentage reduction in the out-of-sample mean absolute prediction error of the model compared with the baseline prediction ····· 56

Table 3.4: Percentage reduction in the out-of-sample mean absolute forecast error of the model compared with the baseline forecast ····· 56

Table 4.1: List of parameter values for the simulation model ····· 69

Chapter 1: Introduction

1.1 Epidemiology of *Aedes*-borne diseases

1.1.1 Dengue

Dengue virus (DENV) is a single-stranded RNA virus belonging to the *Flaviviridae* family [1]. The virus is mainly transmitted by *Aedes aegypti*, and to a lesser extent, *Ae. albopictus* [2]. There are five antigenically distinct serotypes (DENV-1 to DENV-5) discovered to date, with DENV-5 being a new serotype that is maintained in a sylvatic transmission cycle and has yet to be found circulating among humans [3]. Most dengue infections are clinically inapparent (i.e., either asymptomatic or mildly symptomatic), and these hard-to-detect infections have been estimated to contribute to over 80% of DENV transmission [4]. Individuals with apparent symptomatic infections experience high fever, accompanied by other symptoms such as severe headache, vomiting, muscle and joint pains, and rash [2]. Some patients may further develop potentially fatal complications upon defervescence, such as thrombocytopenia, hemorrhage, shock, and major organ failure [2]. No specific treatment is available for dengue, and standardized clinical management is vital for reducing the case fatality rate [5].

1.1.1.1 Global burden

Globally, an estimated 105 (95% CI: 95—114) million dengue infections occur each year, of which over half are concentrated in South and Southeast Asia [6]. Both of these regions play an important role in seeding viral lineages to the other regions of the world such as Latin America and the Caribbean [7,8], where dengue burden is also high [9]. Sporadic autochthonous cases have been reported in southern Europe [10,11], southern United States [12], and northern Queensland in Australia [13], although sustained local transmission is not currently present. In addition, the frequency of dengue outbreaks has been rising in the Eastern Mediterranean Region [2], and recently the first autochthonous dengue cases in Afghanistan have been detected [14]. To date, little is known about the dengue burden in sub-Saharan Africa due to the lack of diagnostic capacities, but returning traveler reports and other supplementary evidence suggest that DENV transmission is present in most countries within the region [15].

1.1.1.2 Immune interactions between dengue serotypes

Primary dengue infection is generally thought to create strong immunity against the infecting serotype and temporary cross-immunity against the other serotypes [16–18]. Despite being uncommon, homologous re-infection can occur and may result in symptomatic disease [19,20]. Upon subsequent infection with a heterologous serotype, disease symptoms tend to be more severe than primary infection [21–23]. This phenomenon has been hypothesized to be predominantly caused by antibody-dependent enhancement (ADE), whereby cross-reactive antibodies are unable to neutralize the virus of a different serotype but instead assist with viral entry into Fc receptor-bearing cells such as dendritic cells, macrophages, and monocytes, thus increasing the severity of infection [24,25]. ADE has also been argued to be responsible for the development of dengue hemorrhagic fever in infants who had previously received anti-DENV antibodies from their mothers [26]. Besides ADE, other proposed mechanisms underlying the enhanced disease severity in secondary infections include the production of pro-inflammatory cytokines by cross-reactive memory T cells that ultimately leads to plasma leakage [27]. Previous studies have also shown that certain sequences of infections (e.g., DENV-1 followed by DENV-2) can result in a higher risk of clinical illness [28], and moderate antigenic variation exists within each serotype, which may play a role in determining disease severity [29]. After a secondary dengue infection, the antibody response becomes broadly neutralizing and has been shown to significantly decrease the risk of disease during tertiary and quaternary infections [30]. However, it is still unclear to what extent secondary infection can prevent subsequent infections from occurring instead of solely protecting against clinical disease [31]. Likewise, uncertainty remains with regard to the nature of transient cross-immunity obtained following a primary infection [31].

While there is general agreement on the clinical significance of immunological serotype interactions, little consensus has been reached as to how these interactions affect the incidence patterns of dengue [31]. Both ADE and cross-immunity have been put forward in earlier studies as important drivers of dengue's complex transmission dynamics [32–34]. However, results of Recker *et al.* suggest that ADE alone can produce the inter-epidemic periods and desynchronized oscillations of the four dengue serotypes observed in real-world settings, provided its effects are decomposed into two forms of enhancement: increased transmissibility during secondary infection and increased susceptibility to secondary infection [35]. More recently, Lourenco & Recker used a spatially explicit individual-based model to demonstrate that stochasticity underlying the interactions between hosts and vectors alone can desynchronize the serotypes and produce multi-annual cycles of dengue

outbreaks even in the absence of immune interactions, suggesting that ADE and cross-immunity are unlikely to be the sole mechanisms underlying the epidemiological dynamics of dengue [36]. Estimating the strengths of these immune interactions, however, can be highly challenging, and more research is required to address this knowledge gap, which has important implications for vaccine impact predictions [37].

1.1.2 Zika

Zika virus (ZIKV) was first isolated from a rhesus monkey in the Zika Forest of Uganda in 1947 [38], and the first human infections were detected five years later [39]. The virus is primarily transmitted by the *Aedes* mosquitoes, but can also spread from mother to fetus or through sexual contacts [40–43]. Although ZIKV has been shown by early serological studies to be endemic in a number of African and Asian countries, it had received little attention for 60 years since its discovery owing to the paucity of reported cases and the self-limiting nature of most infections [40]. In 2007, ZIKV expanded to Yap Island in the Federated States of Micronesia and caused the first recorded Zika outbreak in history [44]. Thereafter, another outbreak was reported in French Polynesia during 2013—2014, where severe complications such as Guillain-Barré syndrome in adults and central nervous system malformations in neonates were first documented [45]. In May 2015, a large Zika outbreak was reported in Brazil, and a few months later the virus quickly spread across Latin America and the Caribbean region [40]. In February 2016, the World Health Organization (WHO) declared the association of ZIKV infection with cases of microcephaly and Guillain-Barré syndrome to constitute a Public Health Emergency of International Concern [40]. Although the number of Zika cases has dramatically declined after 2016 [46], much remains to be understood concerning the virus's epidemiology.

To date, a lot of progress has been made in understanding ZIKV's phylogenetic history [47], the association between microcephaly and ZIKV infection [48], neurodevelopment in infants with prenatal ZIKV exposure [49], as well as the asymptomatic proportion of ZIKV infections [50]. Previous studies have also generated important insights into the potential epidemiological impact of cross-reactivity between ZIKV and DENV. For example, using surveillance data from Brazil and Colombia, Borchering *et al.* demonstrated that ZIKV infection may have conferred cross-protection against DENV and thus resulted in the unusually low dengue incidence in 2017 in both countries [51]. By contrast, Katzelnick *et al.* studied a long-standing pediatric dengue cohort in Nicaragua and found intermediate anti-ZIKV antibody titers to increase the future risk of dengue disease [52]. It has also been hypothesized that cross-reactive anti-DENV antibodies may play an important role in enhancing the

pathogenicity of ZIKV infections [53]. Although enhancement of ZIKV by anti-DENV antibodies has been observed *in vitro*, convincing evidence *in vivo* is not yet available [53]. Besides immunological interactions between ZIKV and DENV, recent studies have also found rapid waning of anti-ZIKV antibodies following the epidemic [54–56], suggesting that ZIKV infection may not confer lifelong immunity. The duration of ZIKV immunity, however, is still unclear and requires further research. Another important yet unresolved question is in which locations ZIKV may have circulated without being detected during the recent global epidemic. Addressing this research gap will have important implications for improving outbreak preparedness and response.

1.1.3 Other *Aedes*-borne diseases

In addition to dengue and Zika, other *Aedes*-borne diseases include yellow fever, chikungunya, and Mayaro fever. Like DENV and ZIKV, yellow fever virus (YFV) is a *flavivirus* [57], in comparison to chikungunya virus (CHIKV) and Mayaro virus (MAYV), which are *alphaviruses* belonging to the *Togaviridae* family [58]. The clinical symptoms of the diseases caused by these arboviruses are highly similar especially in the early phase of infection [59], and hence misdiagnosis can often occur in the absence of laboratory testing. A small proportion of patients with yellow fever may develop jaundice [57], a distinct clinical symptom not shared by the other *Aedes*-borne diseases, and both CHIKV and MAYV infections may cause long-term arthralgia that severely impairs the quality of life [58,60].

YFV is endemic in over 40 countries in Africa and Central and South America [57], but has yet to be found circulating in Asia [61]. By contrast, the global burden of chikungunya is heavily concentrated in Asia and the Americas [62]. Most of the Mayaro fever cases reported to date have been confined to the Amazon region and were caused by spillover from the virus's sylvatic transmission cycle involving non-human primates and the *Haemagogus* mosquitoes [63]. However, both *Ae. aegypti* and *Ae. albopictus* have been found to be able to transmit MAYV in laboratory settings [64], suggesting that the virus may one day become urbanized and pose a serious threat to global health.

1.2 Ecology of *Aedes*-borne diseases

1.2.1 Vector behavior

As the primary vector of dengue, Zika, yellow fever, and chikungunya, *Ae. aegypti* preferentially resides in shady areas inside human dwellings and feeds almost exclusively on humans [65,66]. The mosquito is active during the daytime [67], rendering insecticide-treated bed nets ineffective in preventing infections. Moreover, daytime biting increases the likelihood of long-distance viral transmission caused by host movement, despite the limited mosquito flight distance [68,69]. The *Ae. aegypti* mosquitoes are also known to oviposit in a wide range of containers, especially artificial ones such as pails, vases, and roof gutters, and their breeding sites can often be cryptically located and hard to reach [70]. In addition, studies have revealed a high incidence of insecticide resistance in the *Ae. aegypti* population, which creates additional challenges for vector control [71]. In contrast to *Ae. aegypti*, *Ae. albopictus* is predominantly found in rural and suburban areas, feeds on a variety of hosts, and has an overall lower level of insecticide resistance [72,73]. Nonetheless, both mosquito species are capable of migrating with humans (e.g., their eggs are drought-resistant and can travel in dry containers), which is believed to be an important mechanism for their rapid expansion worldwide [70,74].

1.2.2 Climatic influences on disease transmission

There are numerous pathways whereby climate can influence the transmission of *Aedes*-borne diseases [75]: Temperature directly affects the viral replication rate within the *Aedes* mosquitoes and has been demonstrated in laboratory studies to have a strong impact on the extrinsic incubation periods of DENV [76], ZIKV [77], and YFV [78]. Furthermore, a decrease in diurnal temperature range can increase the susceptibility of *Ae. aegypti* to DENV infection given the same mean temperature [79]. Temperature also has a significant influence on the development and survival of immature *Ae. aegypti* and hence the rate of adult mosquito emergence [80]. The mortality rate of adult *Aedes* mosquitoes is temperature-dependent as well, which not only affects mosquito abundance but also determines the proportion of infected mosquitoes that can survive long enough to become infectious [75]. In view of the important influence of adult vector survival time on arboviral transmission, Brady *et al.* used data collected from 351 laboratory and 59 field experiments to estimate the mortality rate of adult female *Ae. aegypti* and *Ae. albopictus* respectively as a function of temperature and mosquito age [81]. The result of their study can be used in a dynamic transmission model to study the spatio-temporal variation in transmission potential for different *Aedes*-borne diseases on a global scale [82].

Unlike temperature, precipitation mainly influences *Aedes*-borne disease transmission via creating vector breeding habitats [75], although excessive rainfall can lead to breeding site flushing and hence may reduce the

risk of disease outbreaks [83]. A recent study conducted in Brazil has also found extreme drought to increase the risk of dengue transmission in areas with water supply shortages, which is likely due to changes in household water storage practices in response to water scarcity [84,85]. Moreover, both precipitation and temperature can affect humidity, which in turn influences mosquito survival and frequency of biting [86]. In recent years, there has been a growing interest in developing statistical models to evaluate the added value of weather data to dengue epidemic forecasting, with mixed results obtained by previous studies [85,87–90], suggesting that the strength of association between weather and dengue incidence can be highly heterogeneous across locations [75,89]. Previous work has also developed mechanistic models to predict the *Aedes* population dynamics under perturbed conditions such as severe weather fluctuations and vector control [91,92]. These models are highly complex, with a lot of assumptions made about the development of *Ae. aegypti* and its relation to the environmental conditions [93]. Comparison of multiple modelling approaches can be useful for identifying the subsystem of each model that needs further improvement, and high-quality entomological field data are required for model calibration and validation [93].

1.2.3 Mapping environmental suitability for *Aedes*-borne disease transmission

Another important area of research is mapping the environmental suitability for *Aedes* mosquitoes and the viruses they transmit. In their seminal work, Kraemer *et al.* compiled the most comprehensive *Aedes* occurrence database currently available [94], which was then combined with a set of environmental and land-cover variables to map the global distribution of *Ae. aegypti* and *Ae. albopictus* [95]. A similar approach was also employed by Messina *et al.* to estimate the global environmental suitability for ZIKV [96]. More recently, Kraemer *et al.* produced forecasts of the possible geographic expansion of *Ae. aegypti* and *Ae. albopictus* within the next 60 years due to human travel, urban growth, and climate change [97], which can be used to predict the future population at risk of dengue and other *Aedes*-borne diseases, and to guide entomological surveillance and vector control [98].

Despite the significant progress made in predicting the global distribution of *Aedes* mosquitoes, the spatio-temporal dynamics of the *Aedes* population remains poorly understood in urbanized local settings. Longitudinal and fine-scale measurements of *Aedes* abundance are seldom available [99], with few studies, if any, having successfully identified key variables that explain the geographic variation of adult *Aedes* abundance within a

single city. Addressing this research gap will have important implications for the improvement of local vector control strategies.

1.3 Control of *Aedes*-borne diseases

1.3.1 Existing *Aedes* control measures

The existing *Aedes* control measures can be broadly classified into two categories: those affecting the aquatic stages of the mosquitoes (e.g., container larvicide treatment, source reduction, public education, and law enforcement) and those targeting the adult mosquito stages (e.g., space spraying, indoor residual spraying, and personal protection) [100]. Regrettably, there have been very few vector control programs that have successfully suppressed dengue transmission: in Singapore, sustained larval source reduction and community engagement have substantially reduced dengue incidence during 1975—1990 [101]. Although the resulting lowered herd immunity subsequently led to the resurgence of dengue transmission, the annual force of infection has remained at a low level to date [101,102]. In Cuba, an intensive *Aedes* eradication campaign was launched in 1981, which caused dengue transmission to be nearly eliminated throughout the country during 1982—1996 [103].

Apart from the success stories mentioned earlier, the evidence base for the effectiveness of the existing dengue vector control measures remains poor [104]. In particular, no randomized controlled trials (RCTs) examining the epidemiological impact of outdoor fogging have been found [104,105], although this approach has been widely adopted due to its high visibility [100]. While an RCT conducted in Mexico and Nicaragua has used serological data to demonstrate a significant impact of community engagement on dengue transmission [106], most trials assessing the effectiveness of community-based environmental management focused exclusively on entomological outcomes and did not generate any evidence for the reduction in dengue incidence [104]. Besides, an RCT conducted in Santiago de Cuba reported an insignificant increase in dengue incidence due to the addition of indoor residual spraying to the routine *Aedes* control activities [107], and similar results were also obtained by a case-control study in Taiwan [108]. These results were in contrast to the findings from an observational study in Australia, which showed a significant reduction in the odds of secondary dengue infections due to the high coverage of indoor residual spraying [13]. Taken all together, the paucity of high-

quality evidence for the effectiveness of dengue vector control [109] highlights the need for more research to be conducted to assess the epidemiological impact of different *Aedes* control measures in various dengue transmission settings.

To date, there have been even fewer studies evaluating the effectiveness of *Aedes* control for the reduction of Zika or chikungunya transmission [110]. This can be primarily explained by the longer inter-epidemic period and more unpredictable epidemic timing of these diseases, which pose additional challenges to the design and implementation of intervention trials [111]. Hence, during a Zika or chikungunya epidemic, it may be reasonable to adopt *Aedes* control approaches that have already been proven effective at reducing dengue transmission in each particular setting. It should be noted, however, that there is one class of *Aedes* control measures whose epidemiological impact can only be evaluated in a Zika epidemic, namely personal protection (e.g., mosquito repellent) targeting pregnant women in the first trimester of pregnancy with the aim of reducing the incidence of congenital Zika syndrome instead of the overall level of viral transmission within the community [111]. Rigorously designed RCTs will be required in future Zika epidemics to assess the effectiveness of this approach.

1.3.2 Novel *Aedes* control measures

Recently, there has been an increasing number of novel *Aedes* control measures under development. One of the most promising approaches involves releasing mosquitoes carrying the *wMel* strain of *Wolbachia pipientis* to replace the wild population, which has been shown to reduce the incidence of virologically confirmed dengue (VCD) by 77% (95% CI: 65%—85%) in Yogyakarta, Indonesia in a cluster randomized trial [112]. The same approach has also been tested in other countries such as Australia, Vietnam, Brazil, and Colombia, with encouraging results reported by preliminary studies [113]. Besides blocking DENV transmission [114], the *wMel* strain has been demonstrated to lower the vector competence of *Ae. aegypti* for ZIKV, CHIKV, YFV, and MAYV as well [115–118], and more research is needed to understand the exact cellular mechanisms underlying this viral blocking effect [119].

Similar to the aforementioned *Wolbachia* approach, gene-drive technologies (e.g., CRISPR-Cas9) has the potential to enable prolonged suppression of pathogen transmission to be achieved in the absence of frequent releases of engineered mosquitoes [120]. In early 2020, Li *et al.* developed the first gene-drive system in *Ae. aegypti* [121], and during the same year, Buchman *et al.* successfully engineered the first transgene that

protects *Ae. aegypti* against all four dengue serotypes [122]. These studies have laid an important foundation for the future development of technologies to spread anti-dengue genes into the wild *Aedes* population. In addition to vector competence reduction, gene-drive systems can be used for sex ratio distortion to reduce the mosquito population, although such a transgenic method can be much more challenging to develop for *Ae. aegypti* in comparison to *Anopheles* since it does not have a distinct sex chromosome [100]. In view of the rapid advances in the development of genetically modified mosquitoes (GMMs), the WHO has recently updated its guidance framework for the testing of GMMs, where quality standards for efficacy and safety evaluations were provided and the relevant ethical and regulatory issues were discussed [123].

Besides mosquito population replacement, other promising *Aedes* control measures include incompatible insect technique and sterile insect technique, the combination of which has led to a 88% (95% CI: 57%—99%) reduction in dengue incidence within the core release zones compared to the control areas in a field trial conducted in Singapore [124]. A common drawback of all the *Aedes* control measures involving releases of engineered mosquitoes is their delayed impact on the adult vector populations [100]. Hence, the implementation of these novel approaches alone could be inadequate for reducing disease transmission in an outbreak setting, where swift deployment of reactive interventions such as indoor residual spraying may still be required [100].

1.3.3 Design of large-scale field trials to assess *Aedes* control measures

Given the importance of evidence-based policymaking for *Aedes* control, a number of studies have provided key considerations on the design of large-scale vector control trials [109,125–127]. Recommended study designs include randomized controlled before-and-after study, randomized interrupted time series, and randomized step-wedge study, and epidemiological endpoints are required to demonstrate the impact of vector control on disease transmission [109]. A longitudinal pediatric cohort may be useful to maximize transmission detection and minimize movement of trial participants between intervention and control areas, and cross-reactivity should be accounted for in estimating seroconversion rates in settings where multiple closely-related arboviruses co-circulate [125]. Besides, the determination of cluster size and cluster spacing needs to take into account the movement range of the study participants as well as the dispersal range of the engineered mosquitoes (provided they are released) to reduce spillover effects [126], and the period of follow-up needs to be sufficiently long (e.g., two or more transmission seasons) given that the incidence of *Aedes*-borne diseases

can be highly variable over time [109]. It should also be noted that the magnitude of reduction in disease transmission observed in a vector control trial conducted for only several years should not be equated with the long-term intervention effectiveness [111]. Unfortunately, conducting large-scale field trials over a decade or even longer is unlikely to be economically feasible especially for low- and middle-income countries. While mechanistic models can be utilized to predict the long-term effectiveness of sustained *Aedes* control, such studies are rare [128,129] and the current understanding of how the time-varying epidemiological impact of *Aedes* control changes across different transmission settings remains poor.

1.3.4. Vaccines

The only dengue vaccine commercially available to date is CYD-TDV, developed by Sanofi-Pasteur. The vaccine is tetravalent and has been shown to protect against severe VCD in seropositive vaccine recipients in phase-III trials [130]. However, an increased risk of severe VCD has been observed in participants who had not experienced any dengue infections prior to vaccination [130], and the WHO recommends that serological testing be conducted before the vaccine is administered [131].

One particular challenge in pre-vaccination screening is the poor sensitivity of the Immunoglobulin G (IgG) enzyme-linked immunosorbent assay (ELISA) test in those with a monotypic dengue profile: the benefit of the vaccine is the largest among this sub-population, and yet the cutoff value of the test cannot be further lowered to increase sensitivity without causing a sizeable number of false positives [132,133]. A more accurate, rapid, and reasonably cheap pre-vaccination screening test is urgently needed to maximize the vaccine's public health impact [131]. Previous work has also shown that the effect of the test accuracy on the population-level benefits of vaccination depends on the transmission intensity, with specificity being more important in low-transmission settings and sensitivity more important in high-transmission settings [134]. In addition, projections of the CYD-TDV's public health impact need to account for the waning vaccine effectiveness [135] as well as the uncertainties about breakthrough infections and immune interactions between serotypes [37,136].

More recently, Biswal *et al.* reported data from the first part of a phase-III RCT that assesses the efficacy of a second dengue vaccine, TAK-003, developed by Takeda [137]. The vaccine was efficacious against VCD among children aged between 4 and 16 years regardless of the serostatus at baseline, although the efficacy was estimated to be slightly higher among seropositive participants [137]. Furthermore, the efficacy estimates

varied by dengue serotype, and long-term data will be critical to shed more light on this vaccine's efficacy and safety profile [137].

The first yellow fever vaccine was developed in the 1930s [138], and to date four have been prequalified by the WHO [139]. No vaccine protecting against ZIKV or CHIKV infections is commercially available at the time of writing, but a few candidates have entered early-phase clinical trials and shown promising results [140,141].

1.4 Aims of thesis

Although significant progress has been made in understanding the epidemiology and ecology of *Aedes*-borne diseases as well as developing new vector control tools and vaccines, a lot of critical knowledge gaps still remain. In particular, little is known about the true geographical extent of ZIKV, which is believed to be still circulating in many locations in the tropics and subtropics without being detected. The identification of these locations has important implications for the enhancement of global Zika outbreak preparedness and response capacity. To achieve this, however, requires knowledge about the geographical areas that should be prioritized for ZIKV surveillance strengthening. In chapter 2, I aim to develop a mathematical modeling framework integrating diverse sources of information to identify geographical areas at a high risk of undetected ZIKV circulation during the 2015—2016 global epidemic. It is hoped that this work will provide valuable evidence to guide the allocation of finite public health resources to mitigate the risk of another large-scale Zika outbreak in the future.

To date, numerous studies have been conducted to map the distribution of *Aedes* mosquitoes at the global, continental, and national scales. While substantial spatial variation in *Aedes* abundance also exists within the confines of a single city, its associated determinants have rarely been studied in detail. This knowledge gap hinders the development of spatially targeted interventions to reduce the *Aedes* population at a city scale. In addition, it remains unclear to what extent weather influences the temporal fluctuation of *Aedes* abundance inside and nearby residential zones in cities with a tropical climate such as Singapore, although vector control is typically intensified during the wet season. In chapter 3, I aim to build regression models using longitudinal data generated from Singapore's Gravitrap surveillance system to assess a wide range of environmental and anthropogenic variables in their contribution to explaining the spatio-temporal variation of *Ae. aegypti* and *Ae.*

albopictus abundance in Singapore's high-rise public residential zones. The results generated from this study can be used to guide the refinement of *Aedes* control strategies in Singapore.

In recent years, a lot of novel technologies for *Aedes* control have been developed, with the potential to achieve long-term dengue elimination as demonstrated by small-scale field trials. However, large-scale implementation of these novel interventions remains extremely challenging, and it is still essential to understand how the sustained and intensive application of the currently available tools for *Aedes* control would impact dengue transmission both in the short and long-run. Furthermore, few studies, if any, have been conducted to identify the main drivers for the variation in vector control effectiveness across different dengue endemic settings. In chapter 4, I aim to develop an agent-based model to simulate the time-varying impact of sustained *Aedes* control in diverse dengue endemic settings over a 30-year time period. It is hoped that this work will generate useful insights to guide the development of *Aedes* control strategic plans to achieve sustained reduction in the burden of dengue.

Chapter 2: Mapping the cryptic spread of the 2015–2016 global Zika virus epidemic

This work has been accepted for publication as:

Sun, H., Dickens, B.L., Jit, M. *et al.* Mapping the cryptic spread of the 2015–2016 global Zika virus epidemic. *BMC Med* 18, 399 (2020).

2.1 Abstract

Background

Zika virus (ZIKV) emerged as a global epidemic in 2015–2016 from Latin America with its true geographical extent remaining unclear due to widely presumed underreporting. The identification of locations with potential and unknown spread of ZIKV is a key yet understudied component for outbreak preparedness. Here, we aim to identify locations at a high risk of cryptic ZIKV spread during 2015–2016 to further the understanding of the global ZIKV epidemiology, which is critical for the mitigation of the risk of future epidemics.

Methods

We developed an importation simulation model to estimate the weekly number of ZIKV infections imported in each susceptible spatial unit (i.e., location that did not report any autochthonous Zika cases during 2015–2016), integrating epidemiological, demographic, and travel data as model inputs. Thereafter, a global risk model was applied to estimate the weekly ZIKV transmissibility during 2015–2016 for each location. Finally, we assessed the risk of onward ZIKV spread following importation in each susceptible spatial unit to identify locations with a high potential for cryptic ZIKV spread during 2015–2016.

Results

We have found 24 susceptible spatial units that were likely to have experienced cryptic ZIKV spread during 2015–2016, of which 10 continue to have a high risk estimate within a highly conservative scenario, namely, Luanda in Angola, Banten in Indonesia, Maharashtra in India, Lagos in Nigeria, Taiwan and Guangdong in China, Dakar in Senegal, Maputo in Mozambique, Kinshasa in Congo DRC, and Pool in Congo. Notably, among the 24 susceptible spatial units identified, some have reported their first ZIKV outbreaks since 2017, thus adding to the credibility of our results (derived using 2015–2016 data only).

Conclusion

Our study has provided valuable insights into the potentially high-risk locations for cryptic ZIKV circulation during the 2015–2016 pandemic and has also laid a foundation for future studies that attempt to further narrow this key knowledge gap. Our modelling framework can be adapted to identify areas with likely unknown spread of other emerging vector-borne diseases, which has important implications for public health readiness especially in resource-limited settings.

2.2 Introduction

Zika virus (ZIKV) is a *flavivirus* that is mainly transmitted by the *Aedes* mosquitoes [142]. Since its first isolation in a Ugandan forest in 1947 [143], the virus has until recent years only caused regional outbreaks [144,145]. In February 2016, a Public Health Emergency of International Concern was declared by the World Health Organization, as ZIKV swept through the majority of the countries in Latin America and the Caribbean, coinciding with an unusual increase in the number of microcephaly cases and other neurological disorders [40]. The reported ZIKV incidence declined substantially after 2016 [46], but in the meantime, new scientific evidence continued to emerge and reveal new locations where ZIKV circulation had never been identified before [146]. More recently, the first autochthonous Zika case in Europe was reported in October 2019 with its source remaining unknown at the time of writing [147], which underscores our limited understanding of the virus' epidemiology. Although ZIKV is no longer a public health emergency, the potential reoccurrence of future large-scale epidemics remains a concern, which necessitates continued investments in ZIKV research and surveillance in preparation for such an event [148].

To date, modelling studies have yielded important insights into the virus' transmission dynamics [149–153], ecological niche [96,154,155], and at-risk population size [156], but little has been done to understand the gap between where cases may have already occurred and where they have been reported [157]. Due to the high proportion of ZIKV infections that are asymptomatic and its similarity in clinical presentation to other diseases such as dengue fever [144], undetected or unreported spread of ZIKV was widely presumed, especially in countries with limited public health resources [155]. The identification of these locations is critical, as their healthcare systems may be overwhelmed during possible future waves of epidemics if ill prepared, and due to the risk of onward dissemination via air travel. Whilst a recent study used travel surveillance and viral genomes data to detect an unreported ZIKV outbreak in Cuba [158], our understanding of the geographical areas that are likely to have experienced cryptic ZIKV spread is still seriously lacking at a global scale. To narrow this knowledge gap requires a novel modelling framework integrating a wide range of factors that determine the worldwide introduction of the virus and the subsequent autochthonous transmission.

In this study, we aim to answer the following key questions, focusing on all countries or first-level subdivisions where no indigenous Zika cases were reported during the 2015–2016 global epidemic: (i) At least how many ZIKV infections were imported in each country or subdivision during 2015–2016 and to what extent were

imported ZIKV infections underreported? (ii) Which countries or subdivisions were most likely to experience cryptic ZIKV spread during the 2015–2016 global epidemic based on currently available data?

2.3 Methods

2.3.1 Data

2.3.1.1 Epidemiological data

The weekly number of reported autochthonous Zika cases during 2015–2016 for each country in the Americas was published as bar charts by the Pan American Health Organization (PAHO) [159]. These data were digitized by the Andersen Lab for a genomic epidemiological study [160], and made publicly available [161]. We used the Web Plot Digitizer to extract the weekly number of notified Zika cases in Brazil during 2015 published by Lourenço *et al.* [162], since these data were not published by the PAHO. To achieve a higher spatial resolution, we also obtained weekly cumulative case counts for each first-level subdivision of Colombia between Eweek 38 and Eweek 52 of 2016 from the Colombian National Institute of Health's website [163]. Each time series was then differenced to derive the weekly case counts and combined with data compiled by Siraj *et al.* [164], which included weekly-notified case data prior to Eweek 38 of 2016. For weekly autochthonous Zika cases reported by countries outside the Americas, we manually reviewed information compiled by HealthMap alongside additional data sources such as ReliefWeb [165,166] (refer to the Appendix A [167–171] for more details). Each reported case was located to a first-level country subdivision whenever possible, and both confirmed and suspected autochthonous cases were included in our study (Appendix A: Data S2.1).

In addition to autochthonous case data, we obtained the reported total number of imported Zika cases during 2015–2016 for each US state, which was published by the Centres for Disease Control and Prevention (collection of imported case data for Florida and Texas was not needed for the study, as discussed in later sections) [172]. For all the other countries or subdivisions that did not report any autochthonous cases during 2015–2016, the imported case data were collected using HealthMap [165].

2.3.1.2 Travel data

We requested the yearly average length of stay (LOS) on inbound tourism trips up to 2016 from the United Nations World Tourism Organization (UNWTO) [173]. For each country, we took the most recently available estimate for the analyses. In most cases, estimates were derived based on the check-in dates from arrival and departure cards or border survey data, but if such data were unavailable for a country, the estimate based on the Hotel Occupancy Survey was used instead [173]. In the rare event that the data were completely missing for a given country, we took the conservative approach and assumed the average LOS to be equal to the minimum value among all the countries under study, to avoid overestimating the number of imported ZIKV infections (refer to section 2.3.2.2 for more details).

The monthly number of air ticket bookings during 2015–2016 was obtained from the Official Airline Guide (OAG) for every origin-destination route with up to two connections. This was used to derive the weekly number of bookings assuming the daily number of bookings was uniform within each month for each route. For each first-level country subdivision with no incoming air passengers during 2015–2016, we performed an estimation of the most likely airport (within the same country) that the population would rely on when they returned home (details shown in Appendix A [174]). The subdivision was subsequently merged with the one where the identified airport was located, and they were modelled as a single unit from then on. For each country where autochthonous cases could not be located to the subdivision level, the entire country was treated as a single unit of analysis. Hence, the spatial unit of analysis in our study can be a single first-level country subdivision, a combination of subdivisions, or an entire country (hereinafter referred to as “spatial unit”). Countries with zero incoming air passenger during 2015–2016 were excluded from the analysis (refer to the Appendix A for more details).

2.3.1.3 Demographic and ecological data

We obtained the global estimated 2015-population counts at a ~1km×1km resolution from the Socioeconomic Data and Applications Centre (SEDAC) [175], which was used to derive the total population within each spatial unit. The daily temperature at 2 meters between 2015 and 2016 was obtained from the European Centre for Medium-Range Weather Forecasts at ~30km×30km resolution [176]. Each temperature map was resampled using bilinear interpolation based on the cell size of the SEDAC population map. Subsequently, the resampled temperature pixel values were averaged within each spatial unit for each day, using the corresponding 2015-population counts as weights. We did not calculate raw average values because the population distribution can

be highly uneven for some spatial units such as the Sichuan province of China, where the majority reside within the basin instead of the mountain or plateau regions.

The methods employed to quantify the global environmental suitability of *Ae. aegypti* and *Ae. albopictus*, as well as the associated uncertainties, were described in Dickens *et al.* [177]. For each species, we obtained 250 vector suitability maps at a ~5km×5km resolution via bootstrapping. Our models performed reasonably well based on the out-of-sample prediction accuracy, with a median true skill statistic of 0.84 (0.76–0.86) for *Ae. aegypti* and 0.71 (0.66–0.78) for *Ae. albopictus* [177]. Following the same procedure as above, each vector suitability map was resampled according to the resolution of the SEDAC population map to derive the average suitability value weighted by the human population counts for each spatial unit.

2.3.2 Statistical analyses

2.3.2.1 Overview

To map the cryptic spread of ZIKV during the 2015–2016 global epidemic, we first performed simulations to estimate the weekly number of ZIKV infections imported into each spatial unit that did not *report* any autochthonous cases during 2015–2016. Hereinafter, we will refer to these spatial units as *susceptible spatial units*, and the rest as *donor spatial units*. As a by-product of the simulation, a reporting index was derived to quantify the probability of reporting a case per importation for each susceptible spatial unit. Next, we estimated the virus' weekly basic reproduction number (R_0) for each susceptible spatial unit, which was then used to compute the probability that no onward spread following ZIKV importation occurred during 2015–2016. These probability estimates derived from our model, together with the local evidence of *Aedes*-borne disease transmission potential based on the existing literature, were used to identify susceptible spatial units with a high chance of cryptic ZIKV spread during 2015–2016. (Refer to Figure 2.1 for the schematic overview of methods)

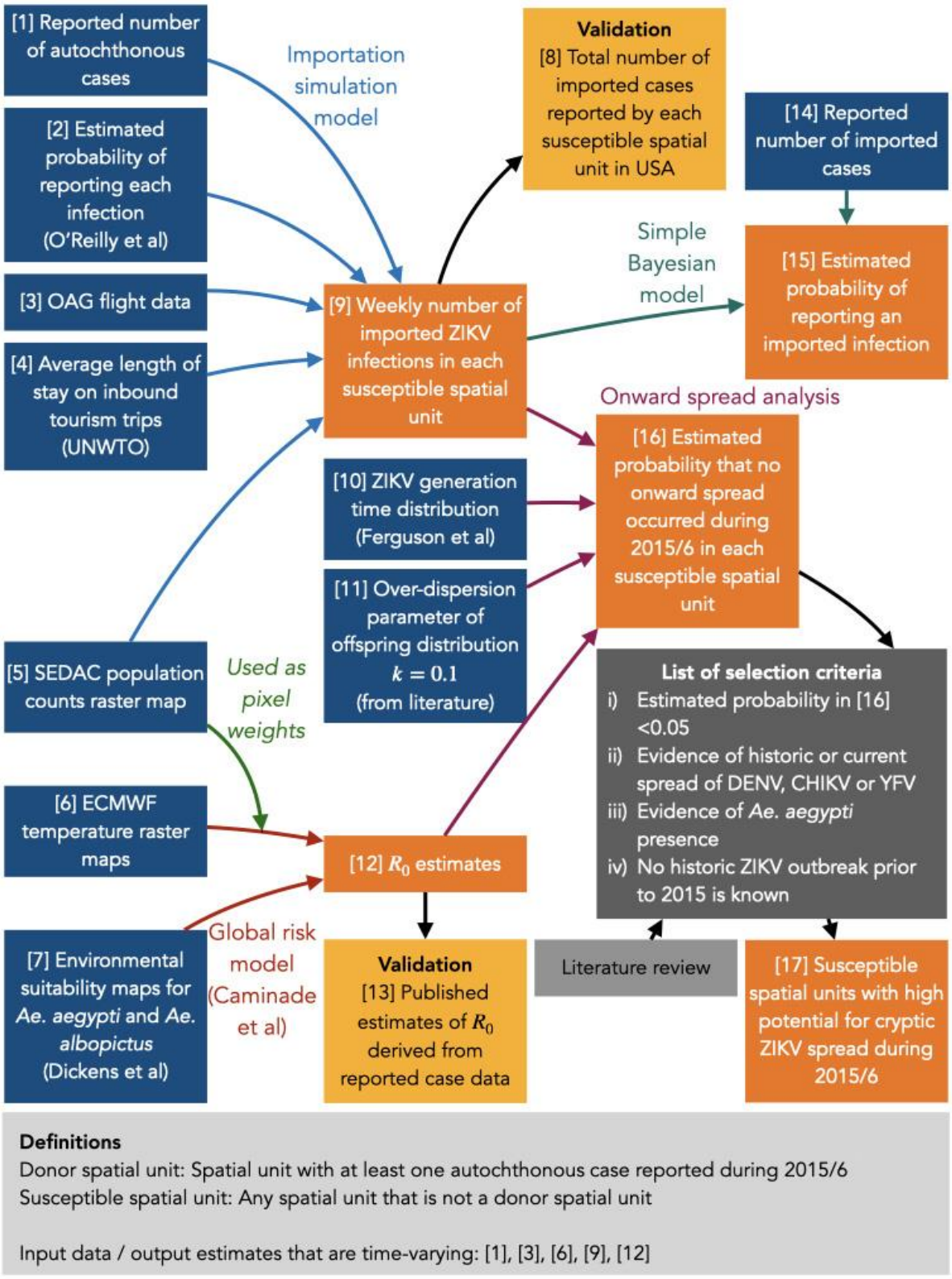


Figure 2.1: Schematic overview of the methods. Blue boxes denote input data, and dark orange boxes output estimates. Note that in the onward spread analysis, we further imposed thermal restrictions for ZIKV transmission and applied a threshold value for the estimated environmental suitability of *Ae. aegypti* to minimize false positives. Refer to section 2.3.2.5 for more details.

2.3.2.2 Simulation of the number of imported infections

To simulate the weekly number of imported ZIKV infections for each susceptible spatial unit, we first made the following definitions and assumptions. First, the *local population* of a spatial unit was defined as all who live in that spatial unit regardless of citizenship status. At any point in time, the size of a spatial unit i's local population who were visiting other spatial units was assumed to be approximately equal to the total number of spatial unit i's visitors. Hence, the total number of individuals located in a spatial unit at any point in time can be approximated by the total local population size, which was given by the SEDAC population estimates. Second, we assumed that air travellers with origin airport located in spatial unit i and destination airport located in spatial unit j were only made up of people belonging to the local population of spatial unit i or those of spatial unit j. Third, people who acquired ZIKV infection while visiting a spatial unit were assumed to remain asymptomatic by the end of their visits, so that all the autochthonous cases reported by a spatial unit can be assumed to come from its local population. Fourth, for any individual who belonged to the local population of a susceptible spatial unit and visited a donor spatial unit during 2015–2016, no immunity against ZIKV developed prior to his/her visit.

A total of 10,000 simulations were performed, where in each simulation $s = 1, \dots, 10000$, we generated an estimate of the number of ZIKV infections imported from a *donor* spatial unit i to a *susceptible* spatial unit j during Eweek t, denoted by $M_{i\to j,t}^{(s)}$. Of these, $M_{i\to j,t}^{(s)}[i]$ belonged to the local population of spatial unit i and $M_{i\to j,t}^{(s)}[j]$ the local population of spatial unit j, which were estimated separately as follows.

To begin with, we divided the autochthonous case count reported in spatial unit i during Eweek t, by the country-specific reporting rate based on O'Reilly *et al.*'s study to obtain the actual number of autochthonous infections $I_{i,t}$ [153]. Here, we used c_i to denote the country to which spatial unit i belonged, and ρ_{c_i} to denote the percentage of autochthonous infections in c_i that were reported, which was assumed to be time-invariant. In each simulation s, the reporting rate estimate $\rho_{c_i}^{(s)}$ was drawn from a beta distribution with parameters a_{c_i} and b_{c_i}, under which the 2.5th and 97.5th percentiles of the resulting distribution equal the endpoints of the 95% credible interval produced by O'Reilly *et al.* [153]. For countries where the reporting rate estimate was unavailable, we assumed the parameters for the reporting rate uncertainty distribution to be equal to those for French Guiana, which had the highest reporting rate estimate among all the countries included in O'Reilly *et*

al.'s study [153]. We considered this as a conservative approach, as it sought to avoid overestimating the weekly number of imported infections $M_{i\to j,t}$. Overall, countries not included in O'Reilly *et al.*'s study only accounted for less than 3% of the total number of autochthonous cases reported worldwide during 2015–2016. In the equations below, $U_{i,t}$ refers to the autochthonous case count reported by spatial unit i in Eweek t, and the same realization $\rho_{c_i}^{(s)}$ applied to all spatial units within country c_i and for all Eweeks within each simulation s:

$$\rho_{c_i}^{(s)} \sim Beta\left(a_{c_i}, b_{c_i}\right),$$

$$I_{i,t}^{(s)} = \frac{U_{i,t}}{\rho_{c_i}^{(s)}}.$$

Next, of the $v_{i\to j,t}$ air travellers with origin airport located in spatial unit i and destination airport located in spatial unit j during Eweek t, $v_{i\to j,t}[i]$ belonged to the local population of spatial unit i. In each simulation s, $v_{i\to j,t}^{(s)}[i]$ was drawn from a binomial distribution, with number of Bernoulli trials $v_{i\to j,t}$, and success probability π_{c_i,c_j}, where c_i and c_j referred to the countries to which spatial units i and j belonged respectively. Whilst for each country the UNWTO provided information on the arrivals of non-resident tourists/visitors at the national borders by country of residence, these data were incomplete and only allowed us to estimate the success probability π_{c_i,c_j} for less than 1.6% of the country pairs in our study. In addition, we did not find country-level indicators such as gross domestic product per capita to be informative for predicting π_{c_i,c_j} based on the very few estimates constructed from the UNWTO data. Hence, in each simulation s, $\pi_{c_i,c_j}^{(s)}$ was drawn from a beta distribution with a substantial amount of variation around 0.5 to capture parameter uncertainty, followed by $v_{i\to j,t}^{(s)}[i]$ that was drawn from a binomial distribution as previously described. We then subtracted $v_{i\to j,t}^{(s)}[i]$ from $v_{i\to j,t}$ to obtain $v_{i\to j,t}^{(s)}[j]$:

$$\pi_{c_i,c_j}^{(s)} \sim Beta(10,10),$$

$$v_{i\to j,t}^{(s)}[i] \sim Bin\left(v_{i\to j,t}, \pi_{c_i,c_j}^{(s)}\right),$$

$$v_{i\to j,t}^{(s)}[j] = v_{i\to j,t} - v_{i\to j,t}^{(s)}[i].$$

It should be noted that within each simulation s, the realization $\pi_{c_i,c_j}^{(s)}$ also applied to all other routes with the same origin and destination countries. In other words, given a pair of origin and destination countries, the aforementioned probability value was generated only once in each simulation.

Therefore, in simulation s, the first component of $M_{i\to j,t}^{(s)}$ was given by:

$$M_{i\to j,t}^{(s)}[i] \sim Bin\left(v_{i\to j,t}^{(s)}[i], \frac{I_{i,t}^{(s)}}{POP_i - \sum_{k\neq i} v_{i\to k,t}^{(s)}[k] \cdot LOS_{c_i}}\right).$$

The justification for the above calculation was as follows. In the denominator, POP_i denoted the total number of people located in spatial unit i at Eweek t (or any point in time, as previously discussed). Here, we assumed a stable system in which the weekly arrival rate of visitors from the local population of any spatial unit $k \neq i$ was equal to the rate at which they exited spatial unit i, $v_{i\to k,t}[k]$, for any Eweek t. On average, these individuals stayed in spatial unit i for LOS_{c_i} weeks during their visits based on the UNWTO's estimates. Applying Little's law originally developed in queueing theory [178], we multiplied $\sum_{k\neq i} v_{i\to k,t}^{(s)}[k]$ by LOS_{c_i} in each simulation s to obtain the estimated total number of people located in spatial unit i at Eweek t who did not belong to the local population of spatial unit i, which was then subtracted from POP_i. Therefore, $M_{i\to j,t}^{(s)}[i]$ can be considered as a realization of a random variable following a hypergeometric distribution, where $v_{i\to j,t}^{(s)}[i]$ random draws were obtained without replacement from a total number of $\left\{POP_i - \sum_{k\neq i} v_{i\to k,t}^{(s)}[k] \cdot LOS_{c_i}\right\}$ individuals, of whom $I_{i,t}^{(s)}$ carried ZIKV. Here, to ease the computation, we approximated $M_{i\to j,t}^{(s)}[i]$ by drawing from a binomial distribution instead. Note that we only included $I_{i,t}^{(s)}$ into the numerator of the success probability, since most autochthonous infections occurring prior to Eweek t had recovered by Eweek t, given a recovery rate of around 1/7–1/5 per day [150,152]. In addition, the aforementioned $I_{i,t}^{(s)}$ individuals were presumably still able to travel because the vast majority were unreported and likely to be asymptomatic.

Of the total (unobserved) number of incident autochthonous infections occurring in spatial unit i during Eweek t (denoted by $L_{i,t}$), $M_{i\to j,t}[j]$ were visitors belonging to the local population of spatial unit j. Conditioning on $L_{i,t}$, $M_{i\to j,t}[j]$ followed a binomial distribution with the number of Bernoulli trials $L_{i,t}$ and a success probability as a function of i, j, and t. Since $L_{i,t}$ was reasonably large (i.e. greater than 20 in most cases), we modelled $M_{i\to j,t}[j]$ as a Poisson random variable with mean parameter proportional to the total person-time at

risk of the visitors from spatial unit j, and similarly for the number of autochthonous infections occurring in spatial unit i during Eweek t that belonged to the local population of spatial unit i ($I_{i,t}$). Hence, the ratio between their expected values was given by:

$$\frac{\mathbb{E}\{M_{i\to j,t}[j]\}}{\mathbb{E}\{I_{i,t}\}} \cong \frac{v_{i\to j,t}[j]\cdot LOS_{c_i}}{(POP_i - \sum_{k\neq i} v_{i\to k,t}[k]\cdot LOS_{c_i})(1-\eta_{i,t})}. \quad (*)$$

At any point in time during Eweek t, the number of people located in spatial unit i was POP_i, of whom $(v_{i\to j,t}[j]\cdot LOS_{c_i})$ were visitors from spatial unit j by Little's law [178], and likewise $(POP_i - \sum_{k\neq i} v_{i\to k,t}[k]\cdot LOS_{c_i})$ belonged to the local population of spatial unit i as derived earlier. In the denominator of equation $(*)$, we included an additional factor $(1-\eta_{i,t})$, which denoted the percentage of the local population of spatial unit i that were still susceptible to ZIKV infection in Eweek t (i.e. $\eta_{i,t} = \sum_{t'<t} I_{i,t'} \,/\, POP_i$, assuming protective immunity to last at least until the end of our study period after primary ZIKV infection).

Hence, in each simulation s, given the "observed data" $I_{i,t}^{(s)}$, it can be shown that the posterior predictive distribution of $M_{i\to j,t}^{(s)}[j]$ was as follows, if we impose a uniform prior over the positive real line for $\mathbb{E}\{I_{i,t}\}$:

$$M_{i\to j,t}^{(s)}[j] \mid I_{i,t}^{(s)} \sim NB\left(I_{i,t}^{(s)}+1, \left[\left(\frac{\mathbb{E}\{M_{i\to j,t}[j]\}}{\mathbb{E}\{I_{i,t}\}}\right)^{(s)}+1\right]^{-1}\right).$$

In other words, $M_{i\to j,t}^{(s)}[j]$ was drawn from a negative binomial distribution that modelled the number of failures before the $\left(I_{i,t}^{(s)}+1\right)^{\text{th}}$ success in a sequence of independent Bernoulli trials with equal success probability $\left[\left(\frac{\mathbb{E}\{M_{i\to j,t}[j]\}}{\mathbb{E}\{I_{i,t}\}}\right)^{(s)}+1\right]^{-1}$, where we derived $\left(\frac{\mathbb{E}\{M_{i\to j,t}[j]\}}{\mathbb{E}\{I_{i,t}\}}\right)^{(s)}$ by plugging the previously simulated values $\left\{v_{i\to k,t}^{(s)}[k]\right\}_{k\neq i}$ and $\left\{I_{i,t'}^{(s)}\right\}_{t'<t}$ into equation (*).

Finally, the estimated weekly number of ZIKV infections imported from a *donor* spatial unit i to a *susceptible* spatial unit j in simulation s was given by the sum of its two components. Note that we treated the $M_{i\to j,t}^{(s)}[j]$ individuals as imported infections in Eweek t despite the possibility that a certain percentage may have returned to spatial unit j slightly after Eweek t, because no information was available regarding the distribution of LOS. Given that the average LOS was very short in general, the effect of this decision upon the subsequent analyses of the onward ZIKV spread for each susceptible spatial unit was negligible:

$$M_{i\to j,t}^{(s)} = M_{i\to j,t}^{(s)}[i] + M_{i\to j,t}^{(s)}[j].$$

To validate our importation simulation model, we plotted the median estimate of the total number of ZIKV infections imported in each susceptible spatial unit in the US in 2015–2016 against the corresponding reported case count, both on a $\log_{10}$ scale. If our simulation results were reasonably accurate and the reporting rates were similar among these susceptible spatial units, we would expect to see all the data points to be close to a straight line with a slope of one.

2.3.2.3 Reporting index

As a by-product of the importation simulation model, a reporting index was derived to quantify the probability of reporting a case per imported infection for each susceptible spatial unit. *Only within this section* did we further merge all the susceptible spatial units of a country to be analysed as a single unit, provided the reported *imported* cases could not be located to the subdivision level.

For each susceptible spatial unit j, we denoted the actual total number of imported ZIKV infections during 2015–2016 as n_j, where its uncertainty distribution can be approximated by $\sum_{i,t} M_{i\to j,t}^{(s)}$ $(s = 1, \ldots, 10000)$ as obtained from the importation simulation model. The reporting index p_j was assigned an uninformative prior $Beta(1,1)$ and assumed to be independent from variable n_j. Thus, given a specific value of n_j, the reported imported case count x_j followed a beta-binomial distribution with uniform density:

$$f(x_j|n_j) = \frac{1}{n_j + 1}, \qquad \text{where } x_j = 0, \ldots, n_j.$$

Using Bayes' theorem, we can compute $f(n_j|x_j)$ as follows, where $I(\cdot)$ was the indicator function:

$$f(n_j|x_j) = \frac{f(n_j)f(x_j|n_j)}{f(x_j)} \propto f(n_j) \cdot \frac{1}{n_j + 1} \cdot I(n_j \geq x_j).$$

The posterior density of p_j was given by the equation below:

$$f(p_j|x_j) = \int f(n_j|x_j) \cdot f(p_j|n_j, x_j)\, dn_j.$$

Hence, for each susceptible spatial unit j, we generated 10,000 values from the posterior distribution of the reporting index. Specifically, for $s^* = 1, \ldots, 10000$:

i. Draw $n_j^{(s^*)}|x_j$ from the set of simulated numbers of imported infections $\sum_{i,t} M_{i\to j,t}^{(s)}$ $(s = 1, \ldots, 10000)$, with probabilities proportional to the weights $\frac{1}{\sum_{i,t} M_{i\to j,t}^{(s)}+1} \cdot I(\sum_{i,t} M_{i\to j,t}^{(s)} \geq x_j)$.

ii. Draw $p_j^{(s^*)}|n_j^{(s^*)}, x_j$ from $Beta\left(x_j + 1, n_j^{(s^*)} - x_j + 1\right)$.

It should be noted that we found a total of four countries that contained at least one donor spatial unit and one susceptible spatial unit simultaneously. For these countries, we estimated that only an average of 0.47 infections per susceptible spatial unit belonged to "within-country importation", with New York having the largest number (6.40, which was less than 0.03% of its estimated total number of imported ZIKV infections). Hence, this did not affect the estimation of reporting index, where the reported case data presumably only included Zika cases imported from abroad.

2.3.2.4 Estimation of basic reproduction number

We applied a global risk model developed by Caminade *et al.* to estimate the daily R_0 of ZIKV within each spatial unit [152], which was subsequently averaged across each Eweek to be used for the analysis of onward ZIKV spread in section 2.3.2.5. The model inputs included the human recovery rate (r), as well as a set of species-specific parameters: biting rates (h_1, h_2), vector preferences (ϕ_1, ϕ_2), vector-to-host and host-to-vector transmission probabilities ($\tau_1^{V\to H}$, $\tau_2^{V\to H}$ and $\tau_1^{H\to V}$, $\tau_2^{H\to V}$), mortality rates (μ_1, μ_2), extrinsic incubation rates (ν_1, ν_2), and vector-to-host ratios (m_1, m_2) [152], where the subscripts 1 and 2 referred to *Ae. aegypti* and *Ae. albopictus* respectively (Table 2.1). The expected number of secondary *human* ZIKV infections generated by a primary human infection in a fully susceptible population was given by:

$$R_0 = \sum_{q=1}^{2} \left(\frac{\tau_q^{V\to H}\tau_q^{H\to V}h_q^2}{\mu_q(T)}\right)\left(\frac{\nu_q(T)}{\nu_q(T)+\mu_q(T)}\right)\left(\frac{\phi_q^2 m_q}{r}\right).$$

Symbol	Description	Calculation
h_1, h_2	Biting rates (per day) [#]	$h_1 = 0.67$ [179], $h_2 = h_1/2.$
ϕ_1, ϕ_2	Vector preferences	$\phi_1 = 1$, $\phi_2 = \phi_1/2.$
$\tau_1^{V\to H}$, $\tau_2^{V\to H}$	Vector-to-host transmission probability	$\tau_1^{V\to H} = 0.5$, $\tau_2^{V\to H} = \tau_1^{V\to H}.$
$\tau_1^{H\to V}$, $\tau_2^{H\to V}$	Host-to-vector transmission probability	$\tau_1^{H\to V} = 0.1$, $\tau_2^{H\to V} = 0.33\tau_1^{H\to V}.$
μ_1, μ_2	Mortality rates (per day)	$\mu_1 = \frac{1}{1.22 + exp(-3.05 + 0.72T)} + 0.196, \quad (T < 22\,°C)$ $\mu_1 = \frac{1}{1.14 + exp(51.4 - 1.3T)} + 0.196, \quad (T \geq 22\,°C)$ $\mu_2 = \frac{1}{1.1 + exp(-4.04 + 0.576T)} + 0.11883, \quad (T < 15\,°C)$ $\mu_2 = 0.000339T^2 - 0.0189T + 0.336, \quad (15°C \leq T < 26.3°C)$ $\mu_2 = \frac{1}{1.065 + exp(32.2 - 0.92T)} + 0.073079. \ (T \geq 26.3\,°C)$
ν_1, ν_2	Extrinsic incubation rates (per day)	$\nu_1 = \frac{1}{4 + exp(5.15 - 0.123\,T)}$, $\nu_2 = \nu_1 \,/\, 1.03.$
m_1, m_2	Vector-to-host ratios	Assumed to be proportional to the vector suitability values. The scaling factor was re-calibrated using the R_0 estimate produced by Zhang *et al.* [150].
r	Human recovery rate (per day)	$r = 1/7.$

Table 2.1: Parameters for the global risk model. Subscripts 1 and 2 referred to *Ae. aegypti* and *Ae. albopictus* respectively. Unless otherwise stated, parameter values were calculated following Caminade *et al.* [152].

[#] Given the relatively minor contribution of the biting rates' temperature-dependence to the variation in R_0 [180], here we treated the biting rates as constant, and the uncertainties in $\frac{h^2\phi^2\tau^{V\to H}\tau^{H\to V}}{r}$ were absorbed by re-calibrating the scaling factor that converted vector suitability values to vector-to-host ratios.

We followed Caminade *et al.* [152] and estimated vector mortality rates and extrinsic incubation rates as a function of the population-weighted average of temperature for each spatial unit and each day. We re-calibrated the scaling factor that transformed vector suitability values to vector-to-host ratios, so that our median R_0 estimate at the temperature 25 °C in French Polynesia would equal to that obtained by Zhang *et al.* via the fitting of a deterministic model to the 2013 French Polynesia outbreak data [150]. All the other parameters were assumed to be constant. Despite the uncertainty in these parameter values, the

factor $\frac{h_1^2\phi_1^2\tau_1^{V\to H}\tau_1^{H\to V}}{r}$ can be viewed as a single constant whose uncertainty was absorbed by the aforementioned model re-calibration (Similarly for *Ae. albopictus*, assuming the estimated ratio between the two species for each of the model parameters $h, \phi, \tau^{V\to H}$, and $\tau^{H\to V}$ in Table 2.1 to be reasonably accurate).

Instead of producing a single daily R_0 estimate for each spatial unit, we generated 250 values of R_0 per spatial unit and per day, where each value was based on a bootstrap estimate of the population-weighted average of vector suitability within the corresponding spatial unit. To validate the model, we took published estimates of R_0 and compared them with our model estimates to assess the agreement (Refer to the Appendix A [179] for the inclusion and exclusion criteria for the published R_0 estimates used for the model validation) [162,171,181,182].

2.3.2.5 Susceptible spatial units with cryptic spread of ZIKV

We calculated the probability that no onward spread of ZIKV occurred following importation during 2015–2016, for each susceptible spatial unit with at least one imported infection throughout the 10,000 simulations. Specifically, in each iteration $s = 1, \ldots, 10000$, we first computed the weekly effective number of imported infections for each susceptible spatial unit j, as defined by Perkins *et al.* [183]. This accounted for the fact that the generation time of ZIKV can be much longer than that of directly transmitted viruses, and hence each human ZIKV infection can be caused by an earlier infection that had occurred one to five weeks before [183].

$$\omega_u = \frac{1}{7}\int_{7(u-1)}^{7u} \{F_{ZIKV}(x+7) - F_{ZIKV}(x)\}\, dx,$$

$$\widetilde{M^{(s)}_{\bullet\to J,t}} = \sum_{u=1}^{5} \delta_j[t-u, t] \cdot \omega_u \cdot M^{(s)}_{\bullet\to j,t-u}.$$

In the first equation, the cumulative distribution function of the ZIKV generation time was denoted by F_{ZIKV}. This was estimated by Ferguson *et al.* and subsequently used by Harris *et al.* to obtain the weight parameters ω_u's [149,184]. In the second equation, $M^{(s)}_{\bullet\to j,t-u}$ referred to the number of ZIKV infections imported in spatial unit j during Eweek $(t-u)$ in simulation s, summed over all the donor spatial units. The extra factor $\delta_j[t-u, t]$ was defined to be one if the weekly mean temperature in spatial unit j remained within the thermal range for ZIKV transmission from Eweeks $(t-u)$ to t, and zero otherwise. In other words, sustained temperature

suitability for ZIKV transmission was required from the time of each imported infection to its secondary human infection. Here, we used Tesla *et al.*'s thermal range estimate (22.7°C–34.7°C), which took into account the temperature constraints for different stages of mosquito development and ZIKV transmission, and has been validated using Zika case data reported by different municipalities in Colombia [77]. We additionally created a highly conservative scenario where a much narrower interval (26°C–29°C) was considered based on Mordecai *et al.*'s estimated temperature range for maximal ZIKV transmission [76], to identify susceptible spatial units with a particularly high likelihood of cryptic ZIKV spread.

The analysis of onward ZIKV spread presented below was largely inspired from Churcher *et al.* [185]. If we assumed the entire local population to be immunologically naïve, the number of infections generated in Eweek t by the $\widetilde{M_{\bullet\to J,t}^{(s)}}$ imported infections followed a negative binomial distribution with expectation $\widetilde{M_{\bullet\to J,t}^{(s)}} \cdot R_{0_{j,t}}^{((s-1)\,\%\,250+1)} \cdot I\left[aeg_j^{((s-1)\,\%\,250+1)} \geq 0.24\right]$ and size $\widetilde{M_{\bullet\to J,t}^{(s)}} \cdot k$. Here, we additionally required the bootstrap estimate of the environmental suitability for *Ae. aegypti* (i.e., the main vector for ZIKV) in spatial unit j, denoted by $aeg_j^{((s-1)\,\%\,250+1)}$, to be at least 0.24. This threshold was derived by Dickens *et al.* that maximized the true skill statistic, which can be used to convert suitability value into binary species presence/absence [177]. The parameter k referred to the over-dispersion parameter of the offspring distribution, and a smaller value would lead to a larger variation in the number of secondary infections produced by each primary infection, and a higher chance of zero offspring. Due to the unavailability of the over-dispersion parameter estimates derived from Zika case data, we took a conservative approach and set k to be 0.1 (i.e. highly over-dispersed offspring distribution) following previous work by Grubaugh *et al.* [160], to minimize false positives in our final results. In each iteration $s = 1, \ldots, 10000$, we derived the probability that no onward spread of ZIKV occurred following importation in spatial unit j during 2015–2016, denoted by $\theta_j^{(s)}$. These 10,000 probability values were then averaged to obtain $\bar{\theta}_j$, representing the probability of no onward ZIKV spread in spatial unit j averaged over the uncertainty in the weekly numbers of imported infections and local ZIKV transmissibility:

$$\theta_j^{(s)} = \prod_t \left(1 + \frac{R_{0_{j,t}}^{((s-1)\,\%\,250+1)} \cdot I\left[aeg_j^{((s-1)\,\%\,250+1)} \geq 0.24\right]}{k}\right)^{-\widetilde{M_{\bullet\to J,t}^{(s)}} \cdot k},$$

$$\bar{\theta}_j = \frac{1}{10000} \sum_{s=1}^{10000} \theta_j^{(s)}.$$

It should be noted that in the above calculation, we ignored ZIKV immunity among the local population of susceptible spatial unit j accumulated via weekly ZIKV importation. This was because the imported number of ZIKV infections $\sum_{i,t} M_{i \to j,t}\,[j]$ over 2015–2016 was estimated to account for at most ~0.03% of the entire local population among all the susceptible spatial units j, and hence was negligible.

Under each of the thermal restrictions previously specified, we obtained a list of susceptible spatial units with a high chance of cryptic ZIKV spread during 2015–2016 through the following steps. First, we only retained all the susceptible spatial units with $\bar{\theta}$ below 0.05, and considered any $\bar{\theta}$ values below $0.05/|\{\bar{\theta}\}|$ as highly significant findings (provided the spatial unit was also retained in the subsequent filtering process). The calculation of $0.05/|\{\bar{\theta}\}|$ can be considered as a Bonferroni correction to control for the Type I error, where $|\{\bar{\theta}\}|$ referred to the total number of spatial units for which $\bar{\theta}$ was computed. Next, we conducted a literature review to remove any remaining susceptible spatial units where no evidence of historical or current spread of dengue, Chikungunya, or yellow fever was found, as these spatial units were assumed to have very limited potential for ZIKV transmission. Susceptible spatial units without evidence of *Ae. aegypti* establishment based on the existing literature were also excluded even if *Ae. albopictus* was present, since the former is known to be the primary vector for ZIKV transmission.

Finally, we additionally excluded French Polynesia, where a large proportion of the local population had developed immunity against ZIKV due to a large outbreak there in 2013 [186], as our model implicitly assumed the ZIKV seroprevalence for each susceptible spatial unit to be zero at the start of 2015. Any remaining susceptible spatial units with evidence of historical ZIKV transmission but no outbreaks were intentionally retained, since the population-level immune landscape at the beginning of 2015 in these spatial units was still poorly understood in general due to incomplete or outdated serological data.

2.4 Results

2.4.1 Model validation results

We observed a high correlation between the estimated total number of imported ZIKV infections and the reported case count during 2015–2016 for each susceptible spatial unit in the US (Pearson $r = 0.913$ for $\log_{10}$ transformed data). Except two data points, where either the reported or the estimated value was zero, the rest of the observations were close to a straight line with a slope of one (Figure 2.2A), suggesting our simulation results were reasonably accurate provided the reporting rates were not drastically different among these susceptible spatial units within the same country. In addition, there was overall a good agreement between the median R_0 estimates derived from the global risk model and those obtained from the literature, with the majority falling within the ±0.3 error band (Figure 2.2B). Refer to Figures 2.3 and 2.4 for visualizations of the global risk model outputs.

2.4.2 Estimated number of imported ZIKV infections & reporting index

The majority of the susceptible spatial units in Africa and Asia presented relatively low numbers of imported ZIKV infections based on our simulations, which were generally consistent with the reported case counts during 2015–2016 (Figure 2.5). However, a few exceptions within these continents were found. For example, we estimated that 432 (332–582) ZIKV infections were imported in Shanghai, China, although not a single imported case was reported by the city. Similarly, an estimated number of 473 (322–734) ZIKV infections were imported in the Luanda spatial unit in Angola during 2015–2016 without being detected. As a by-product of the importation simulation model, we estimated a reporting index for each susceptible spatial unit, to quantify the probability of reporting a case with each importation (Appendix A: Table S2.1). Note that for some countries presented in Figure 2.5 and Appendix A: Table S2.1, we were only able to find the total reported number of imported Zika cases at a country level (e.g., Australia, New Zealand). To facilitate the comparison of the reported and simulated data, each of these countries was treated as a single unit (only in Figure 2.5 and Appendix A: Table S2.1).

2.4.3 Mapping the cryptic spread of ZIKV during 2015–2016

A total of 24 susceptible spatial units were identified to have a high chance of cryptic ZIKV spread during 2015–2016 under our primary scenario, of which 21 (87.5%) are located in Asia or Africa (Table 2.2, Figure 2.6). In

particular, there were 10 susceptible spatial units estimated to have a significant risk of cryptic ZIKV spread during 2015–2016 even under the highly conservative scenario (i.e., thermal restriction for ZIKV transmission being 26°C–29°C), namely, Luanda in Angola, Banten in Indonesia, Maharashtra in India, Lagos in Nigeria, Taiwan and Guangdong in China, Dakar in Senegal, Maputo in Mozambique, Kinshasa in Congo DRC, and Pool in Congo (Table 2.2, Figure 2.6). Of all the 24 susceptible spatial units identified, we found 8 where there exists evidence of historical ZIKV circulation prior to 2015, although in most cases the serological data were collected many decades ago (Table 2.2) [187–192]. Refer to Appendix A: Table S2.2 for the estimated probability that no onward spread of ZIKV occurred following importation during 2015–2016 in each susceptible spatial unit.

Note that some spatial units in our study may represent a group of subdivisions within a country. Within each of these spatial units, only one subdivision had nonzero incoming air passengers during 2015–2016 (see Methods for more details), which was assigned as the name for that spatial unit. For example, the susceptible spatial unit named “Luanda” in Table 2.2 contained four other subdivisions in Angola in addition to the Luanda subdivision itself. The list of all the first-level country subdivisions within each susceptible spatial unit was provided in Appendix A: Table S2.3.

Country	Spatial unit name	Estimated probability ($\bar{\theta}$) that no onward spread of ZIKV had occurred during 2015–2016	Evidence of historical ZIKV circulation prior to 2015 (Y/N)
Angola	Luanda*	<0.0001	Y [187]
Indonesia	Banten*	<0.0001	Y [188]
India	Maharashtra*	<0.0001	Y [189]
Nigeria	Lagos*	<0.0001	Y [190]
China	Taiwan*	<0.0001	N
China	Guangdong*	<0.0001	N
India	Karnataka	0.0002	N
India	Tamil Nadu	0.0003	Y [189]
India	Delhi	0.0004	N
United States	Hawaii	0.0004	N
Senegal	Dakar*	0.0004	Y [191]
Uruguay	Canelones	0.0004	N
Réunion	Réunion	0.0004	N
Australia	Queensland	0.0005	N
Indonesia	Bali	0.0007	N
Mozambique	Maputo*	0.0007	Y [192]
Ghana	Greater Accra	0.0018	N
Congo DRC	Kinshasa*	0.0019	N
India	Kerala	0.0050	N
Saudi Arabia	Makkah	0.0115	N
India	Goa	0.0130	N
India	Telangana	0.0190	Y [189]
Congo	Pool*	0.0281	N
Côte d'Ivoire	Abidjan	0.0470	N§

Table 2.2: Susceptible spatial units with a high chance of cryptic ZIKV spread during 2015–2016 (Refer to Appendix A: Table S2.3 for the list of first-level country subdivisions belonging to each susceptible spatial unit).

* $\bar{\theta}$ remained lower than 0.05 even within the highly conservative scenario.

§ There was evidence of historical ZIKV circulation prior to 2015 in Côte d'Ivoire, but it was not clear whether any of the positive serum samples were collected from the Abidjan spatial unit or not.

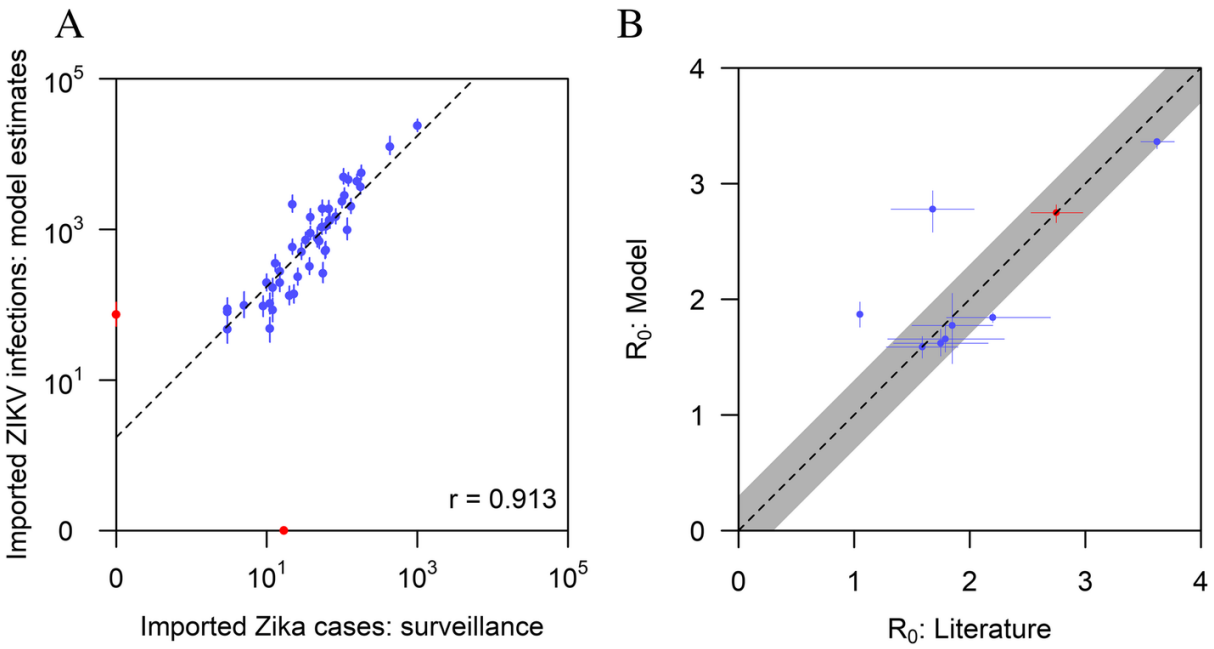


Figure 2.2: Model validation results. A) Validation of the importation simulation model. The dashed line was based on a linear regression model with the slope fixed to be one fitted to all observations on a $\log_{10}$ scale excluding the two red dots, which show that the reported or the estimated value was zero. B) Comparison of R_0 values obtained from our estimation versus the existing literature. The grey polygon denotes the ±0.3 error band and the red dot refers to the R_0 estimate at the temperature 25 °C in French Polynesia by Zhang *et al.*, which was used to calibrate the model.

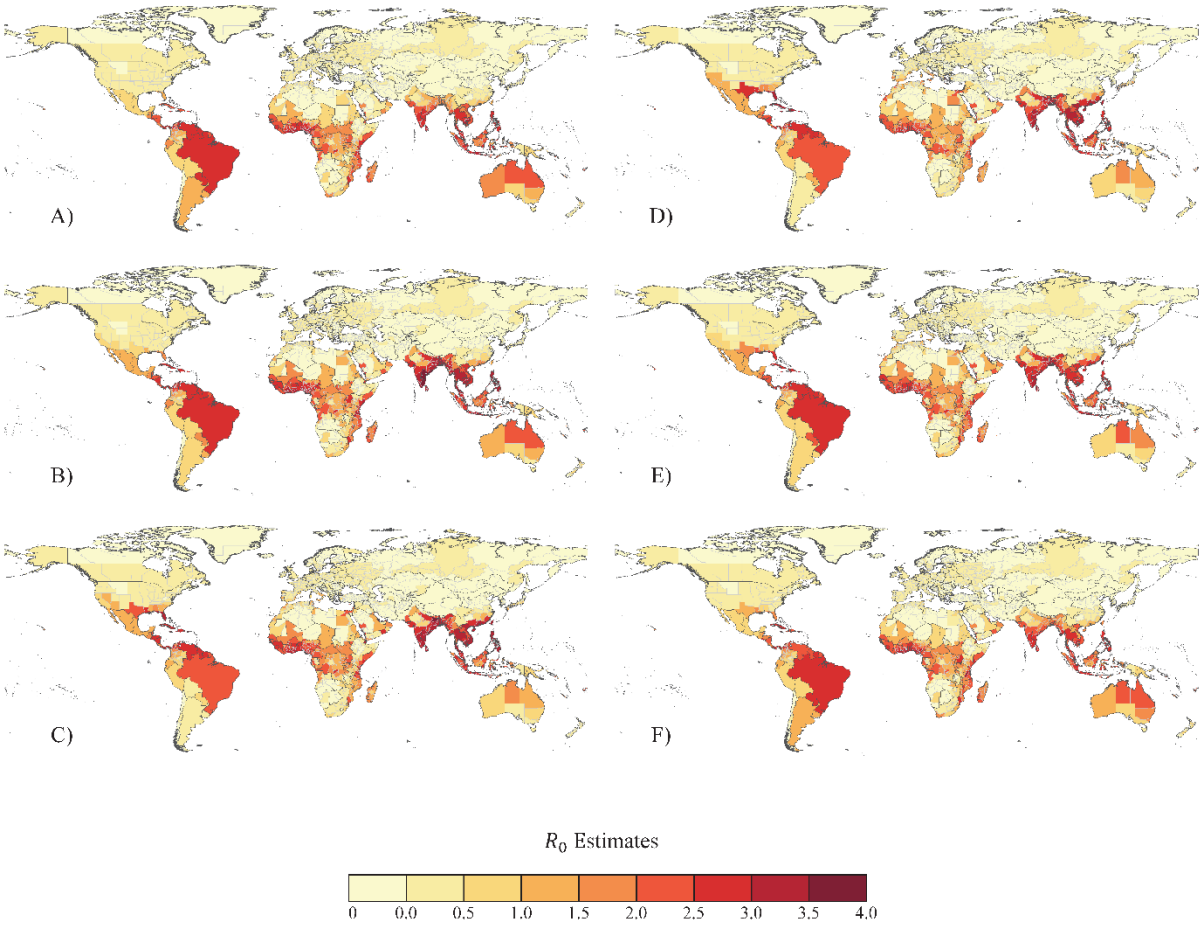


Figure 2.3: Median estimate of ZIKV R_0 for each spatial unit obtained from the global risk model at Eweeks (A) 2, (B) 12, (C) 22, (D) 32, (E) 42, and (F) 52 in 2016 respectively.

Note that these estimates did not incorporate local evidence of *Aedes*-borne disease transmission potential or thermal restrictions for ZIKV transmission under different scenarios. Additional adjustment steps were implemented in the analysis of onward ZIKV spread to minimize false positive rates (Refer to the Methods section for more details).

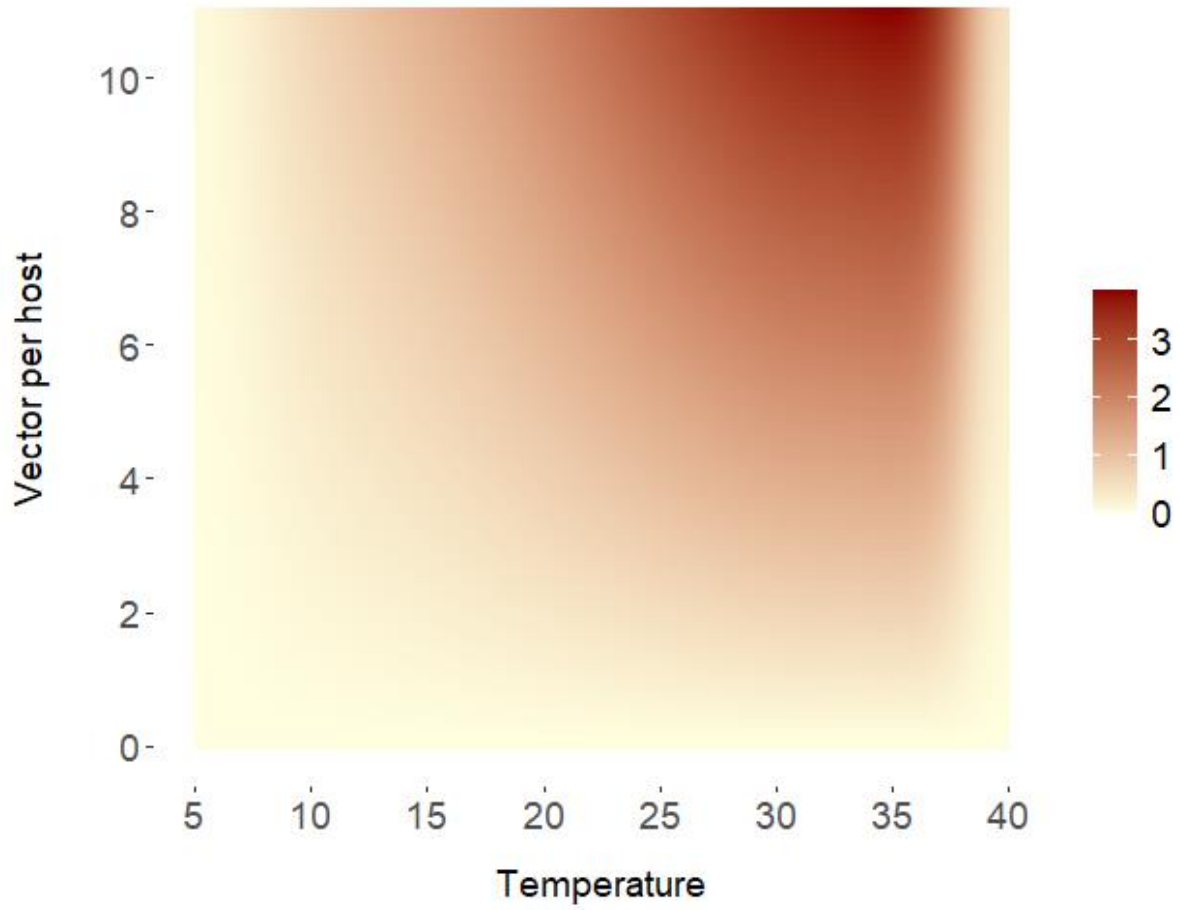


Figure 2.4: Estimated ZIKV R_0 presented as a function of temperature and vector-to-host ratio*.

For demonstration purposes, we assumed an equal vector-to-host ratio for *Ae. aegypti* and *Ae. albopictus*, and the y-axis value refers to the vector-to-host ratio for *each* species. Note that these estimates were preliminary results only, which did not incorporate the thermal restrictions for ZIKV transmission under different scenarios. An additional adjustment step was implemented in the analysis of onward ZIKV spread to minimize false positive rates (Refer to the Methods section for more details).

* The 2.5th, 25th, 50th, 75th, and 97.5th percentiles of the estimated vector-to-host ratios for all the spatial units were: 0.00, 0.71, 2.00, 6.41, and 10.70 respectively for *Ae. aegypti* and 0.00, 0.63, 1.63, 5.52, and 10.17 respectively for *Ae. albopictus*.

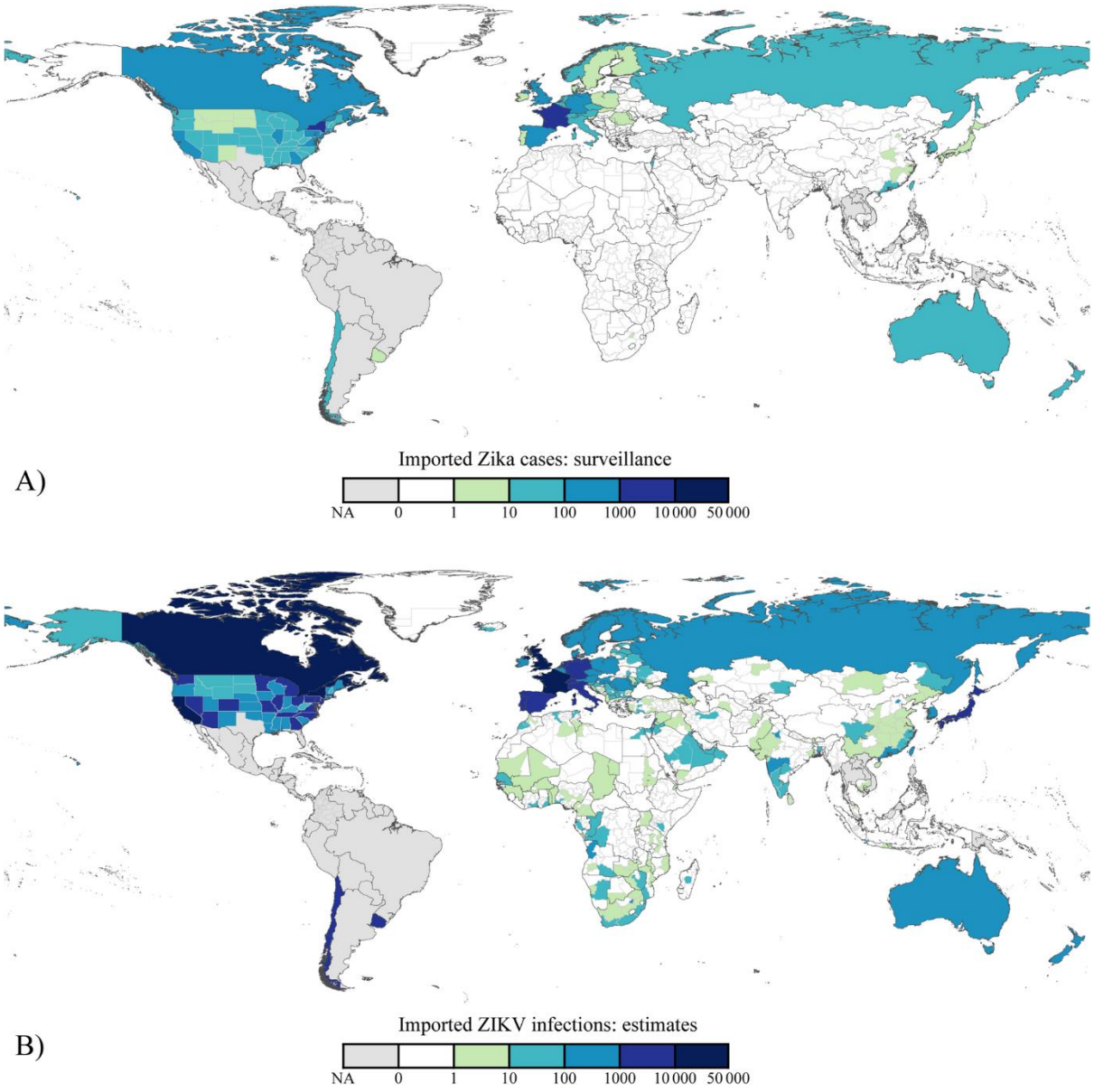


Figure 2.5: ZIKV importation in each susceptible spatial unit during 2015–2016: A) reported case count vs B) median estimate of the total number of imported infections derived from the simulation model. Each country was treated as a single unit if the imported case count was reported at a country level (e.g., Australia, New Zealand), to facilitate comparison of the two maps. All donor spatial units were coloured in grey.

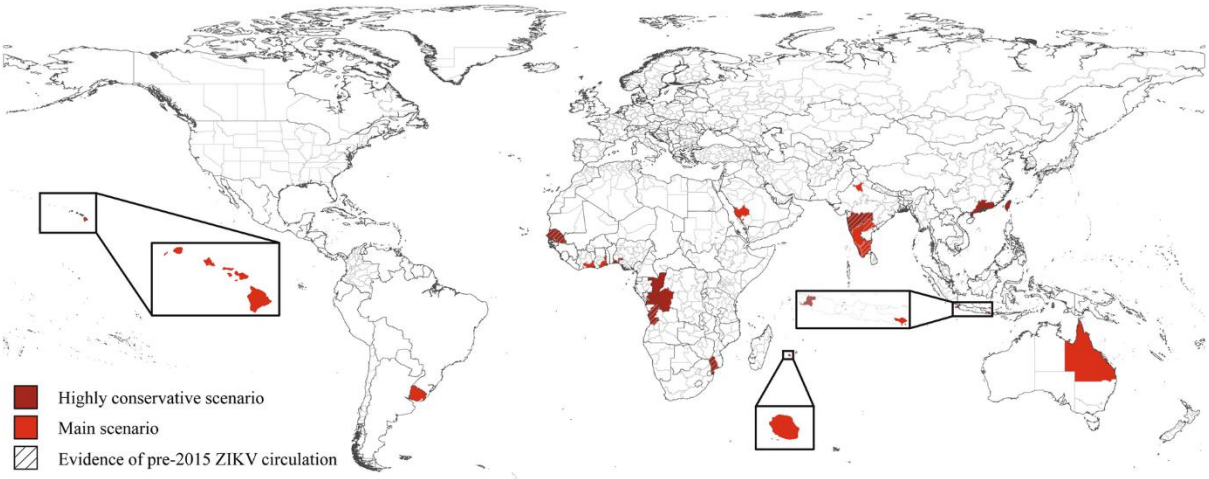


Figure 2.6: Susceptible spatial units with a high chance of cryptic ZIKV spread during 2015–2016 (shown in red). If evidence of historical ZIKV circulation prior to 2015 was found in any of these locations, the corresponding polygon is striped. Susceptible spatial units coloured in dark red refer to those that continue to have high potential for cryptic ZIKV spread during 2015–2016 under the highly conservative scenario (i.e., thermal restriction for ZIKV transmission being 26°C–29°C).

2.5 Discussion

This study unveils the potential cryptic spread of ZIKV during 2015–2016 to further our understanding of the virus' circulation worldwide. This was achieved via integrating heterogeneous data sources to model ZIKV's transmissibility, importation, and onward spread, combined with the evidence of *Aedes*-borne disease transmission potential from the existing literature. Although ZIKV has received much less media attention ever since the epidemic waned, our results highlight the geographical areas where future studies may be needed to investigate if past or ongoing ZIKV circulation is present, in preparation for possible epidemics in the future.

Overall, the global distribution of the number of imported ZIKV infections based on our simulations was consistent with the importation risk estimates derived by Nah *et al.* [193], where countries in Western Europe and the Americas experienced a higher risk of importation in contrast to the majority of Asian and African countries. Several spatial units in Asia and Africa, however, were estimated to have a large number of imported ZIKV infections, such as Shanghai in China and Luanda in Angola, in agreement with the findings of Bogoch *et al.* [155]. Different from the previous studies, we directly estimated the absolute number of imported infections at a reasonably high spatial resolution, and our results were found to correlate very well with the imported case counts reported by different susceptible spatial units in the US.

To date, new evidence of ZIKV transmission continues to accumulate, which enables our estimated geographical extent of cryptic ZIKV spread to be partially validated. On 26 December 2016, the first Zika case in Angola was identified in a 14-year-old boy residing in Luanda, immediately after the initiation of ZIKV RT-PCR testing in the country [194]. Correspondingly, our model successfully detected autochthonous ZIKV transmission in Luanda, with the estimated probability of no indigenous cases occurring during 2015–2016 being the lowest (8.53×10^{-31}) among all the 24 susceptible spatial units listed in Table 2.2. Note that we classified Luanda as a susceptible spatial unit in our study, since the first detected case was not announced until the early January of 2017 [195]. Besides Luanda, Tamil Nadu in India reported its first autochthonous Zika case in July 2017 [196]. Based on our model, 31 (19–46) ZIKV infections were imported in Tamil Nadu during 2015–2016, with the chance of no onward spread following these imported infections being low (0.0003). Thus, it was possible that ZIKV had already circulated silently in Tamil Nadu by the end of 2016, which later gave rise to the first reported case in 2017. Moreover, in a recent study published in October 2019, low ZIKV seroprevalence was observed in serum samples collected from Southern Taiwan at the end of 2015, providing evidence of possible undetected autochthonous transmission [197]. Overall, the abovementioned epidemiological evidence adds to the credibility of our results, and underscores the necessity of local investigation in the rest of the susceptible spatial units that were found to have a high potential of cryptic ZIKV spread during 2015–2016.

In Africa, information on the ZIKV seroprevalence and incidence remains limited and largely outdated, but recent evidence suggests that the virus has been silently spreading in the continent for at least two decades [191]. According to Herrera *et al.*, of the 387 serum samples collected from febrile patients in Nigeria and Senegal during 1992–2016, 6.2% were positive for IgM to ZIKV and negative for dengue reactivity, with four samples from which ZIKV envelope was amplified [191]. Thus, there are at least two plausible interpretations for the 9 susceptible spatial units in Africa listed in Table 2.2. These spatial units could have indeed experienced ZIKV spread during 2015–2016, which was undetected due to limited surveillance capacity. Alternatively, the level of herd immunity prior to 2015 may have been higher than currently known (due to paucity of serological data), thereby successfully preventing ZIKV transmission during 2015–2016. In either case, a significant knowledge gap remains to be filled, and this is particularly critical in light of the past or current spread of dengue, Chikungunya, or yellow fever viruses in these spatial units, which shows that the future risk of ZIKV epidemics should not be ignored. For example, a dengue outbreak occurred in Abidjan city, Côte d'Ivoire in

2019, where a total of 1,776 suspected dengue cases were reported between 1 January and 25 June [198]. Recent entomological investigations carried out in Abidjan also revealed a very high larval index of *Ae. aegypti*, the primary vector species of ZIKV [199]. Elsewhere, a cross-sectional study conducted between November 2015 and June 2016 in Kinshasa, Democratic Republic of Congo, found that 30.2% and 26.4% of the participants had experienced past dengue and Chikungunya infections respectively [200]. Although no acute ZIKV infections were found, of note is that much fewer blood samples (n=80) were tested for ZIKV, with no urine samples collected and only RT-PCR performed [200]. Thus, the possibility of low-level cryptic transmission of ZIKV in Kinshasa remains to be investigated.

In South Asia, the presence of ZIKV antibodies in humans was first discovered in India in 1952 [189]. More recently, India reported its first three laboratory-confirmed indigenous Zika cases in Gujarat on 15 May 2017, months after they were detected [201]. Subsequently, another case was found in Tamil Nadu during the same year [196]. After testing over 35,000 febrile illness serum samples, the Indian government had only identified the aforementioned four cases by 2017, indicating a very low level of ZIKV circulation [202]. Interestingly, we have found seven spatial units in the country with a high potential of cryptic ZIKV spread during 2015–2016, and these included Tamil Nadu but not Gujarat. Based on the representative partial sequences, the virus from Gujarat seems to belong to an old Asian strain, in contrast to the ZIKV in Tamil Nadu, which was classified as the then-current Asian outbreak strain [203]. It is thus possible that the unreported cases in Gujarat were not imported from any of the donor spatial units in our study, and hence were not detected by our model. The four Zika cases reported in 2017, together with the following 2018 outbreaks in Rajasthan and Madhya Pradesh [203] demonstrates the importance of strengthening ZIKV surveillance in other states as well, especially the ones where we found a high potential for cryptic spread of ZIKV. Given the considerably high intensity of dengue transmission in India [204], any low-level spread of ZIKV could be easily undetected due to misdiagnosis, until the virus causes a major outbreak possibly in the future.

In Southeast Asia, data on ZIKV epidemiology is scarce despite the long-term circulation of the virus [146]. Our study has identified two provinces in Indonesia—Bali and Banten—where cryptic ZIKV spread was likely to have occurred during 2015–2016. The earliest serological evidence of ZIKV transmission within the country was obtained in Central Java and Lombok around the 1970s [205,206], and the virus was found to remain circulating in the early 2010s [188]. Specifically, among the 662 samples collected from healthy children aged 1–4 years in

2014, 9.1% were ZIKV-seropositive, who came from 11 provinces including Banten but not Bali [188]. The total number of serum samples obtained from these two provinces was limited (67 combined) [188], and the level of ZIKV immunity among the general population was still unclear. Besides Southeast Asia, we also found two spatial units in East Asia that may require future risk assessment, namely Guangdong and Taiwan. Both spatial units have reported dengue outbreaks within the past few years [207,208], as well as imported ZIKV infections as recently as the early 2020 [209,210]. These, combined with the latest discovery of low ZIKV seroprevalence in southern Taiwan and a neighbouring province of Guangdong [197,211], suggest that the potential risk of future ZIKV outbreaks should not be overlooked.

In West Asia, our results highlight only one spatial unit that requires special attention, namely Makkah. In 2009, a dengue epidemic occurred in the city, with a 20-fold increase in the incidence rate compared with the year before, and the disease became endemic since then [212], suggesting the local environment could also be suitable for ZIKV transmission. We estimated that at least 22 (12–36) ZIKV infections were imported in Makkah during 2015–2016, with a very low chance of no onward transmission (0.012). Notably, the city receives ~200,000 pilgrims from Indonesia annually [213], where ZIKV is presumed to be endemic [188], thus facing a potentially ongoing risk of ZIKV importation every year. To date, no indigenous Zika cases in Makkah have been reported, but serological data indicate that the virus may have circulated elsewhere within the country. In a recent study involving 410 asymptomatic pregnant women in Najran, 24 tested positive for anti-ZIKV IgM and 52 were anti-ZIKV IgG positive, none of whom had travelled abroad [214]. Although no acute infections were detected using RT-PCR and cross-reactivity with dengue virus could not be excluded, the study points to the possibility of silent ZIKV spread in Saudi Arabia, which needs to be further assessed by larger studies at the nationwide level [214].

Outside of Africa and Asia, we found high potential for cryptic ZIKV spread during 2015–2016 in the following spatial units: Canelones in Uruguay (which covers the vast majority of the country), Hawaii in the United States, and Queensland in Australia. Being one of the very few countries in the Americas without any ZIKV transmission reported to date [146], Uruguay also did not find any autochthonous dengue cases until an outbreak which occurred in the early 2016 [215]. The high estimated risk of cryptic ZIKV spread in the Canelones spatial unit during 2015–2016 was likely to be attributed to the very large number of imported ZIKV infections, which we estimated to be 2,043 (1,537–2,913). In contrast, both Hawaii and Queensland were estimated to have much

fewer imported ZIKV infections but meanwhile higher environmental suitability for *Ae. aegypti*, with multiple historical dengue epidemics reported to date [216,217], and hence may face a higher risk of local ZIKV outbreak in the future.

Zika has not gone away. It is thought that low-level transmission is still ongoing in many countries in the world [146]. The identification of locations with likely cryptic spread of ZIKV during the 2015–2016 global epidemic can have significant public health implications, as evidenced by the previous silent circulation of the virus in northeast Brazil that eventually caused a huge epidemic nationally and internationally [218]. Even in spatial units where winter months exist (e.g., Guangdong), ZIKV can be transmitted vertically in *Ae. aegypti* as a means of survival under adverse conditions, and thus the epidemiological importance of cryptic ZIKV spread in these areas should not be discounted [219]. In addition, improving knowledge about global ZIKV spread could also contribute to a better understanding of dengue epidemiology, given the immune interactions between these closely-related flaviviruses as demonstrated by a recent study [52].

Our study must be interpreted in light of the following limitations. For each susceptible spatial unit, we were unable to quantify the population-level immunity immediately prior to the 2015–2016 global ZIKV epidemic, largely owing to the paucity of updated serological data, which also include antibody titre data that are needed to account for the effect of pre-existing dengue immunity on the susceptibility to ZIKV infection in each location [220]. This is compounded by the complex effect of vector-virus genetic interactions on ZIKV transmissibility [221], which is still poorly understood in general and outside the scope of our study. In addition, we did not account for land or sea transport in our analysis, and assumed their impact on our estimates to be limited given that most ZIKV infections that were imported from *donor* to *susceptible* spatial units involved air travel, although short-range commuting flows could play a more important role in the epidemic invasion path once the virus was introduced to a new country or continent [222]. We also did not consider importation caused by infected passengers in transit who entered a country as short-term visitors before reaching their final destinations, due to the generally short LOS presumed for these relatively small number of individuals. Besides, our study did not account for the impact of weather variables such as rainfall on the seasonal variation in vector-to-host ratios, which is challenging to model on a global scale and can be further explored in future work.

Finally, we relied on only the locations with reported indigenous Zika cases as sources of importation, and hence both the estimated number of imported infections and the geographical extent of cryptic ZIKV spread during

2015–2016 were conservative. In other words, it is not possible to reveal all the locations at a high risk of cryptic ZIKV spread in a single modelling study, and the findings generated in this study were conditional on the currently available data only. For instance, it is thought that ZIKV may be endemic in many parts of Southeast Asia, but epidemiological data are seriously lacking [146], thereby hindering a more comprehensive analysis of ZIKV spread within the region. Nonetheless, this study has made an important contribution to narrowing the knowledge gap of the global ZIKV epidemiology and has also laid a foundation for future studies that attempt to further explore this important topic. For example, once previously-undetected circulation of ZIKV is confirmed by serological / RT-PCR testing at locations that we estimated as having a high potential of cryptic ZIKV spread, future modelling studies can then aim to identify geographical areas with (1) high connectivity to the aforementioned locations, and (2) local evidence of *Aedes*-borne disease transmission potential, to further narrow this key knowledge gap.

2.6 Conclusion

In conclusion, our study has provided valuable insights into the potentially high-risk locations for cryptic ZIKV circulation during the 2015–2016 pandemic. Enhanced surveillance is recommended in these geographical areas to mitigate the risk of future epidemics—locally, nationally, and even globally—given the world is increasingly vulnerable to pandemic threats due to the expansion of air traffic networks. In the context of the growing importance of enhanced vigilance and epidemic preparedness in today's world, our modelling framework can also be adapted to identify areas with likely unknown spread of other emerging vector-borne diseases, which has important implications for public health readiness especially in resource-limited settings.

In addition to surveillance for emerging infections in human populations, entomological surveillance can provide an indicator of the transmissibility of viruses once they are introduced into a country or region. Singapore has been lauded for its vector control, one element of which is to identify areas with high *Aedes* mosquito abundance to be targeted for measures. In the next chapter, we will develop regression models using Singapore's island-wide vector surveillance data, to identify factors underlying the spatio-temporal variation of adult *Aedes* abundance in the country's high-rise public residential zones.

Chapter 3: Spatio-temporal analysis of the main dengue vector populations in Singapore

This work has been accepted for publication as:

Sun, H., Dickens, B.L., Richards, D. *et al.* Spatio-temporal analysis of the main dengue vector populations in Singapore. *Parasites Vectors* 14, 41 (2021).

3.1 Abstract

Background

Despite the licensure of the world's first dengue vaccine and the current development of additional vaccine candidates, successful *Aedes* control remains critical to the reduction of dengue virus transmission. To date, there is still limited literature that attempts to explain the spatio-temporal population dynamics of *Aedes* mosquitoes within a single city, which hinders the development of more effective citywide vector control strategies. Narrowing this knowledge gap requires consistent and longitudinal measurement of *Aedes* abundance across the city, as well as examination of relationships between variables on a much finer scale.

Methods

We utilized a high-resolution longitudinal dataset generated from Singapore's islandwide Gravitrap surveillance system over a 2-year period, and built a Bayesian hierarchical model to explain the spatio-temporal dynamics of *Aedes aegypti* and *Ae. albopictus* in relation to a wide range of environmental and anthropogenic variables. We also created a baseline during our model assessment to serve as a benchmark to be compared with the model's out-of-sample prediction / forecast accuracy as measured by the mean absolute error.

Results

For both *Aedes* species, building age and nearby managed vegetation cover were found to have a significant positive association with the mean mosquito abundance, with the former being the strongest predictor. We also observed substantial evidence of a nonlinear effect of weekly maximum temperature on the *Aedes* abundance. Our models generally yielded modest but statistically significant reductions in the out-of-sample prediction / forecast error relative to the baseline.

Conclusion

Our findings suggest that public residential estates with older buildings and more nearby managed vegetation should be prioritized for vector control inspections and community advocacy to reduce the abundance of *Aedes* mosquitoes and the risk of dengue transmission.

3.2 Introduction

Dengue fever is a rapidly emerging vector-borne disease mainly transmitted by *Aedes aegypti* and *Ae. albopictus* [2], causing an estimated number of 105 million infections per year worldwide [6]. Clinical manifestations of dengue infection range from mild fever, to potentially lethal complications such as dengue shock syndrome [2]. Despite the licensure of the world's first dengue vaccine and the current development of additional vaccine candidates [223], successful vector control remains critical to the reduction of dengue virus transmission [224]. Moreover, the benefits of *Aedes* population control extend beyond dengue infection prevention alone, given the multiple diseases that can be transmitted by these mosquito species, such as Zika, chikungunya, and yellow fever.

Previous work has yielded important insights into the behaviors and ecology of the main dengue vectors. Both *Aedes* species could easily disperse throughout areas with radius ~300 meters to seek oviposition sites [225]. The *Ae. aegypti* mosquitoes, in particular, have become a highly efficient vector for dengue transmission owing to its skip oviposition behavior (i.e. deposit eggs from the same batch in multiple sites), desiccation-resistant eggs, preference for human biting, multiple feeds per gonotrophic cycle, and adaptation to reside and breed in human habitats, among other factors [68]. The *Ae. albopictus* mosquitoes were found to have a relatively lower contribution to the reported dengue cases overall despite their high competence for dengue transmission, which is primarily attributed to aspects of their ecology [82]. Both environmental and anthropogenic factors can exert an important influence on the distribution of *Aedes* mosquitoes [75,95,97,177,226], and modelling studies have been carried out to map the suitability and distribution of the main dengue vectors at a global scale [95,97,177]. However, there is still very limited literature that attempts to explain the spatio-temporal population dynamics of *Aedes* mosquitoes within a single city [99], which hinders the development of more effective citywide vector control strategies. To bridge this knowledge gap requires consistent and longitudinal measurement of *Aedes* abundance across the city [99], as well as examination of relationships between variables at a much finer scale.

As an island city-state lying one-degree north of the equator, Singapore faces regular dengue outbreaks with the four dengue virus serotypes co-circulating all year round [227]. The low herd immunity [102], coupled with the tropical climate and highly urbanized environment, poses challenges to the nation's dengue control program [228]. As part of Singapore's vector control program, the National Environment Agency conducted regular

inspections of homes and surrounding areas all year round to remove mosquito-breeding habitats, and mobilized the community and stakeholders to minimize instances of stagnant water [229]. Vector control activities were also ramped up in dengue cluster areas, with space sprays used for adulticiding. To monitor the spatio-temporal trend of the adult *Aedes* abundance in Singapore, the National Environment Agency also established an islandwide Gravitrap surveillance system since 2017, with over 50,000 Gravitraps deployed in the public housing estates across the island [230,231], which accommodate ~80% of the resident population [232]. The weekly mean catch per trap for each species provides an indication of the *Aedes* abundance around each specific residential location and each time point, which is presumed to be closely associated with an individual's risk of exposure to mosquito bites inside or around homes and also much less susceptible to the measurement bias encountered in non-systematic breeding sites inspection [233]. To facilitate resource planning for Singapore's vector control, we used the longitudinal dataset generated from the islandwide Gravitrap surveillance system during 2017–8 as well as a wide range of environmental and anthropogenic variables acquired from various sources to (1) explain the spatio-temporal dynamics of the *Ae. aegypti* and *Ae. albopictus* population in Singapore's high-rise residential zones and (2) assess our model's ability to predict *Aedes* abundance across space and generate forecasts up to three weeks ahead.

3.3 Methods

3.3.1 Data

Aedes mosquito data were collected fortnightly for each of the 552 sites from 2017–8 [231], with odd-numbered blocks inspected one week and even-numbered blocks the next. Fortnightly collections were then halved to obtain the weekly numbers of *Ae. aegypti* and *Ae. albopictus* caught at each site respectively, which contained roughly equal numbers of odd and even-numbered blocks. We created a 300m buffer around each block based on Liew et al [225], and for each site, all buffers were merged into a single polygon, to be used for deriving zonal statistics for the environmental and anthropogenic variables.

The Singapore land classification map was generated at a resolution of 10m using seven separate images from the Sentinel-2 satellite of the European Space Agency [234]. The collected images were taken on different dates to ensure the existence of cloud-free pixels for the whole of mainland Singapore based on a cloud cover

classification algorithm [234]. With 309 labelled data points obtained manually using Google Earth, a random forest algorithm was used to produce seven land cover maps, excluding cloudy areas for each of the collected images [234]. The final classified land cover for each pixel was set to be the majority vote out of all the predictions, with an out-of-bag classification accuracy of 81% [234]. For each site, we derived the percentage of the buffer area covered by water, grass, forest, and managed vegetation (i.e., trees and shrubs with structure dominated by human management) respectively, setting "urban" as the reference level.

Data on waterbodies were extracted from OpenStreetMap [235]. We measured the distance to the nearest waterway from each block using ArcMap 10.6, which was then averaged within each site (similarly for the distance to the nearest water area). We also obtained Singapore's drain line map from the Public Utilities Board. The total drain line density for each site was defined to be the total length of the drain lines falling within the corresponding buffer divided by the buffer area. In addition, the average age of buildings for each site was computed using lease commencement year data collected from the Singapore Land Authority [236].

Weekly mean, maximum, and minimum temperature and mean relative humidity were obtained from a total of 21 weather stations installed by the National Environment Agency. For each climatic variable and each week, we fitted a thin plate spline surface to produce an interpolated value for each site. Weekly raster maps of total precipitation were obtained from the Meteorological Service Singapore at a ~ 500m × 500m resolution, and for each site and each week, all the pixel values within the corresponding buffer were averaged. All the aforementioned explanatory variables were standardized to zero mean and unit variance, and a quadratic term for each of the standardized temperature variables was created to examine nonlinear effects [81].

3.3.2 Statistical analyses

To understand the direction and strength of associations between *Aedes* abundance and different environmental and anthropogenic variables, we first computed the pairwise Pearson correlation coefficients for the full set of covariates and removed redundant variables using a threshold of ±0.6 to avoid collinearity. A Bayesian spatio-temporal model was created, where we assumed that the number of *Ae. aegypti* or *Ae. albopictus* caught at site i during week t (y_{it}) followed a negative binomial distribution with mean μ_{it} and dispersion parameter r, namely

$$y_{it} \sim NB(\mu_{it}, r), \text{ where}$$

$$\log(\mu_{it}) = \log(E_{it}) + b_0 + \sum_j \beta_j x_{ij} + \sum_k \gamma_k w_{itk} + u_i + v_{PA_i} + \phi_t.$$

In the equation above, E_{it} denotes the number of Gravitraps present at site i during week t, b_0 the intercept, x_{ij} the spatial variables of site i, and w_{itk} all the weekly weather measurements of site i between one and three weeks prior to week t. We included an unstructured spatial effect u_i for each site and an extra term v_{PA_i} for the corresponding planning area (each containing 20 sites on average, with an interquartile range of 8–26) to account for additional spatial dependence. The temporally structured effect ϕ_t was assumed to follow a random walk with a maximum order of two. Throughout this paper, we used $t = 1$ to denote the first epidemiological week of 2017, and $t = 104$ the last epidemiological week of 2018. All the model parameters were assigned a minimally informative prior (refer to Table 3.2 caption), and parameter estimation was performed using Integrated Nested Laplace Approximation [237,238], with 95% credible intervals (CrI) computed to summarize the uncertainty in each model parameter.

For each species, both the optimal order of the random walk for the temporally structured effect and the time lag of the weather variables were selected based on the deviance information criterion. Further variable subset selection was not implemented at this stage to avoid biased parameter estimates resulting from sequential comparisons, since the primary aim of our Bayesian spatio-temporal model was to infer the associations between *Aedes* abundance and all the different environmental and anthropogenic variables considered in this study.

Next, we used cross-validation to assess how accurately one can predict *Aedes* abundance across space. Here we treated the total number of *Ae. aegypti* or *Ae. albopictus* caught at each site during 2017–8 as the response variable, with the log transformation of the total number of trap-week units as an offset. Only spatial fixed effects and the planning area random effect were included in the model, and a backwards elimination procedure was implemented for fixed effects variable selection using Akaike information criterion during model training. Two forms of cross-validations were performed, namely, leave-one-site-out and leave-one-planning-area-out, where in each case a baseline prediction of the mean catch per trap per week was generated for each site, to be used as a benchmark to assess whether the model can indeed yield a higher prediction accuracy. In the former case, each time a site i was left out for testing, its baseline prediction was defined as the observed mean catch per trap per week of the site that was geographically the closest to site i. In the latter case, the

baseline prediction for each site i was defined as the observed mean catch per trap per week averaged across all the sites within the planning area that was geographically the closest to site i but did not contain site i.

Finally, we evaluated the contribution of weather variables to the improvement of the weekly *Aedes* abundance forecast accuracy up to three weeks ahead. Using t_C to denote the current time point, we treated the mean catch per trap of each site at week $(t_C + \Delta t)$ $(\Delta t = 1,2, \text{or } 3)$ as the response variable, and each was regressed upon all the weather and/or entomological (i.e. *Aedes* abundance) covariates at weeks t_C, $(t_C - 1)$ and $(t_C - 2)$. Here, the entomological covariates were always included as autoregressive terms, with the additional inclusion of weather covariates in an alternative model to assess the resulting change in the out-of-sample forecast errors. Due to the large number of lagged weather variables included in the alternative model, we used the least absolute shrinkage and selection operator to perform variable selection during model training. For each site and each species, we derived the out-of-sample *Aedes* abundance forecast accuracy at week $(t^* + \Delta t)$ for all combinations of $t^* \in \{55, 60, \ldots, 95, 100\}$ and $\Delta t \in \{1,2,3\}$ respectively, with the model being trained on all the historical data points of that site with $t_C < t^*$. The corresponding baseline forecast was defined as the observed *Aedes* abundance of that site at week t^*, and hence we will also refer to t^* as the baseline time point.

3.4 Results

In total, 4,923,456 trap-week units of observation were obtained, with 495,638 *Ae. aegypti* and 132,533 *Ae. albopictus* caught during the entire study period. For both species, we observed a marked difference in the *Aedes* abundance across space, with the mean catch per trap per week at some sites exceeding 5-fold that at some other sites (Figure 3.1). For example, only an average of 0.05 *Ae. aegypti* mosquitoes were caught per trap per week at a site within Clementi area during the study period, in contrast to 0.28 at a site in Tampines. Operationally, these data are updated on a weekly basis to provide policy makers with an indication of which areas may require more vector control to mitigate the risk of dengue transmission. The mean and standard deviation of the site-level mean catch per trap per week were 0.102 and 0.074 for *Ae. aegypti*, and 0.027 and 0.019 for *Ae. albopictus* (Table 3.1 contains the summary statistics of all the variables in this study, and visualizations of selected variables across space or time can be found in Appendix B).

Variables	Mean (SD)
Mean number of mosquitoes caught per trap per week (*Ae. aegypti*)	0.102 (0.074)
Mean number of mosquitoes caught per trap per week (*Ae. albopictus*)	0.027 (0.019)
Forest cover (% of 300m radius buffer)	5.6 (5.2)
Water cover (% of 300m radius buffer)	2.0 (3.2)
Grass cover (% of 300m radius buffer)	8.6 (4.8)
Managed vegetation cover (% of 300m radius buffer)	25.4 (7.8)
Drain line density (km^{-1})	9.7 (6.9)
Distance to waterway (m)	685 (490)
Distance to water area (m)	613 (390)
Building age (years as of 2018)	28 (10)
Weekly mean temperature (°C)	27.9 (0.9)
Weekly maximum temperature (°C)	33.1 (0.9)
Weekly minimum temperature (°C)	23.9 (1.0)
Weekly mean relative humidity (%)	80.5 (4.1)
Weekly total precipitation (mm)	51.6 (63)

Table 3.1: Summary statistics of all the variables included in the study.

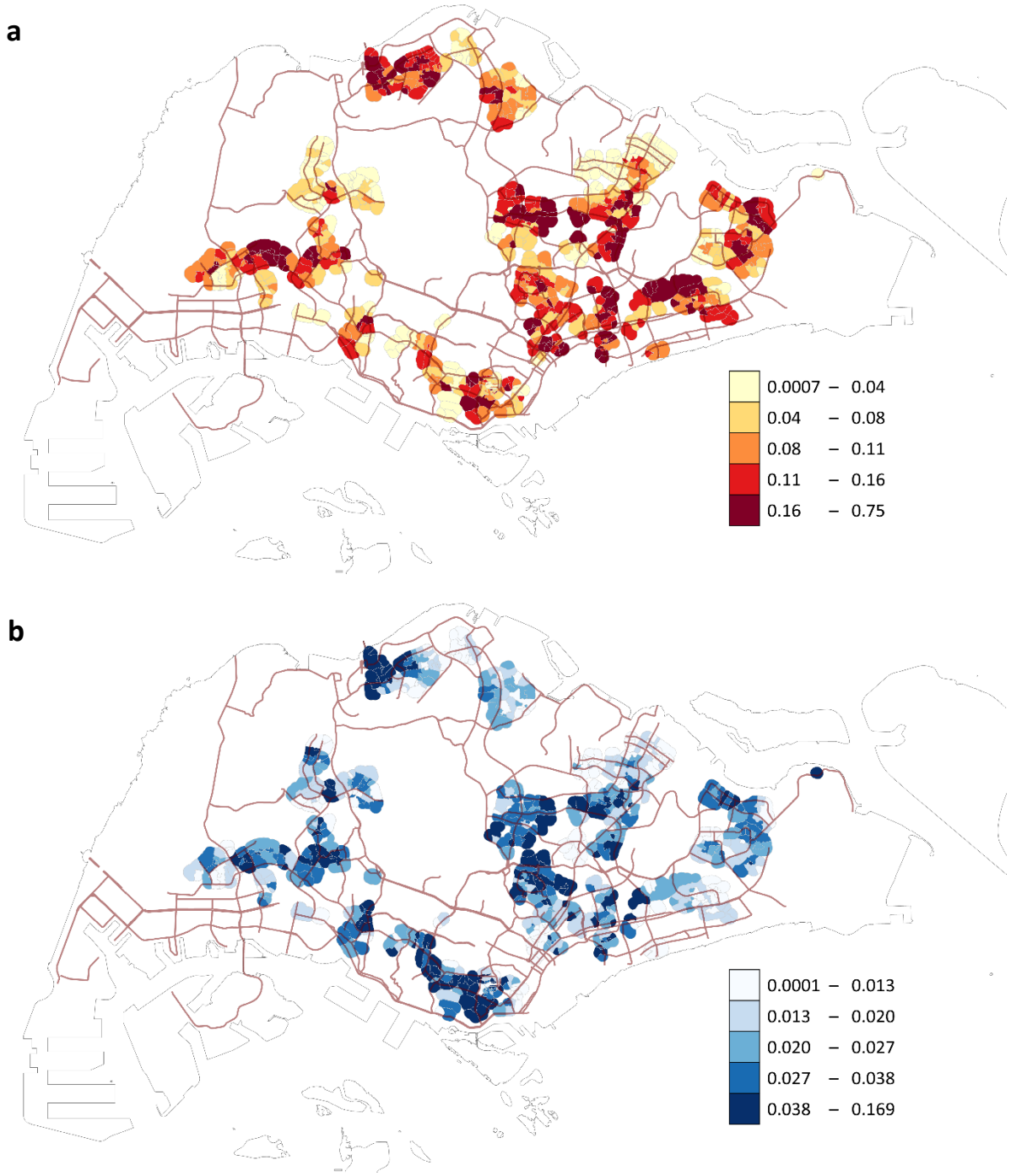


Figure 3.1: Observed *Aedes* abundance (mean catch per trap per week) of each site during 2017–2018: (a) *Ae. aegypti* and (b) *Ae. albopictus*. We first created a 300m buffer around each block, and all buffers for each site were merged into a single polygon, which was then coloured according to the observed *Aedes* abundance value.

Based on the spatio-temporal model estimates, both nearby managed vegetation cover and building age were found to have a direct association with the abundance of both species (Table 3.2 & Figure 3.2). On average, we estimated that a 1-standard deviation (10 years) increase in the average age of buildings was associated with a 52.3% (95% CrI: 42.0%–63.2%) increase in the *Ae. aegypti* abundance, and a 38.1% (95% CrI: 31.0%–45.6%) increase in the *Ae. albopictus* abundance at the site level, when all the other variables were held constant (Figure 3.2). For forest cover and distance to water area, the signs of the *point* estimates were found to be opposite between the two mosquito species, although the 95% credible interval may contain the null effect in some cases (Table 3.2 & Figure 3.2). Even after controlling for all the fixed effects, there remained substantial heterogeneity of *Aedes* abundance both between sites and between planning areas, as shown by the standard deviation estimates of the spatial random effects (Table 3.2).

For both species, inclusion of weather measurements in all the past three weeks together with a random walk model of order 2 for the temporally structured effect yielded the lowest Deviance Information Criterion. However, in comparison with the spatial covariates, the weather covariates were estimated to have a relatively limited impact on the variation of adult *Aedes* abundance in the context of Singapore (Table 3.2 & Figure 3.2). The 95% credible intervals of the quadratic term coefficients for the weekly maximum temperature were away from zero for both species and all time lags (Table 3.2), and the *Aedes* abundance was estimated to first increase and then decrease as we varied the lagged weekly maximum temperature from 28.0°C to 36.6°C while holding the other variables constant (Figure 3.3). Specifically, with all other covariates held constant, the median estimate of *Ae. aegypti* abundance peaked at 30.3°C, 30.0°C, and 31.3°C for weekly maximum temperature measured at 1, 2, and 3 weeks' lag respectively. Similarly, the turning points were 31.9°C, 31.6°C and 31.9°C for *Ae. albopictus*. Out of all the covariates collected in this study, only weekly mean and minimum temperature were removed from the Bayesian spatio-temporal model during collinearity assessment.

In both leave-one-site-out and leave-one-planning-area-out cross validation, which were performed to assess predictive accuracy across space, there was an overall increasing trend in the observed site-level *Aedes* abundance as we moved from the lowest to the highest quintile based on the out-of-sample model predictions (Figure 3.4). Except for the leave-one-site-out cross validation of the model for *Ae. aegypti*, there was a modest and statistically significant reduction in the mean absolute prediction error of the model compared with the baseline prediction (Table 3.3). Likewise, a modest and statistically significant reduction in the out-of-sample

forecast error was observed for models forecasting 2- or 3-week ahead *Aedes* abundance, regardless of species or whether we included weather covariates as additional predictors (Table 3.4). Our model, however, did not outperform the 1-week ahead baseline forecast (Table 3.4), owing to the fortnightly mosquito collection and the subsequent conversion to weekly data (details described in Methods), which caused data points at adjacent weeks to share 50% of the information in common. Notably, we found that in all cases, the additional inclusion of lagged weather covariates did not improve the out-of-sample forecast accuracy compared with a simple model that only included autoregressive terms as predictors (Table 3.4).

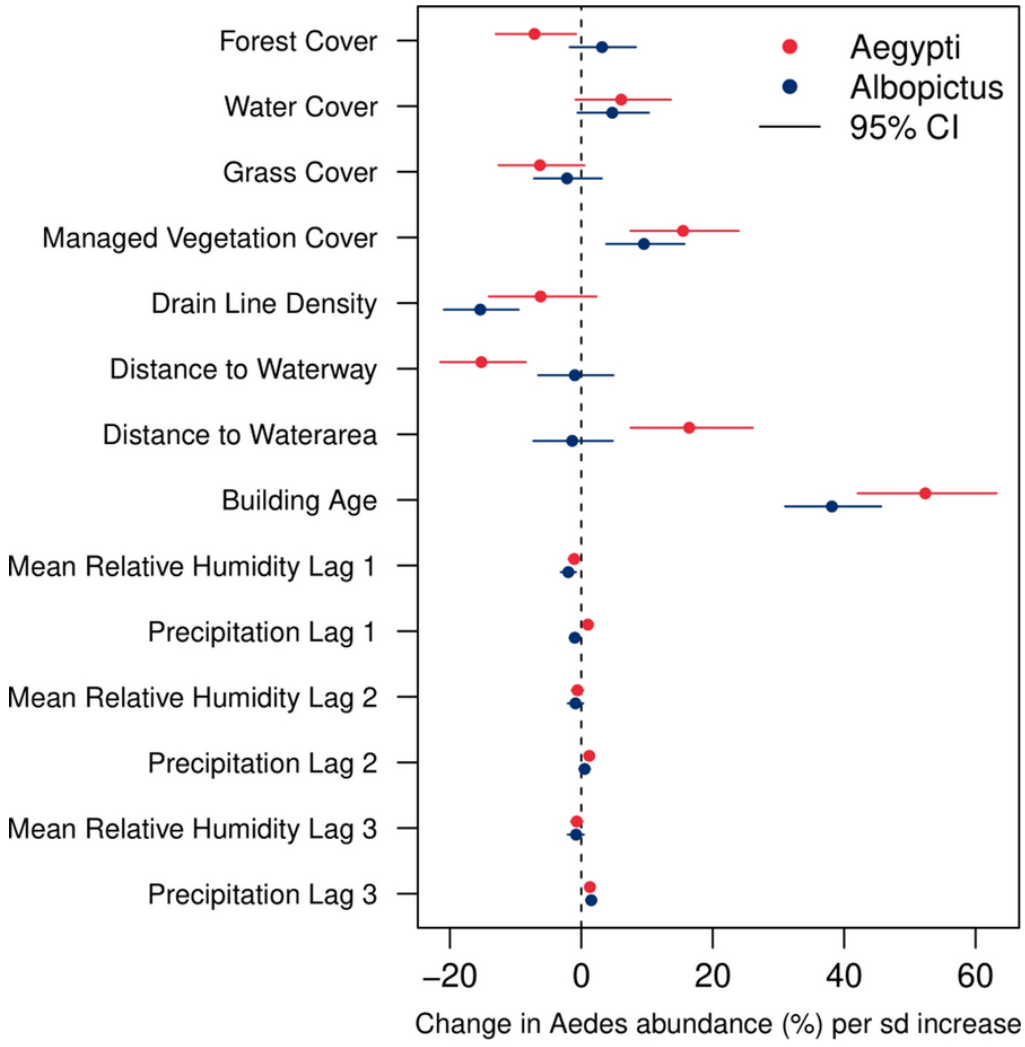


Figure 3.2: Estimated percentage change in the expected value of *Aedes* abundance (weekly mean catch per trap) due to a 1-standard deviation increase in each covariate when all the other variables were held constant. Filled circles denote the posterior median estimates and the solid lines denote the 95% credible intervals. The standard deviation of each covariate can be found in Table 3.1, and the estimated effects of lagged temperature covariates on the predicted *Aedes* abundance were visualized in Figure 3.3.

	Ae. aegypti		
	2.5th percentile	50th percentile	97.5th percentile
Forest Cover (within 300m radius buffer)	-0.140	-0.074	-0.008
Water Cover (within 300m radius buffer)	-0.009	0.059	0.128
Grass Cover (within 300m radius buffer)	-0.135	-0.065	0.005
Managed Vegetation Cover (within 300m radius buffer)	0.072	0.144	0.215
Drain Line Density (within 300m radius buffer)	-0.152	-0.064	0.023
Distance to Waterway	-0.242	-0.165	-0.088
Distance to Water area	0.072	0.152	0.232
Building Age	0.351	0.421	0.490
Max Temperature Lag 1	-0.037	-0.027	-0.018
(Max Temperature Lag 1)^2	-0.008	-0.004	0.000
Mean Relative Humidity Lag 1	-0.020	-0.011	-0.002
Precipitation Lag 1	0.003	0.010	0.016
Max Temperature Lag 2	-0.040	-0.031	-0.021
(Max Temperature Lag 2)^2	-0.008	-0.005	-0.001
Mean Relative Humidity Lag 2	-0.015	-0.006	0.003
Precipitation Lag 2	0.006	0.012	0.018
Max Temperature Lag 3	-0.032	-0.023	-0.013
(Max Temperature Lag 3)^2	-0.009	-0.006	-0.002
Mean Relative Humidity Lag 3	-0.016	-0.007	0.002
Precipitation Lag 3	0.007	0.013	0.020
Unstructured spatial effect (Site) *	0.603	0.642	0.686
Unstructured spatial effect (Planning area) *	0.383	0.514	0.708
Temporally structured effect **	0.048	0.058	0.071
Negative binomial size parameter	6.681	6.863	7.049

Table 3.2A: Posterior estimates of the Bayesian spatio-temporal model parameters (*Ae. aegypti*) §. All the environmental and anthropogenic variables were standardized to zero mean and unit variance prior to model fitting, and a quadratic term for each of the standardized temperature variables was also created to introduce nonlinear effects. Estimates are posterior median (bigger font) and equal tailed 95% credible intervals (smaller font).

* Refers to the posterior estimate of the standard deviation of the spatial random effect.

** Refers to the posterior estimate of the standard deviation of the independent second-order increment in the temporally structured effect.

§ Each regression parameter (i.e., intercept and the coefficients of the fixed effects) was assigned a normal prior $\mathrm{N}(0, 5^2)$. We assumed a $\mathrm{logGamma}(1, 0.01)$ prior on the logarithm of the precision of the spatial random effects and independent second-order increment in the temporally structured effect. The default Penalized Complexity prior in R-INLA was specified for the logarithm of the negative binomial size parameter.

	Ae. albopictus		
	2.5th percentile	50th percentile	97.5th percentile
Forest Cover (within 300m radius buffer)	-0.018	0.031	0.080
Water Cover (within 300m radius buffer)	-0.006	0.046	0.098
Grass Cover (within 300m radius buffer)	-0.075	-0.022	0.031
Managed Vegetation Cover (within 300m radius buffer)	0.037	0.091	0.146
Drain Line Density (within 300m radius buffer)	-0.235	-0.167	-0.100
Distance to Waterway	-0.068	-0.010	0.048
Distance to Water area	-0.076	-0.014	0.047
Building Age	0.270	0.323	0.376
Max Temperature Lag 1	-0.027	-0.016	-0.004
(Max Temperature Lag 1)^2	-0.010	-0.005	-0.001
Mean Relative Humidity Lag 1	-0.032	-0.020	-0.008
Precipitation Lag 1	-0.018	-0.010	-0.002
Max Temperature Lag 2	-0.029	-0.017	-0.005
(Max Temperature Lag 2)^2	-0.010	-0.005	0.000
Mean Relative Humidity Lag 2	-0.021	-0.009	0.003
Precipitation Lag 2	-0.003	0.005	0.013
Max Temperature Lag 3	-0.025	-0.013	-0.002
(Max Temperature Lag 3)^2	-0.010	-0.005	0.000
Mean Relative Humidity Lag 3	-0.021	-0.008	0.004
Precipitation Lag 3	0.007	0.015	0.023
Unstructured spatial effect (Site) *	0.443	0.475	0.508
Unstructured spatial effect (Planning area) *	0.240	0.337	0.487
Temporally structured effect **	0.076	0.092	0.113
Negative binomial size parameter	12.877	13.679	14.597

Table 3.2B: Posterior estimates of the Bayesian spatio-temporal model parameters (*Ae. albopictus*) §. All the environmental and anthropogenic variables were standardized to zero mean and unit variance prior to model fitting, and a quadratic term for each of the standardized temperature variables was also created to introduce nonlinear effects. Estimates are posterior median (bigger font) and equal tailed 95% credible intervals (smaller font).

* Refers to the posterior estimate of the standard deviation of the spatial random effect.

** Refers to the posterior estimate of the standard deviation of the independent second-order increment in the temporally structured effect.

§ Each regression parameter (i.e., intercept and the coefficients of the fixed effects) was assigned a normal prior $\mathrm{N}(0, 5^2)$. We assumed a $\mathrm{logGamma}(1, 0.01)$ prior on the logarithm of the precision of the spatial random effects and independent second-order increment in the temporally structured effect. The default Penalized Complexity prior in R-INLA was specified for the logarithm of the negative binomial size parameter.

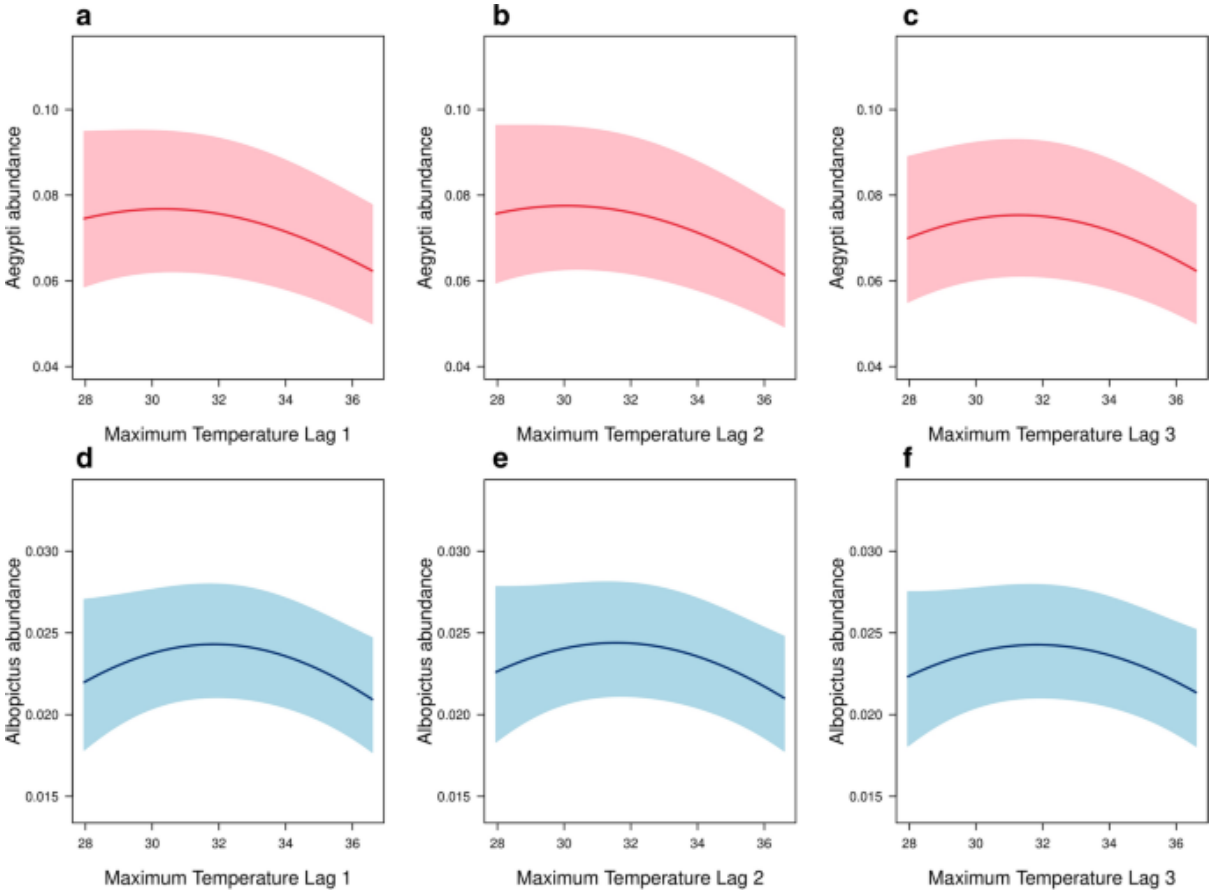


Figure 3.3: Predicted values of *Aedes* abundance (weekly mean catch per trap) at different values of lagged weekly maximum temperature with all the other variables held fixed at their average values. Solid lines denote the posterior median estimates and the shaded areas denote the 95% prediction bands.

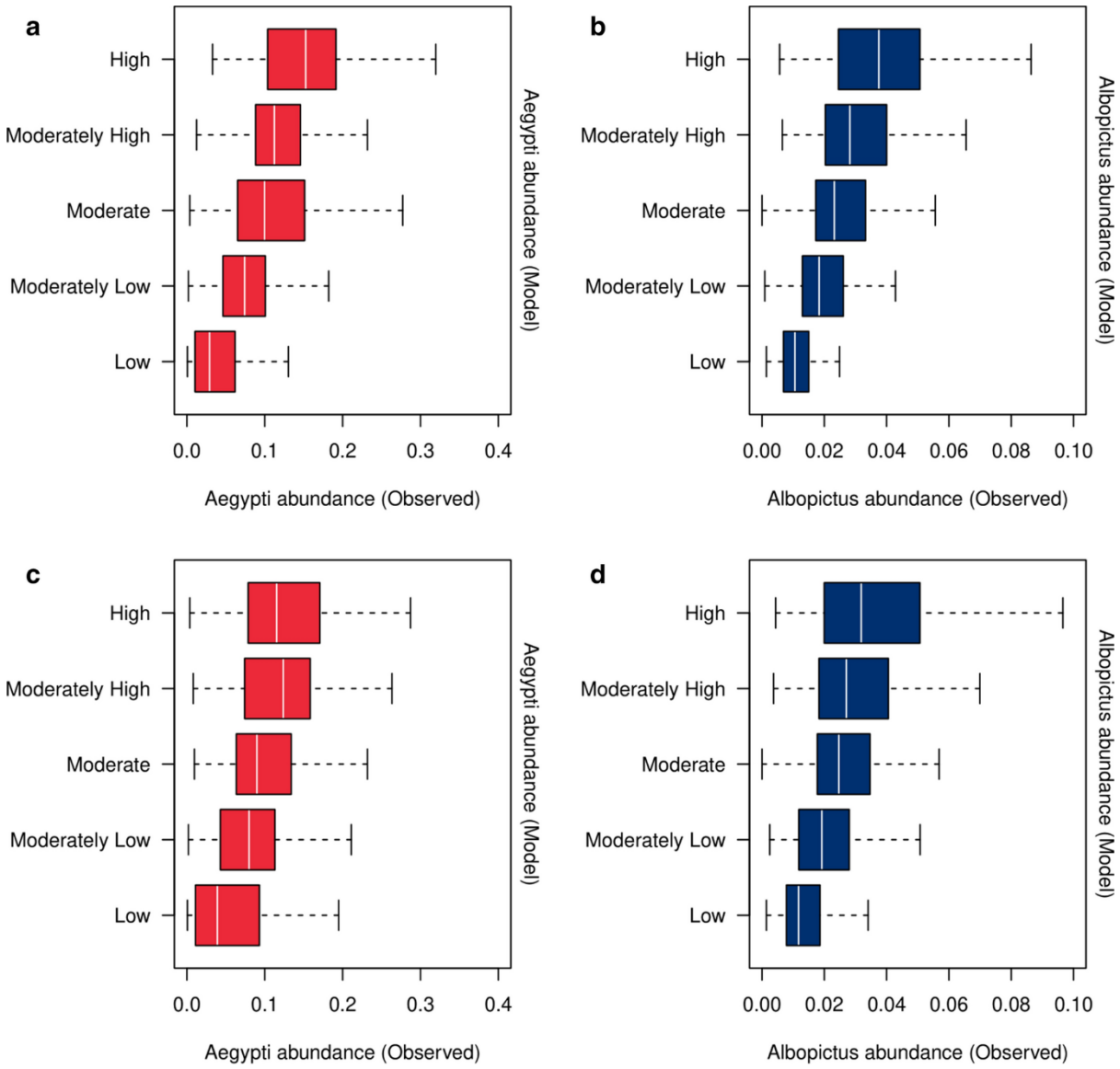


Figure 3.4: Box-plots of the observed site-level *Aedes* abundance (mean catch per trap per week) during 2017–2018 within each quintile based on the out-of-sample model predictions: (a) & (b) leave-one-site-out cross validation, and (c) & (d) leave-one-planning-area-out cross validation. A small number of extreme values were omitted from the graph for clarity.

Table 3.3: Percentage reduction in the out-of-sample mean absolute prediction error of the model compared with the baseline prediction. Under both types of spatial cross validation, the mean absolute prediction error was derived by averaging the absolute prediction errors across all sites. The *p*-values were obtained based on a two-sided paired t-test.

	Leave-one-site-out	Leave-one-planning-area-out
Ae. aegypti	3.6% ($p = 0.477$)	9.3% ($p = 0.006$)
Ae. albopictus	17.5% ($p < 0.001$)	14.1% ($p < 0.001$)

Table 3.4: Percentage reduction in the out-of-sample mean absolute forecast error of the model compared with the baseline forecast. The mean absolute forecast error was derived by averaging the absolute forecast errors across all sites and baseline time points. The *p*-values were obtained based on a two-sided paired t-test.

		1 week ahead	2 weeks ahead	3 weeks ahead
Weather covariates included	*Ae. aegypti*	-10.0% ($p < 0.001$)	6.5% ($p < 0.001$)	10.7% ($p < 0.001$)
	Ae. albopictus	-18.5% ($p < 0.001$)	7.8% ($p < 0.001$)	10.3% ($p < 0.001$)
Weather covariates not included	*Ae. aegypti*	0.3% ($p = 0.764$)	8.3% ($p < 0.001$)	10.8% ($p < 0.001$)
	Ae. albopictus	-4.9% ($p < 0.001$)	9.8% ($p < 0.001$)	11.0% ($p < 0.001$)

3.5 Discussion

This study examined the spatial and temporal variation of the main dengue vectors in Singapore's high-rise public residential zones in relation to a wide range of environmental and anthropogenic variables. The insights derived from this study further add to previous work that aimed to understand *Aedes* ecology in the local context, and can facilitate the formulation of more effective vector control strategies in the future. Our model performance also suggests the potential use of spatio-temporal mapping as a tool to improve the understanding of the *Aedes* distribution in other cities or countries, where intensive entomological surveillance may be harder to achieve.

We found that the majority of our spatial covariates had an at least borderline significant association with the *Aedes* abundance (i.e., the 95% credible intervals did not / barely overlap zero). In particular, building age was shown to be the strongest predictor. This might be due to a combination of factors, including infrastructural degradation and the water storing practices associated with the sociodemographic profile of residents that result in more instances of water stagnation that can breed mosquitoes. For both species, we estimated that an

increase in the managed vegetation cover within the buffer area was associated with a substantial rise in the mean vector abundance, likely owing to the increased availability of water in leaf axils, leaf litter, and discarded receptacles hidden in foliage and tree holes which supports mosquito breeding. Unlike managed vegetation cover, both forest and grass cover were found to be negatively associated with the abundance of *Ae. aegypti*. This was not unexpected given that *Ae. aegypti* prefers highly urbanized areas where it can breed in artificial containers. On the other hand, there was a positive association between forest cover and *Ae. albopictus* abundance based on the point estimate, which is consistent with the existing knowledge of the vector's ecology [239].

Our analysis showed a borderline significant negative association between drain line density and the abundance of *Ae. aegypti* in contrast to the positive correlation reported by Seidahmed *et al.* [240] This discrepancy can be due to a number of reasons: for example, the study conducted by Seidahmed *et al.* was restricted to a small area of Singapore, and results may be confounded by the different housing types with different demography [240], whereas this study used two-year data collected from the same type of housing areas across the island. Moreover, the number of *Aedes* mosquitoes caught inside the high-rise residential buildings and the number of outdoor breeding habitats can be impacted by drain line density in different ways. Since perimeter drains are known to be the most common breeding habitats of *Aedes* in Singapore's public areas according to routine inspections [241], their abundance could simultaneously decrease the per-mosquito probability of looking for oviposition sites inside residential buildings and increase the total number of *Aedes* mosquitoes in the nearby public area. Thus, our estimate is likely to be a reflection of the resulting net effect, and similarly for the estimated effects of other spatial covariates such as nearby managed vegetation cover.

Previous work has highlighted the challenges of mapping the spatial distribution of *Aedes* mosquitoes for operational dengue vector control [242]. In particular, predictors that can be informative across an entire continent or a sufficiently large country may lose predictive power within the confines of a single city [242]. While we did observe substantial evidence of a non-zero association between many spatial covariates and *Aedes* abundance in our analysis, the estimated standard deviations of the spatial random effects remained large, suggesting substantial unexplained heterogeneity across space. Hence, entomological surveillance remains critical to generating knowledge of *Aedes* abundance in the field to inform vector control. On the other hand, we found that in most cases the model performed significantly better than the baseline at predicting

mosquito abundance at new locations, based on the spatial variables at those locations, suggesting that statistical modelling can still serve as a complementary tool to refine our understanding of the *Aedes* abundance at locations where entomological data are unavailable and hence to identify additional locations that may require enhanced vector control. It should be noted that the model's improvement in prediction accuracy over the baseline was found to be smaller for *Ae. aegypti* than *Ae. albopictus*, despite the former being the most important dengue vector in Singapore. This finding may be explained by the ecology of *Ae. aegypti*, which is a container breeder that is subject to the vagaries of human behavior; adherence to household practices to prevent breeding is hard to measure and may vary spatially, rendering spatial modelling much more challenging.

There is abundant literature on how different weather variables regulate the population dynamics of *Aedes* via influencing mosquito habitat availability, development, survival, and reproduction [75]. In this study, however, we estimated the effects of all the lagged weather variables on the observed *Aedes* abundance to be very minimal, which can be owing to the restricted range of the weather variables in the context of Singapore as well as vector control activities that were typically ramped up during higher breeding seasons. The assessment of the out-of-sample forecast errors, too, shows that the additional inclusion of weather covariates did not improve the accuracy of the *Aedes* abundance forecasts, and a simple model with autoregressive terms alone could yield a modest and statistically significant improvement in the 2-week and 3-week ahead forecasts over the baseline. While these results may suggest that weather had a negligible impact on *Aedes* abundance in Singapore, it should be noted that this study was conducted in public housing estates and detected mosquitoes that may have hatched nearby, and the relationship between weather and outdoor breeding habitats may be more complex than was identified herein. A longitudinal entomological study in the Geylang neighbourhood of Singapore found that the outdoor *Aedes* population was likely to be shaped by rainfall through a monsoon-driven sequence of flushing, drying, and return of breeding habitats [243]. Taken together, these results suggest the differential impacts of weather on the *Aedes* population dynamics and hence the potential risk of exposure to mosquito bites in different settings, i.e., *Aedes* abundance inside and nearby public housing estates may be less sensitive to changes in weather compared with outdoor abundance.

Our results need to be interpreted in the light of the following limitations. First, we were unable to account for the effects of vector control programs on the observed *Aedes* abundance across space and time. Regulatory inspections and community efforts aimed at removing larval habitats, as well as chemical control to reduce

adult mosquito populations, usually peak during higher vector breeding / dengue transmission seasons, and this could to some extent mask the true association between different weather variables and the *Aedes* abundance, with the confounding bias difficult to quantify or adjust for. Besides, there could be residual spatial dependence structure in our data due to factors such as potential ongoing expansion of *Aedes* mosquitoes, which may have been coincidentally absorbed by certain spatial covariates because of confounding. Nonetheless, this issue was assessed via a leave-one-planning-area-out cross validation framework with baseline predictions created to serve as a benchmark for model performance comparison, and results suggest that the spatial covariates could indeed enhance the out-of-sample predictive accuracy. In addition, our parameter estimates were derived based on mosquito data collected from Singapore's high-rise public residential zones, and thus should not be used for extrapolation to low-rise houses. Previous work has suggested that the risks of indoor breeding of *Aedes* mosquitoes could be highly dependent on the accommodation type in Singapore [240]. In early 2020, the National Environment Agency extended the deployment of Gravitraps to private landed residential estates [244], and as more data are being generated, this will shed further light on how *Aedes* abundance differs between high-rise and low-rise residential zones across the island.

3.6 Conclusion

Our study has demonstrated the potential and challenges of spatio-temporal modelling for improving the understanding of the main dengue vectors' ecology, and provided empirical evidence to guide the refinement of vector control strategies in the context of Singapore. Our findings suggest that public residential estates with older buildings and more nearby managed vegetation should be prioritized for vector control inspections and community advocacy to reduce the abundance of *Aedes* mosquitoes and the risk of dengue transmission. The insights obtained from this study could also be helpful to inspire future studies that attempt to understand the spatio-temporal dynamics of the dengue vector population at a city scale, particularly in settings where vector surveillance data are less abundant and thus require modelling to further narrow the knowledge gap.

In the next chapter, I will aim to expand what is currently known about the impact of *Aedes* control on dengue transmission. Specifically, I will develop an agent-based model to simulate the time-varying epidemiological impact of sustained *Aedes* control in diverse dengue endemic settings over a 30-year time period. It is hoped that the insights generated from this work will be useful to guide the development of *Aedes* control strategic plans to achieve sustained reduction in the burden of dengue.

Chapter 4: Short-term and long-term epidemiological impacts of sustained vector control in various dengue endemic settings: A modelling study

4.1 Abstract

Background

As the most widespread viral infection transmitted by the *Aedes* mosquitoes, dengue has been estimated to cause 51 million febrile disease cases globally each year. Although sustained vector control remains key to reducing the burden of dengue, current understanding of the key factors that explain the observed variation in the short- and long-term vector control effectiveness across different transmission settings remains limited.

Methods

We used a detailed individual-based model to simulate dengue transmission with and without sustained vector control over a 30-year time frame, under a wide range of transmission scenarios. Vector control effectiveness was derived for different time windows within the 30-year intervention period. We then used the extreme gradient boosting algorithm to predict the effectiveness of vector control given the simulation parameters, and the resulting machine learning model was interpreted using Shapley Additive Explanations.

Results

According to our simulation outputs, dengue transmission would be nearly eliminated during the early stage of intensive interventions, but over time incidence would gradually bounce back to the pre-intervention level. The time point at which intervention ceases to be effective is strongly influenced not only by the intensity of vector control, but also by the pre-intervention transmission intensity and the individual-level heterogeneity in biting risk. Moreover, the impact of many transmission model parameters on the intervention effectiveness is shown to be modified by the intensity of vector control, as well as to vary over time.

Conclusion

Our study has identified the main drivers for the difference in the time-varying effectiveness of sustained vector control across different dengue endemic settings, and the insights obtained will be useful to inform future model-based studies that seek to predict the impact of dengue vector control in their local contexts.

4.2 Introduction

As the most widespread viral infection transmitted by the *Aedes* mosquitoes, dengue has been estimated to cause 51 million febrile disease cases globally each year, of whom 3.6 million may require hospitalization [6]. Unfortunately, the only licensed dengue vaccine to date, CYD-TDV, has been found to enhance the future risk of severe dengue infections for seronegative recipients and hence cannot be used to provide protection for all individuals [245]. Even under the very optimistic assumption that 100% of the CYD-TDV vaccine recipients are seropositive, sustained vector control remains key to reducing the burden of dengue [224]. Recently, there has been a growing interest in applying novel vector control technologies to reduce dengue transmission. For instance, combining incompatible and sterile insect techniques has been demonstrated to dramatically reduce the *Aedes* population size by over 90% under field conditions [124,246,247], and the application of attractive toxic sugar baits has also shown impressive results in both laboratory and small-scale field experiments [248,249]. Other examples include introducing the wMel *Wolbachia* strain to the *Aedes* population with the aim of reducing vector competence instead of suppressing the mosquito population [250].

Although the novel technologies described above have the potential to achieve dengue elimination even in high-transmission settings [111], it is still important to assess the short- and long-term impacts of the current vector control measures before more powerful vector control tools become available to be rolled out on a large scale. At present, evidence supporting the epidemiological impact of existing dengue vector control measures remains limited, due to the lack of surveillance data and the challenge in deploying rigorous study designs to assess intervention effectiveness, as well as insufficient investment in sustained mosquito suppression [111]. In particular, assessing the long-term effectiveness of dengue vector control can be extremely challenging, as this would require substantial investment in monitoring and evaluation. Moreover, long-term dengue vector control trials may violate the principle of equipoise once an intervention under evaluation has been shown to reduce disease incidence in the short to medium term, even though there may be genuine uncertainty regarding its epidemiological impact in the long run.

Although not a replacement for surveillance data, mathematical modelling can serve as a useful tool to compare the predicted dengue incidence with and without interventions and thus to minimize the confounding bias commonly encountered in studies without an appropriate control group. So far, very few studies have been carried out to model the impact of dengue vector control in both the short and long terms [111,128], and to the

best of our knowledge, none has examined how the time-varying impact of dengue vector control differs across a wide range of transmission settings. Narrowing this knowledge gap will yield useful insights into the important details of a transmission context that must be accounted for to predict the short- and long-term epidemiological impacts of a sustained dengue vector control intervention.

In this study, we aim to use mathematical modelling to (i) simulate the change in dengue incidence within a 30-year time period following a sustained reduction in the *Aedes* population size by 30%–70%, and (ii) identify key factors that explain the variation in the short- and long-term vector control effectiveness across different dengue endemic settings.

4.3 Methods

4.3.1 Individual-based model

4.3.1.1 Overview

We developed an individual-based model to simulate dengue transmission within a spatially explicit setting while accounting for the individual-level variation in the risk of exposure to mosquito bites. The flexibility of the modeling framework also allowed us to examine the sensitivity of our vector control effectiveness estimate with respect to changes in assumptions about serotype immune-interactions [36]. Specifically, we first simulated the introduction and autochthonous transmission of dengue virus in the absence of vector control for 100 years ("warm-up period") to establish the baseline immunity in the population. Thereafter, we continued the simulation for another 30 years ("intervention period") with vector control being implemented to maintain the vector density at a substantially reduced level. The discrete time step used to simulate dengue transmission was equal to the length of the mosquito feeding cycle (l_f), during which a mosquito took a blood meal, looked for breeding sites, laid eggs, and then searched for the next blood meal [251]. We created a wide range of scenarios in our simulations by varying the intensity of vector control, pre-intervention mosquito density, dengue importation rate, rate of change in the population age structure, host mobility rate, and the degree of individual-level heterogeneity in the risk of being bitten by the *Aedes* mosquitoes. In addition, we also varied the assumptions regarding the degree of antibody-dependent enhancement and duration of temporary cross-

immunity across different simulation runs to assess their influence on the model output (refer to Figure 4.1 for the schematic overview of the individual-based model).

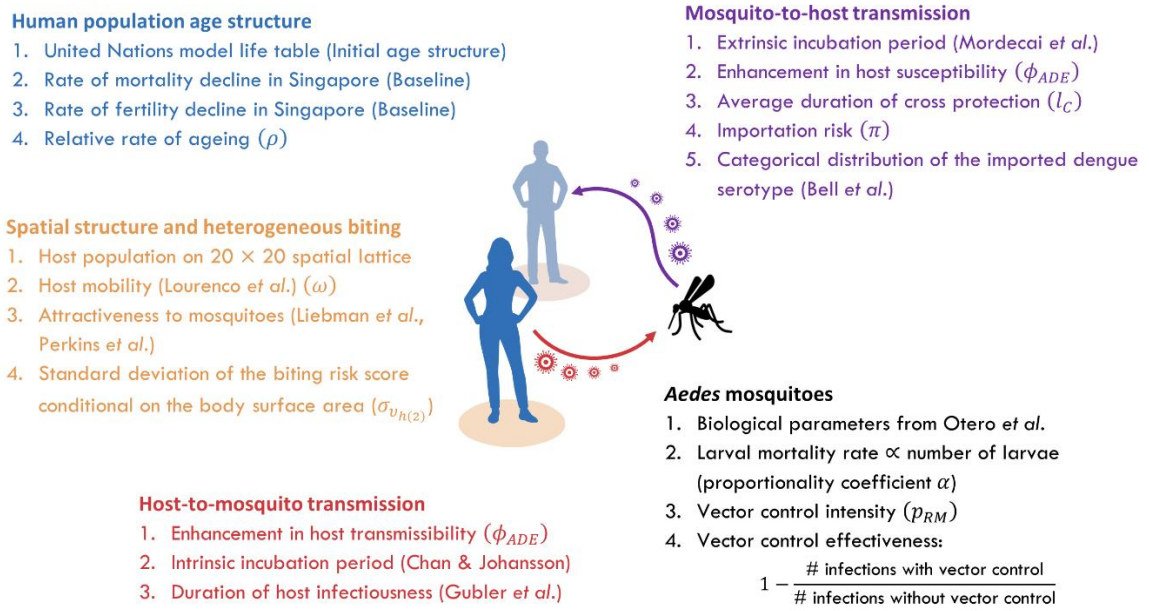


Figure 4.1: Schematic overview of the individual-based model.

4.3.1.2 Dengue infections in humans

Each human was allowed to have at most four sequential infections, with life-long protection against homotypic reinfection and temporary cross-protection against heterotypic infections. Upon receiving an infectious bite, each human had a probability $\tau_{M\to H}^{(k)}$ of becoming infected in the absence of immunity. Here, the vector-to-host transmission probability depended on the number of historical dengue infections an individual had previously experienced ($k = 0,1,2,3$), where we set $\tau_{M\to H}^{(k)}$ to be 0.5 for $k \neq 1$, and $0.5 \cdot (1 + \phi_{ADE})$ for $k = 1$ to account for the enhanced susceptibility to secondary infections. On average, each infected human took l_{LP} days to become infectious and another l_R days to achieve viral clearance, followed by a cross-protection period lasting an average of l_C days (or lifelong immunity against all serotypes after recovery from a quaternary infection); we assumed the length of each of these time intervals to be exponentially distributed.

Similar to the model implemented by Hladish *et al*. to simulate dengue introductions [252], each human had a time-invariant probability π of being exposed to dengue virus from an external source in each feeding cycle, and

the exposure may be resisted by immunity. To simulate the serotype of the imported dengue virus for each exposed individual, we obtained a random draw from a categorical distribution with the probability parameters equal to the relative frequencies of the four dengue serotypes in Southeast Asia estimated by Bell *et al.* [29]. Specifically, we first averaged Bell *et al.*'s estimates across the 45-year study period to be used for the simulation of dengue introductions throughout the 100-year warm-up period [29]. Here, we used a time-invariant probability vector since the length of the warm-up period far exceeded the length of the time period for which data were available. Next, we used linear interpolation to transform Bell *et al.*'s quarterly estimates for the period of 1970–2000 into daily estimates [29]. These estimates were then used to simulate the introduced dengue serotypes at each feeding cycle during the 30 years of the intervention period.

4.3.1.3 Dengue infections in mosquitoes

Each susceptible mosquito had a probability $\tau_{H\to M}^{(k)}$ of becoming infected after biting an infectious human who had historically been infected k times prior to the current infection. We set $\tau_{H\to M}^{(k)}$ to be 0.5 for $k \neq 1$, and $0.5 \cdot (1+\phi_{ADE})$ for $k = 1$ to account for the enhanced transmissibility from humans undergoing secondary infections; we assumed equal degrees of enhancement in transmissibility and susceptibility. Note that in actuality the transmission probabilities between humans and mosquitoes in the absence of enhancement could differ from 0.5. However, the uncertainty in these values can be absorbed by adjusting the vector-to-host ratio to achieve a certain level of transmission intensity. We assumed the extrinsic incubation period to follow an exponential distribution with mean $l_{EIP}(T)$, which was calculated as a function of temperature (T) using a thermal response model estimated by Mordecai *et al*. [76]. Here, we accounted for the seasonal variation in dengue transmission intensity by allowing the temperature to fluctuate between 25°C and 31°C (i.e., similar to the range of the monthly mean temperature in Bangkok or Phnom Penh) within each year using a sine function. We did not explore alternative temperature ranges in this study, since temperature amplitudes were presumed to be reasonably small in most settings with endemic dengue transmission, and any changes in the transmission intensity due to changes in the mean temperature can be reproduced by adjusting the average vector-to-host ratio. Finally, once a mosquito had survived beyond the extrinsic incubation period of the virus, she was assumed to remain infectious until death.

4.3.1.4 Spatial structure and heterogeneous biting

We created a set of communities forming on a squared lattice of size 20×20, following the spatial structure specified by Lourenço & Recker [36]. At the start of the simulation, a total of N_H fully susceptible humans and N_{BS} mosquito breeding sites were uniformly distributed over space. Within each feeding cycle t, each adult female mosquito m randomly bit a human h with probability $p_{Bite}^t(m,h)$, which was calculated as follows:

$$p_{Bite}^t(m,h) = \begin{cases} (1-\omega)\cdot\dfrac{v_h^t}{\sum_{h'} v_{h'}^t \cdot \delta(C[m],C[h'])}, & \text{if } \delta(C[m],C[h]) = 1; \\ \omega\cdot\dfrac{v_h^t}{\sum_{h'} v_{h'}^t \cdot \left(1-\delta(C[m],C[h'])\right)}, & \text{if } \delta(C[m],C[h]) = 0. \end{cases}$$

In the equation above, we used $C[h]$ to denote the community where human h lived (similarly for $C[m]$). The value of the indicator function $\delta(C[m],C[h])$ was defined to be one if the two communities $C[m]$ and $C[h]$ were identical or neighbors to each other, and zero otherwise. Given the limited flight range of *Aedes aegypti*, we followed Lourenço & Recker and assumed each mosquito m was only able to bite humans present in her own community $C[m]$ or the neighboring communities [36]. Long-distance transmission occurred due to human movement, and we allowed each mosquito to randomly bite a human living in a distant community with probability ω [36]. The numerator v_h^t denoted the "biting risk score" of individual h at time t, which was normalized according to the equation above to obtain $p_{Bite}^t(m,h)$. We computed the biting risk score as the product of the following two variables:

$$v_h^t = v_{h(1)}^t \cdot v_{h(2)}.$$

The first variable $v_{h(1)}^t$ denoted the relative attractiveness to mosquitoes, which was estimated as a function of an individual's body surface area according to results from Liebman *et al.*'s field study [253], with the growth in body surface area from birth to adulthood simulated based on the age and sex of that individual following Perkins *et al*. [136]. The second variable $v_{h(2)}$ accounted for the residual variation in the biting risk score due to other factors (e.g., differential duration each person spent near biting mosquitoes). For simplicity, we assumed $v_{h(2)}$ to be independent of age and generated its value from a gamma distribution with mean one and standard deviation $\sigma_{v_{h(2)}}$, and we varied the value of $\sigma_{v_{h(2)}}$ across different transmission settings to study its impact on the short- and long-term effectiveness of vector control (refer to section 4.3.1.7 for more details).

4.3.1.5 Human age structure

We simulated the birth, ageing, and death of humans at a time step of 28 days. The maximum human lifespan was set to be 100 years, and each human's age was treated as an integer variable measured in the unit of 28 days, so that every time step we simulated the demographic changes in the population, each human either had their age increased by one or was removed from the population due to death.

We used the United Nations model life table (General Pattern with life expectancy at birth being 75 years) [254] and assumed a constant mortality rate within each 5-year age interval in the life table. This allowed us to convert the age interval length of the life table to 28 days. Prior to the start of the warm-up period, we initialized the age of each human such that the proportion of individuals falling within each 28-day age interval tallied with the stationary age distribution informed by the life table. Throughout the warm-up period, the number of incident deaths occurring at each age every 28 days was fixed at the corresponding expected value according to the life table, so that the stationary age structure can be maintained. Each death was replaced by a newborn whose sex was randomly generated assuming equal probabilities of male and female births.

For the 30 years of the intervention period, we modelled different rates of changes in the population age structure to examine the associated impact on the vector control effectiveness. Let the annual mortality rate at age a in year y be $\lambda_{a,y}$, with $y = 0$ referring to the first year of the intervention period. We assumed $\lambda_{a,y}$ to decline over time according to the following equation:

$$\log(\lambda_{a,y}) = \log(\lambda_{a,0}) + \rho \cdot \beta_a \cdot y \quad (\rho \geq 0,\ \beta_a < 0).$$

In the equation above, the constant β_a was defined such that $(1 - e^{\beta_a})$ was equal to the average annual percentage reduction in mortality rate at age a in Singapore across the period of 1990–2020 [255]. Hence, the rate of mortality decline can be varied by changing the value of the relative rate of ageing ρ, with $\rho = 1$ indicating that the mortality rate declined at the same speed as observed in Singapore. Once we calculated $\lambda_{a,y}$ based on the values of a, y and ρ, the age-specific probability of death within a 28-day time interval can be derived to simulate mortality events using a binomial distribution.

The same method was applied to create different rates of fertility decline, where we used the same relative rate of ageing ρ described earlier. Note that previously we did not specify the age-specific fertility rates since each death was instantaneously replaced by a newborn during the warm-up period. Here, we derived the baseline age-specific fertility rates (i.e., when y was equal to zero) such that both of the following conditions were

satisfied: (1) the ratios of the baseline fertility rates between different ages tallied with the corresponding ratios observed in Singapore in 1990 [255]; (2) the expected total number of newborns in each 28-day time interval, calculated under the baseline age-specific fertility rates and the stationary age distribution mentioned earlier, tallied with the total number of newborns every 28 days during the warm-up period.

4.3.1.6 Development of mosquitoes

We modelled the development of mosquitoes using a Markov process with four states: eggs, larvae, pupae, and adults. Within each feeding cycle, each adult female mosquito m randomly chose a breeding site located in either her own community $C[m]$ (i.e., the community where she was born) or one of the neighboring communities to lay eggs. The number of eggs laid in one oviposition followed a Poisson distribution with mean θ_E. Male and female adult emergence was assumed to be equally likely, and male mosquitoes were not explicitly modelled upon adult emergence.

The relationship between the mosquito developmental rates and temperature was specified using Schoolfield *et al.*'s model [256]. Following Otero *et al.*, the mortality rates of eggs and adult female mosquitoes were both treated as constant, and the larval mortality rate was decomposed into the sum of the natural mortality rate under optimal conditions and the density-dependent mortality rate [257]. We modelled the natural larval mortality rate using an exponential decay function of temperature [257] and assumed the density-dependent larval mortality rate to be proportional to the number of larvae present in a breeding habitat; we denoted the proportionality coefficient by α. The pupal mortality rate was also assumed to vary with temperature according to an exponential decay function, with an additional "emergence factor" to account for unsuccessful emergence of adult mosquitoes [257]. The discrete time step used to simulate the development of mosquitoes during the aquatic stage was taken to be one tenth of the mosquito feeding cycle length, to keep the probabilities of multiple transitions (i.e., from eggs to pupae, from eggs to adults, or from larvae to adults) low (<0.05).

At the start of the simulation, the number of eggs in each breeding site was randomly drawn from a Poisson distribution with mean n_E, with no larvae, pupae, or adults present. To vary the pre-intervention vector-to-host ratio across different simulation runs, we varied the parameter α which was used to calculate the density-dependent larval mortality rate as described earlier. Here, we chose to vary the productivity instead of the total number of breeding sites for the sake of computational efficiency, since the simulation run-time increased with

the number of breeding sites. Finally, at the end of the 100-year warm-up period, we randomly selected a proportion (p_{RM}) of the breeding sites to be permanently removed to reduce the mosquito population size by roughly the same proportion. Although here we implemented vector control via source reduction, the same effect can be achieved using other strategies such as the incompatible insect technique or the sterile insect technique used in real-world settings.

4.3.1.7 Simulation model parameters

We used Latin hypercube sampling [258] to create 1,000 combinations of the following parameters: π, ω, ρ, $\sigma_{v_{h(2)}}$, α, ϕ_{ADE}, and l_C, to study the impact of dengue importation, human mobility, population ageing, individual-level heterogeneity in the risk of being bitten, mosquito abundance, degree of antibody-dependent enhancement, and the average duration of cross-protection upon vector control effectiveness. Note that although both the values of ϕ_{ADE} and l_C are likely to be relatively stable across different transmission settings, they were jointly varied with the other five parameters to test the sensitivity of our results with respect to changes in the strengths of immune interactions assumed in the simulation model. For each combination of the seven parameters mentioned earlier, dengue transmission during the warm-up period was simulated only once. Thereafter, we simulated dengue transmission during the intervention period ten times, each time with a different level of vector control intensity quantified by p_{RM}. The set of values taken by p_{RM} contained zero, which represented the counterfactual scenario to be used for the calculation of vector control effectiveness as described in the next section (Refer to Table 4.1 for the list of parameter values). The simulation model was implemented using Python 3.9.

Constant parameters			
Notation	Description	Value	Remark
N_H	Total number of humans at initialization	100,000	—
N_{BS}	Total number of breeding sites at initialization	100,000	—
n_E	Average number of eggs per breeding site at initialization	500	Approximately equal to the number of eggs per breeding site at the peak breeding season in Otero *et al.*'s simulation model [259]
l_f	Length of the mosquito feeding cycle (days)	4	[260] Assumed constant within (25°C, 31°C)
l_{LP}	Average duration of the human latent period (days)	6	Approximated using the intrinsic incubation period [261]
l_R	Average duration of human infectiousness (days)	5	[262]
θ_E	Average number of eggs laid by an *Aedes* mosquito in one oviposition	63	[257]
μ_E	Mortality rate of mosquito eggs (per day)	0.011	[257]
ef	Emergence factor (Proportion of pupae reaching maturation that emerged as adults)	0.83	[257]
μ_F	Mortality rate of adult female mosquitoes (per day)	0.091	[257]
Parameters whose values were varied across simulations			
Notation	Description	Range	Remark
π	Probability of being exposed to dengue virus from an external source per human per cycle	$(2^0, 2^4) \times 10^{-5}$	Drawn from a uniform distribution on a $\log_2$ scale
ω	Probability of biting a human living in a distant community per mosquito per bite	$(10^{-4}, 10^{-1})$	Drawn from a uniform distribution on a $\log_{10}$ scale
ρ	Relative rate of ageing	Unif (0,4)	Given the same age structure at the beginning of the intervention period, the median age of the human population after 30 years ranged from 38 to 51 years old across the 10,000 simulations
$\sigma_{v_{h(2)}}$	Standard deviation of the biting risk score controlling for body surface area	Unif (0,1)	The range was restricted to (0,1) to avoid a high degree of collinearity between the baseline PE9 and $\sigma_{v_{h(2)}}$
α	Proportionality coefficient used to compute the density-dependent larval mortality rate	Unif (0.1, 0.4)	—
ϕ_{ADE}	Susceptibility enhancement and transmissibility enhancement (assumed equal)	Unif (0,1)	Sensitivity analysis
l_C	Average duration of cross-protection (days)	Unif (1, 365)	Sensitivity analysis
p_{RM}	Vector control intensity (proportion of breeding sites removed)	(*)	—

Table 4.1: List of parameter values for the simulation model

(*) For each combination of the seven parameters above, we simulated transmission during the intervention period 10 times, each time with a different value of p_{RM} from the set {0} U {0.3, 0.35, …, 0.65, 0.7}

4.3.2 Analyses of the simulation output

To measure the overall dengue transmission intensity prior to the implementation of vector control in each simulation run, we computed the proportion of nine-year olds who had prior dengue exposure at the end of the warm-up period ("baseline PE9"). We also calculated the increase in the median age of the entire population during the 30-year intervention period (Δ_{age}). The baseline PE9, Δ_{age}, and the simulation model parameters $p_{RM},\ \pi,\ \omega,\ \sigma_{v_{h(2)}},\ \phi_{ADE}$, and l_C (hereafter referred to as the "predictor variables") will be assessed in their contribution to explaining the variability in the vector control effectiveness, as explained later in this section.

We extracted the monthly number of dengue infections from each simulation run and defined the vector control effectiveness (provided $p_{RM} > 0$) within a time window $(t_y,\ t_y + \Delta_t]$ as follows:

$$\text{eff}_{(t_y,\, t_y+\Delta_t]} = 1 - \frac{\#\text{ of infections with vector control}_{(t_y,\, t_y+\Delta_t]}}{\#\text{ of infections without vector control}_{(t_y,\, t_y+\Delta_t]}}.$$

In the equation above, we used $t_y \in \left\{0, \frac{1}{12}, \frac{2}{12}, \ldots, \frac{358}{12}, \frac{359}{12}\right\}$ to denote the number of years between the start of the intervention period and the left endpoint of the time window, and $\Delta_t \in \{\frac{1}{12}, \frac{2}{12}, \ldots \frac{360}{12}\}$ to denote the time window length in years, subject to the constraint $t_y + \Delta_t \leq 30$. For each simulation where $p_{RM} > 0$, we calculated the vector control effectiveness within each of the following time windows: (0, 5], (5, 10], (10, 15], (15, 20], (20, 25], (25, 30], and (0, 30]. In addition, we obtained the "critical time point" t_y^* defined as the minimum value of t_y such that the intervention effectiveness within the time window $(t_y, t_y + 5]$ was negative, to represent the time point at which the intervention stopped having an impact:

$$t_y^* = \min\left\{t_y:\ t_y \in \left\{0, \frac{1}{12}, \frac{2}{12}, \ldots, \frac{299}{12}, \frac{300}{12}\right\} \wedge\ \text{eff}_{(t_y, t_y+5]} < 0\right\}.$$

Due to the multi-annual incidence patterns of dengue, we set the length of the time window to be five years in the equation above to reduce noise in the estimated underlying trend in the vector control effectiveness. In the rare event that right censoring occurred (i.e., $\text{eff}_{(t_y, t_y+5]} \geq 0, \forall\ t_y$), we set the value of the critical time point to be 25. Hereafter, the intervention effectiveness calculated for each of the seven time windows mentioned earlier, as well as the critical time point, will be referred to as the "outcome variables", and we will use $\boldsymbol{y_j} = (y_{1\ j}, \ldots, y_{9000\ j})^T$ to denote the vector containing all the values of an outcome variable j $(j = 1, \ldots, 8)$ obtained from the 9,000 simulation runs where vector control was implemented (i.e., $p_{RM} > 0$). Similarly, the predictor variable matrix will be denoted by $\mathbf{X} = (x_{si})$ $(s = 1, \ldots, 9000,\ i = 1, \ldots, 8)$.

We used the extreme gradient boosting (XGBoost) algorithm to capture the nonlinear and interaction effects of the predictor variables on each of the eight outcome variables [263]. To train the XGBoost model for each outcome variable j, we first split our data $(\mathbf{X}, \boldsymbol{y_j})$ into training and test sets using a 9:1 ratio and tuned the following hyper-parameters via 5-fold cross validation performed on the training set: maximum tree depth, the minimum number of observations in each node, the fractions of rows and columns to be randomly sampled to grow each tree, and the learning rate. Given the optimal hyper-parameters, the number of boosting rounds for the final model was determined by monitoring the out-of-sample prediction accuracy evaluated on the test set. Finally, we calculated the "Shapley additive explanations" (SHAP) value S_{ijs} for each combination of predictor variable i, outcome variable j, and simulation run s [264]. Specifically, the SHAP value S_{ijs} quantified the change in the *predicted* value of y_{sj} due to x_{si}, where the comparator was set to be the predicted value of y_{sj} had the predictor variable i been excluded from the XGBoost model. All the analyses of the simulation output were performed using Python 3.9.

4.4 Results

Following a sustained reduction in the mosquito population size, the monthly number of dengue infections immediately dropped to near zero and continued to hover just above zero for a number of years, with a marked difference in the time point at which dengue incidence started to rebound across different transmission settings (refer to Figure 4.2 for a visualization of the general model behavior). We found vector control intensity and the baseline PE9 to be the most important predictors for the critical time point (i.e., the time point at which vector control stopped having an impact), followed by the degree of individual-level heterogeneity in the biting risk score (Figure 4.3). Notably, the higher the level of vector control intensity, the more pronounced the negative impacts of baseline PE9 and heterogeneous biting on the critical time point (Figure 4.3). The above findings also held true for the case of vector control effectiveness across the entire 30 years of the intervention period ("overall intervention effectiveness") (Figure 4.4).

During the first five years of the intervention period, vector control was substantially more effective when implemented at a higher intensity, and the distribution of intervention effectiveness became less dispersed as vector control intensity increased (Figure 4.5). However, the median value of intervention effectiveness diminished towards zero over time and became almost indistinguishable between different levels of vector

control intensity in the end (i.e., within the time window (25y, 30y]) (Figure 4.5). Interestingly, the median value of intervention effectiveness was found to be negative within the time window (10y, 15y] when the vector control intensity was within the range of 30%–45% (Figure 4.5).

Similar to vector control intensity, none of the other predictor variables had a substantial impact on vector control effectiveness in the long term. For example, an increase in the importation risk (π), from $2^0 \times 10^{-5}$, to $2^4 \times 10^{-5}$ per human per cycle was estimated to reduce intervention effectiveness by ~15 percentage points on average within the first five years of the intervention period, but the impact became much less significant beyond year 5 (Figure 4.6). Besides, there was an inverse relationship between the baseline PE9 and intervention effectiveness within the first 15 years of the intervention period, with the impact of baseline PE9 being more pronounced at a relatively low vector control intensity during (0y, 5y], moderate vector control intensity during (5y, 10y], and moderately high intensity during (10y, 15y] (Figure 4.7). Thereafter, an increase in the baseline PE9 was associated with a decrease in intervention effectiveness only at high levels of vector control intensity within the time windows (15y, 20y] and (20y, 25y] (Figure 4.7). The same conclusion held true for the relationship between heterogeneous biting and vector control effectiveness, except that during the last five years of the intervention period, there was an overall weak positive relationship between the degree of individual-level heterogeneity in the biting risk score and intervention effectiveness (Figure 4.8). In addition, neither population ageing nor host mobility was found to have a substantial impact on vector control effectiveness within any of the 5-year time windows considered in this study (Figures 4.9 and 4.10), although a 12-year increase in the median age of the population during the 30-year intervention period was estimated to increase the overall intervention effectiveness by ~8 percentage points on average at the highest level of intervention intensity (i.e., $p_{RM} = 0.7$) (Figure 4.4).

When we increased the degree of antibody-dependent enhancement in our simulations, there was a modest decrease in both the short-term and the overall intervention effectiveness, as well as the value of the critical time point (Figures 4.11 and 4.12). Furthermore, the average duration of cross-protection was found to have a negligible impact on both the intervention effectiveness and the critical time point (Figures 4.11 and 4.13). Overall, varying the strengths of immune interactions did not result in a substantial change in our model results.

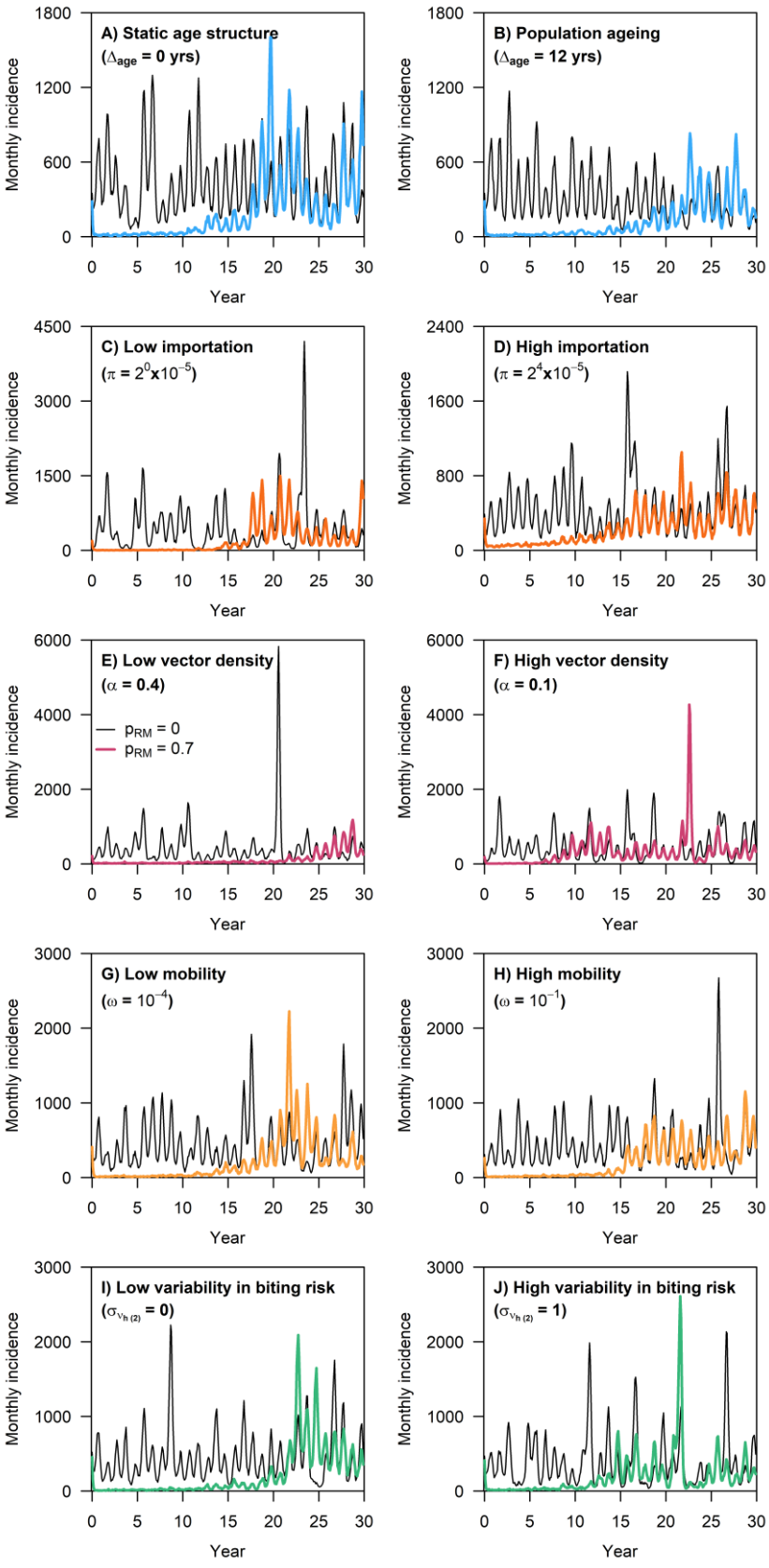


Figure 4.2: Simulated monthly number of dengue infections during the 30-year intervention period in different dengue endemic settings. Each figure panel showed model outputs obtained from two simulation runs with the same starting condition, where one simulation was performed without vector control and the other with the vector population size permanently reduced by 70% at the start of the intervention period. Unless otherwise stated, model parameters were fixed at the following values: $\pi = 4 \times 10^{-5}, \omega = 0.005,\ \rho = 0,\ \sigma_{\upsilon_{h(2)}} = 0.5,$ $\alpha = 0.25,\ \phi_{ADE} = 0.5,\ l_C = 180.$

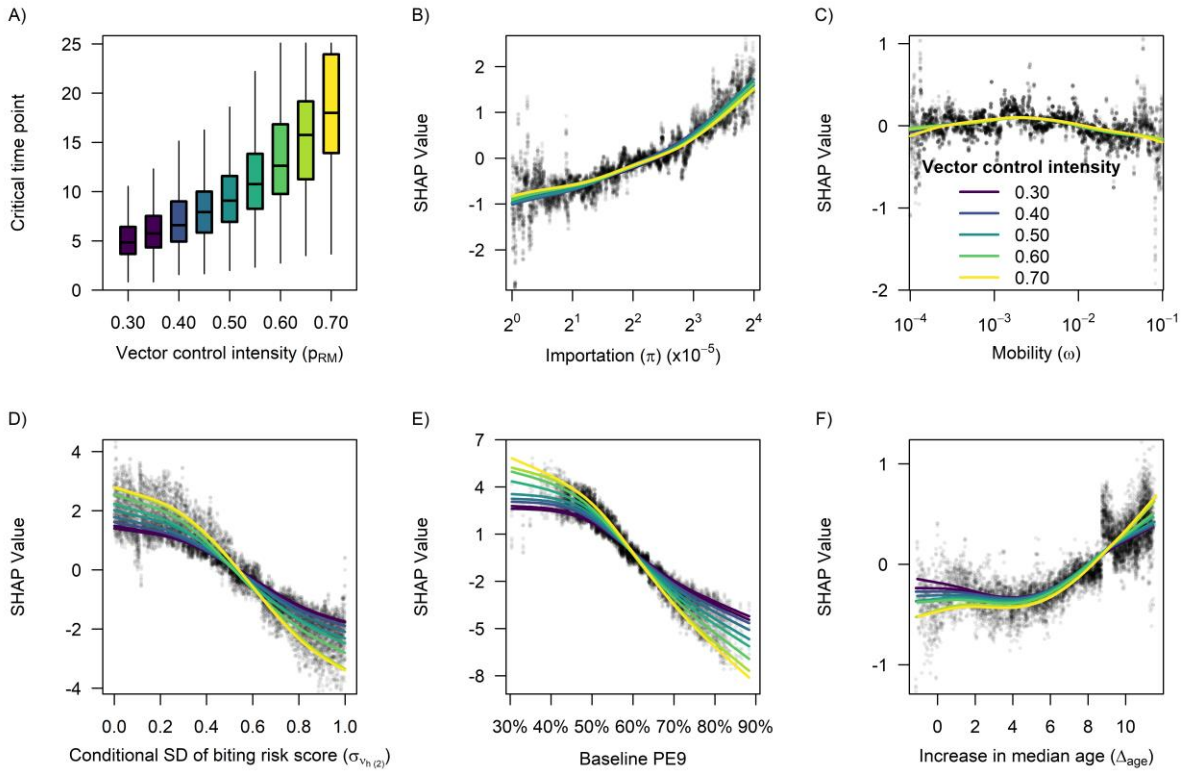


Figure 4.3: Impact of different predictor variables on the critical time point. For (A), the original values of the critical time point obtained from the simulation runs were summarized, with outliers omitted from the graph for clarity. For (B)—(F), we used SHAP values to quantify the change in the predicted critical time point due to the value of each predictor variable, where each point represents a single simulation run and each curve was obtained by fitting a smoothing spline to the set of data points corresponding to a given level of vector control intensity.

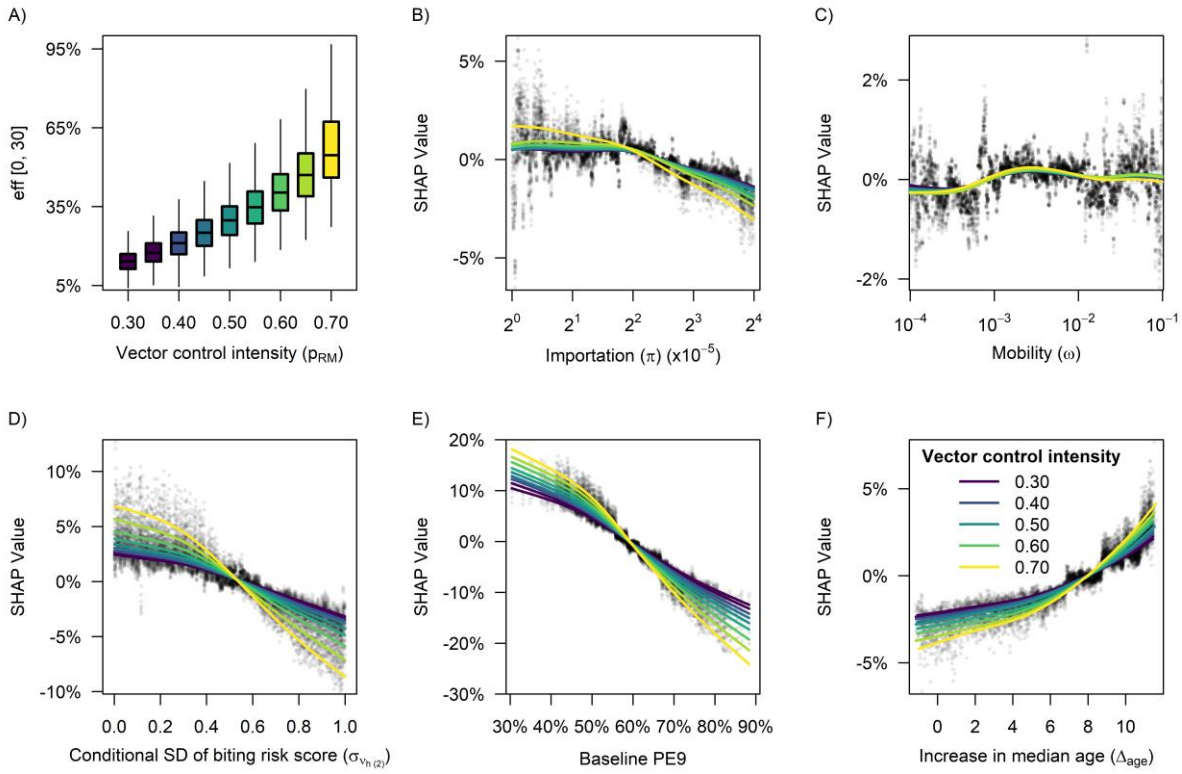


Figure 4.4: Impact of different predictor variables on the vector control effectiveness across the entire 30-year intervention period. For (A), the original values of the intervention effectiveness obtained from the simulation runs were summarized, with outliers omitted from the graph for clarity. For (B)—(F), we used SHAP values to quantify the change in the predicted intervention effectiveness due to the value of each predictor variable, where each point represents a single simulation run and each curve was obtained by fitting a smoothing spline to the set of data points corresponding to a given level of vector control intensity.

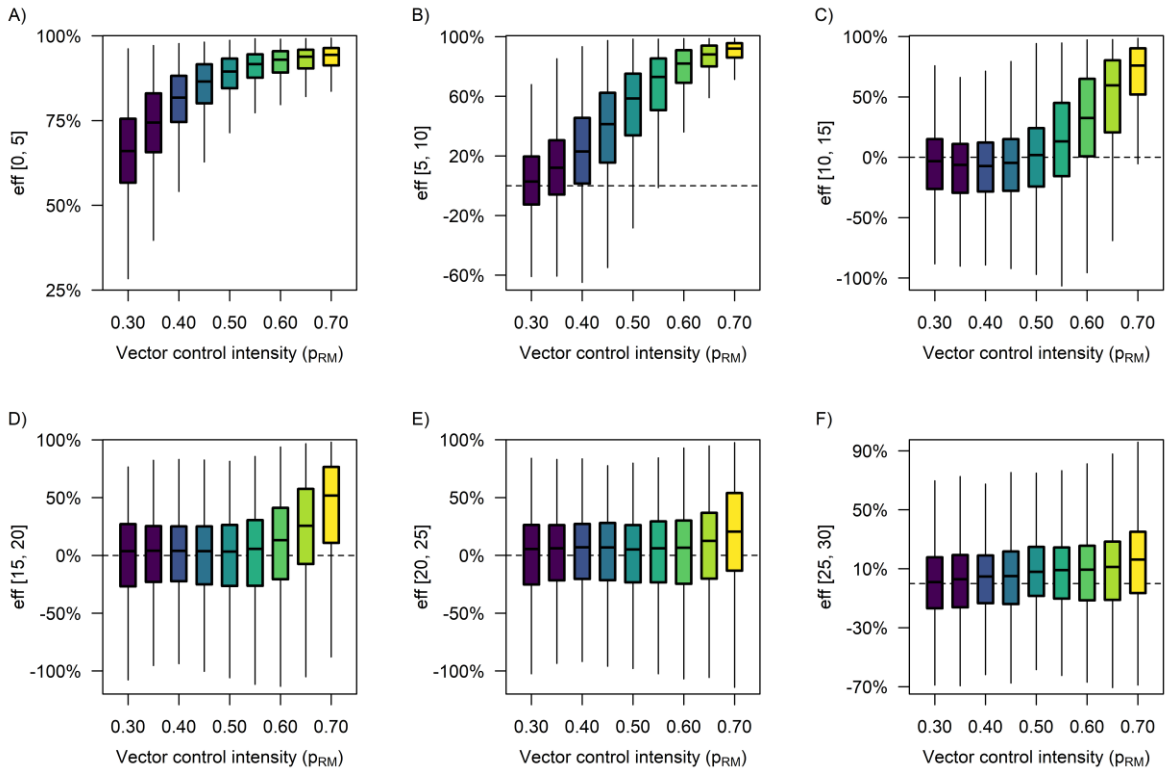


Figure 4.5: Boxplots of the intervention effectiveness under different levels of vector control intensity within each of the following time windows: (A) (0y, 5y], (B) (5y, 10y], (C) (10y, 15y], (D) (15y, 20y], (E) (20y, 25y], (F) (25y, 30y]. Outliers were omitted from the graph for clarity.

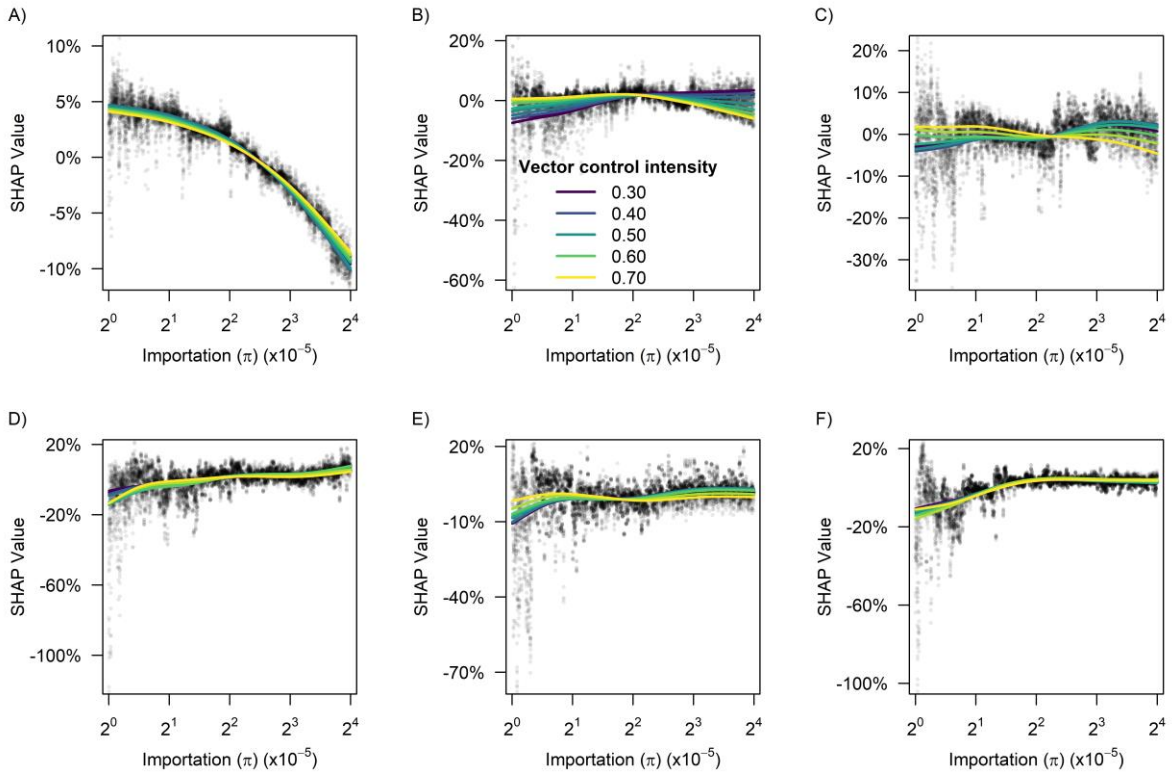


Figure 4.6: Impact of dengue importation on the predicted vector control effectiveness within each of the following time windows: (A) (0y, 5y], (B) (5y, 10y], (C) (10y, 15y], (D) (15y, 20y], (E) (20y, 25y], (F) (25y, 30y]. The parameter π referred to the probability of being exposed to dengue virus from an external source per human per feeding cycle. Each point represents a single simulation run, and the SHAP value quantified the change in the predicted vector control effectiveness due to the value of π. Each curve was obtained by fitting a smoothing spline to the set of data points corresponding to a given level of vector control intensity.

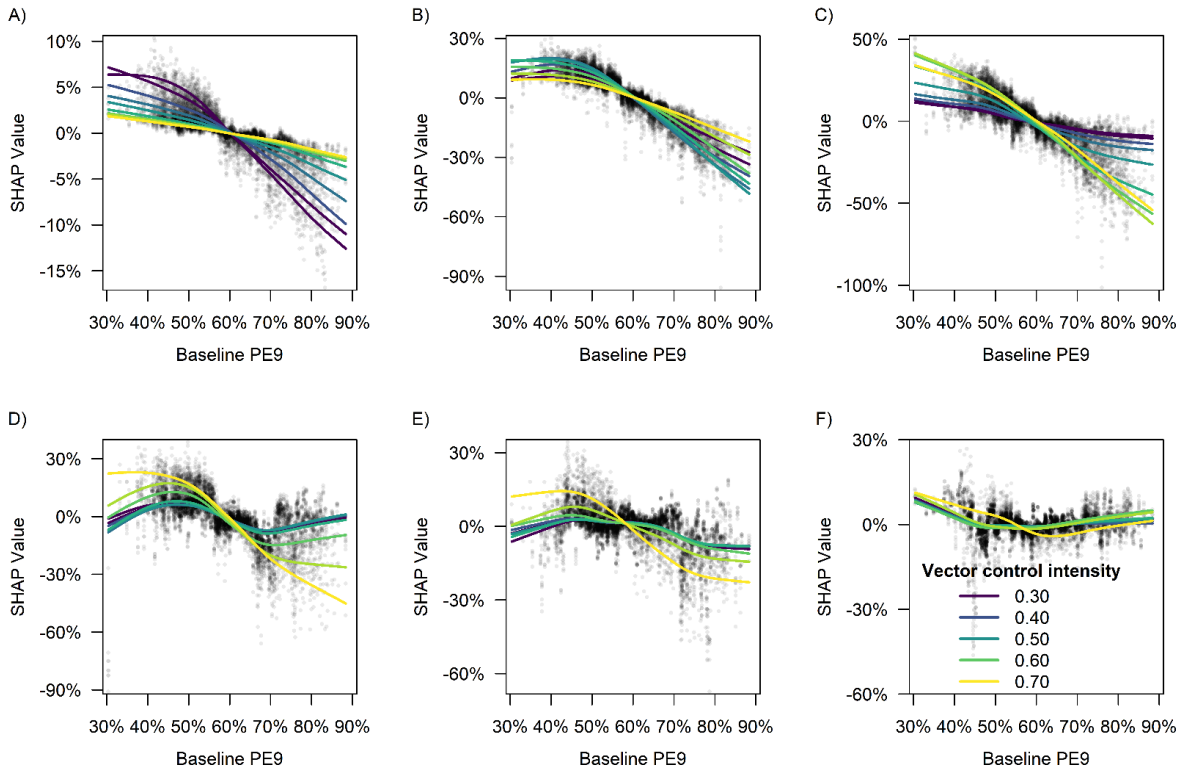


Figure 4.7: Impact of baseline PE9 on the predicted vector control effectiveness within each of the following time windows: (A) (0y, 5y], (B) (5y, 10y], (C) (10y, 15y], (D) (15y, 20y], (E) (20y, 25y], (F) (25y, 30y]. The baseline PE9 referred to the proportion of nine-year olds who had prior dengue exposure at the end of the warm-up period. Each point represents a single simulation run, and the SHAP value quantified the change in the predicted vector control effectiveness due to the value of baseline PE9. Each curve was obtained by fitting a smoothing spline to the set of data points corresponding to a given level of vector control intensity.

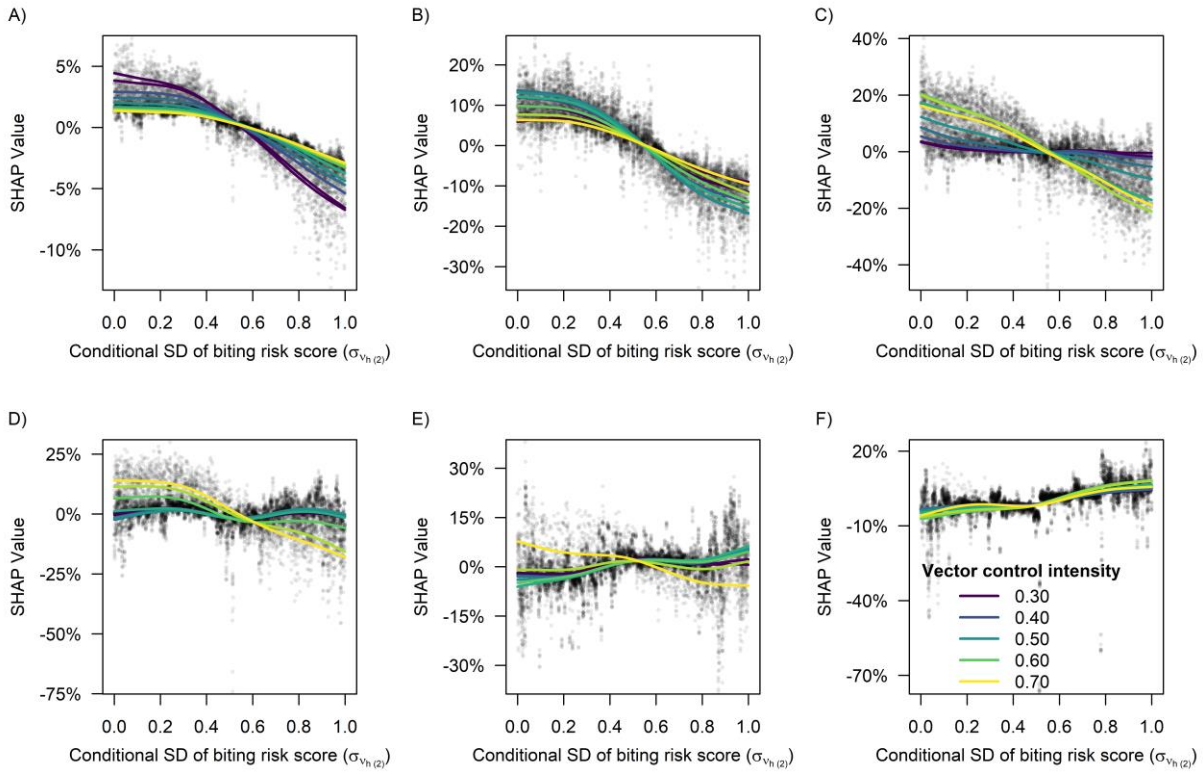


Figure 4.8: Impact of individual-level heterogeneity in the biting risk score on the predicted vector control effectiveness within each of the following time windows: (A) (0y, 5y], (B) (5y, 10y], (C) (10y, 15y], (D) (15y, 20y], (E) (20y, 25y], (F) (25y, 30y]. The parameter $\sigma_{v_{h(2)}}$ referred to the standard deviation of the biting risk score controlling for body surface area. Each point represents a single simulation run, and the SHAP value quantified the change in the predicted vector control effectiveness due to the value of $\sigma_{v_{h(2)}}$. Each curve was obtained by fitting a smoothing spline to the set of data points corresponding to a given level of vector control intensity.

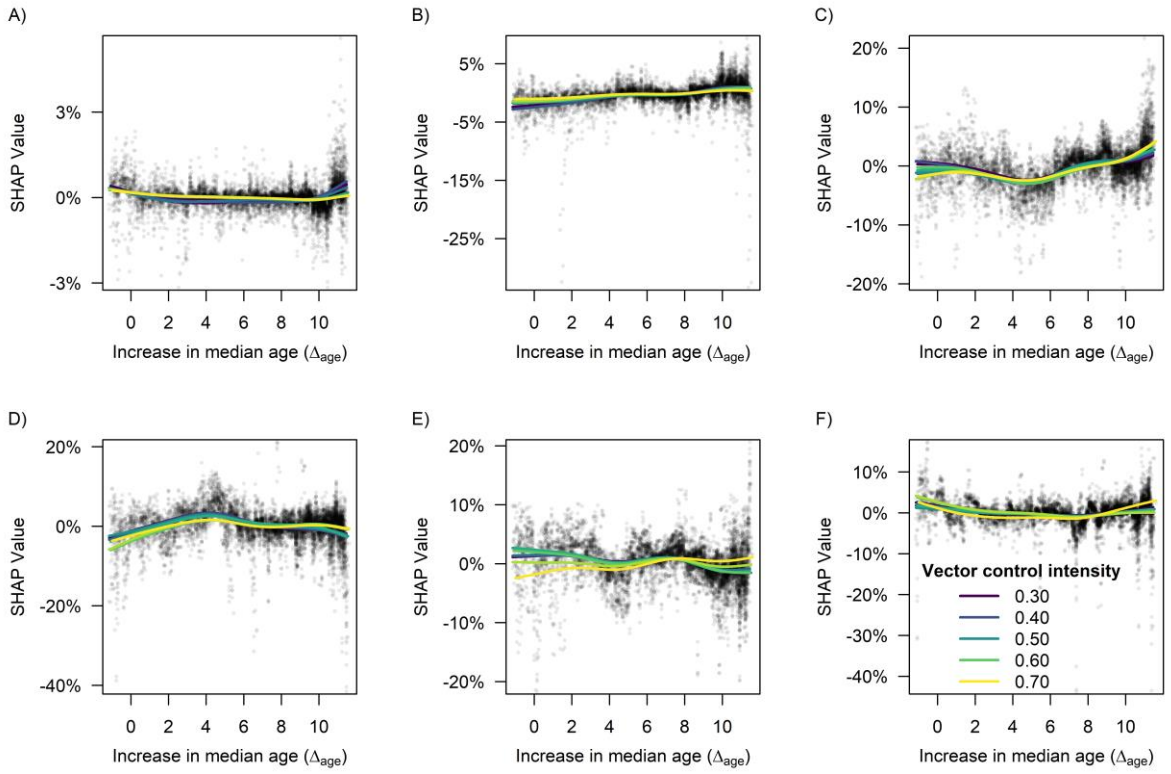


Figure 4.9: Impact of population ageing on the predicted vector control effectiveness within each of the following time windows: (A) (0y, 5y], (B) (5y, 10y], (C) (10y, 15y], (D) (15y, 20y], (E) (20y, 25y], (F) (25y, 30y]. The parameter Δ_{age} referred to the increase in the median age of the population during the 30-year intervention period. Each point represents a single simulation run, and the SHAP value quantified the change in the predicted vector control effectiveness due to the value of Δ_{age}. Each curve was obtained by fitting a smoothing spline to the set of data points corresponding to a given level of vector control intensity.

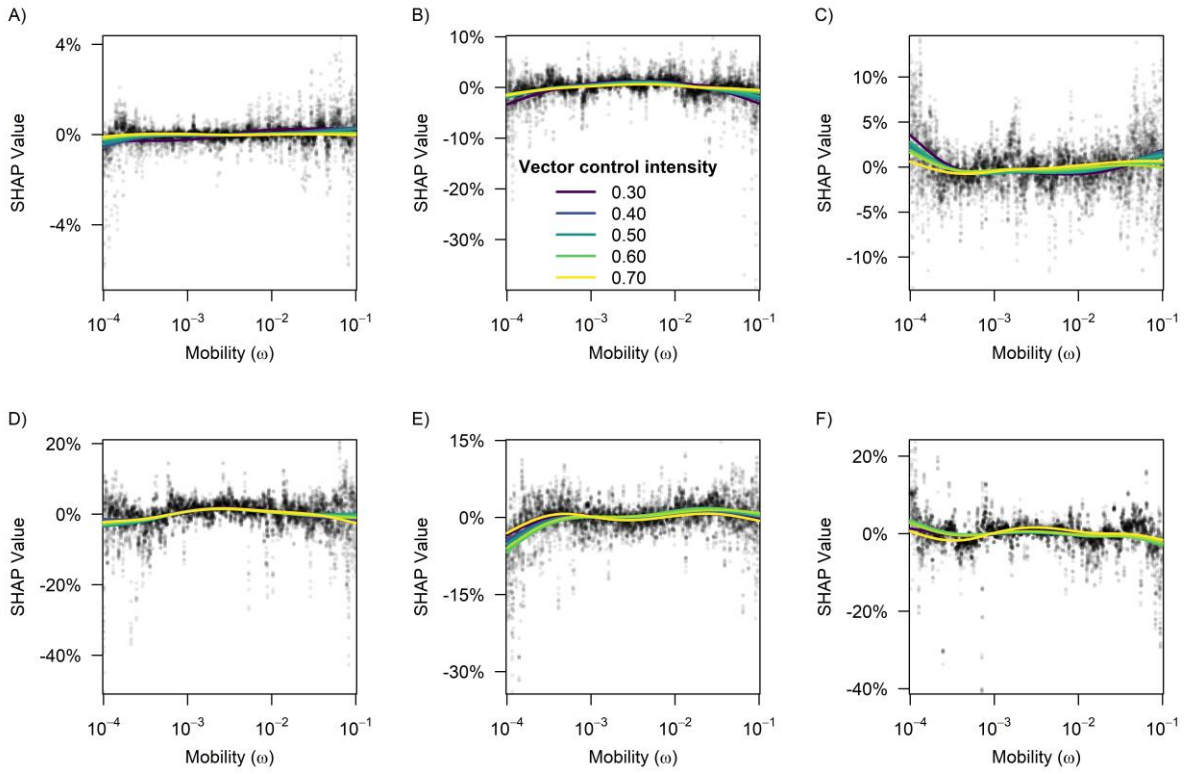


Figure 4.10: Impact of host mobility on the predicted vector control effectiveness within each of the following time windows: (A) (0y, 5y], (B) (5y, 10y], (C) (10y, 15y], (D) (15y, 20y], (E) (20y, 25y], (F) (25y, 30y]. The parameter ω referred to the probability of biting a human living in a distant community per mosquito per bite. Each point represents a single simulation run, and the SHAP value quantified the change in the predicted vector control effectiveness due to the value of ω. Each curve was obtained by fitting a smoothing spline to the set of data points corresponding to a given level of vector control intensity.

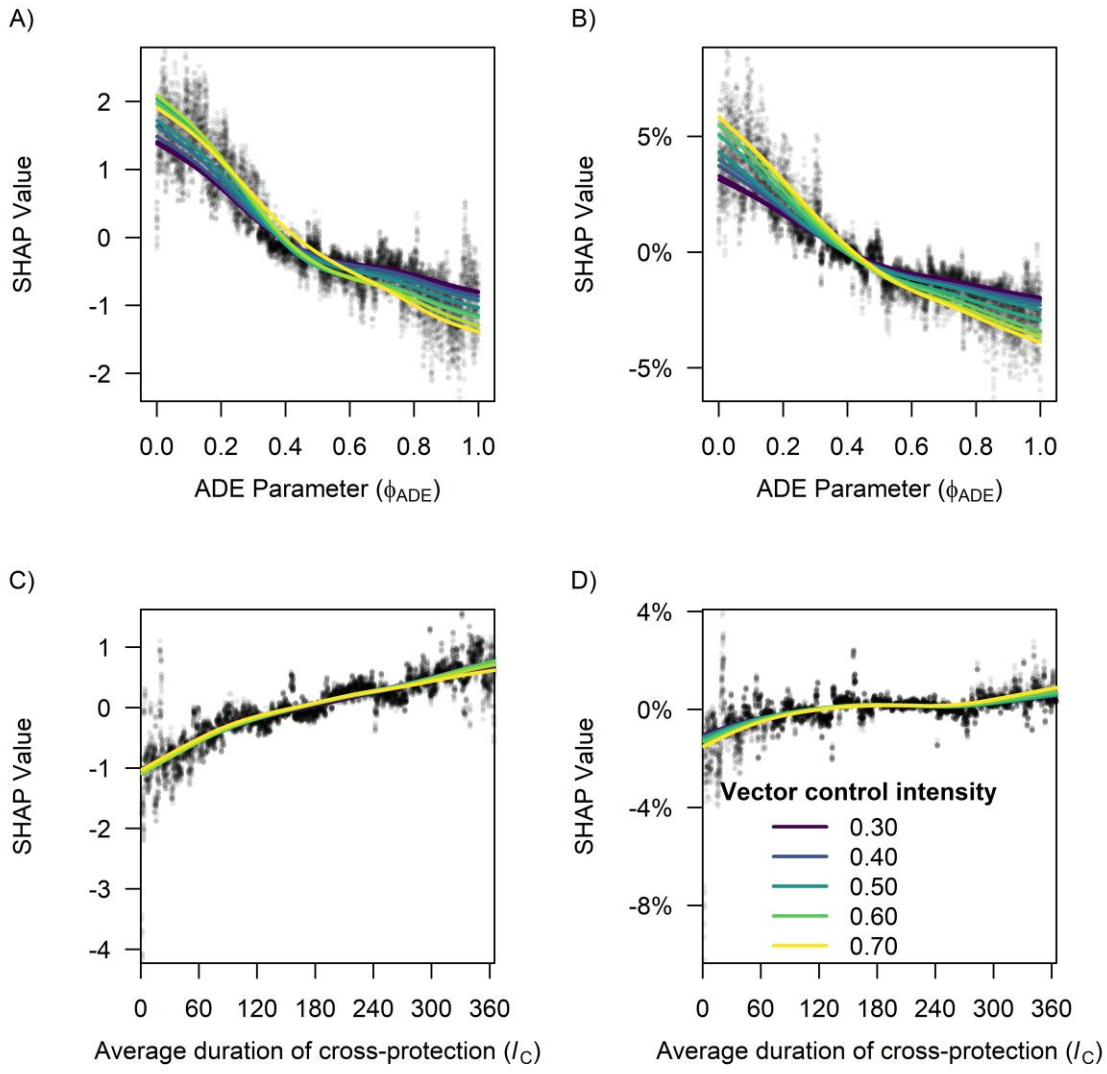


Figure 4.11: Impact of the strengths of immune interactions assumed in the simulation model on the (A, C) critical time point, and (B, D) vector control effectiveness across the entire 30-year intervention period. Each point represents a single simulation run, and the SHAP value quantified the change in the predicted value of the outcome variable due to the value taken by the predictor variable. Each curve was obtained by fitting a smoothing spline to the set of data points corresponding to a given level of vector control intensity.

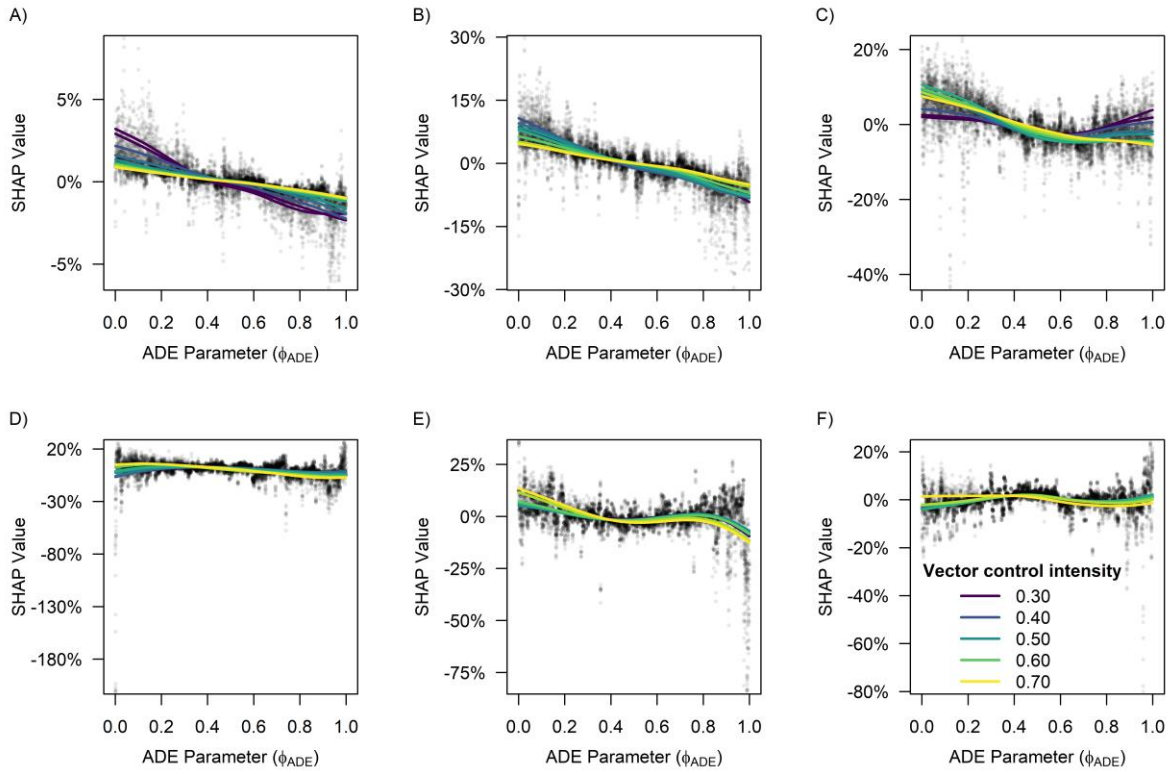


Figure 4.12: Impact of antibody-dependent enhancement on the predicted vector control effectiveness within each of the following time windows: (A) (0y, 5y], (B) (5y, 10y], (C) (10y, 15y], (D) (15y, 20y], (E) (20y, 25y], (F) (25y, 30y]. The parameter ϕ_{ADE} quantified both the susceptibility enhancement and transmissibility enhancement.

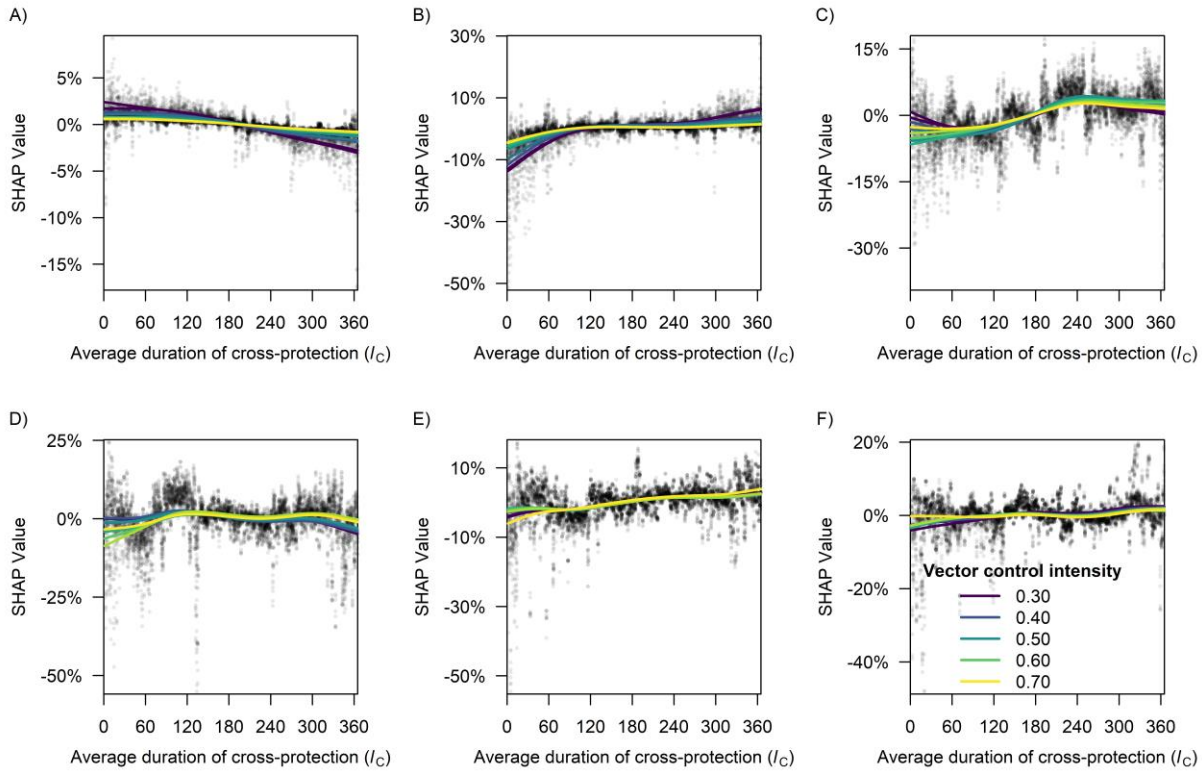


Figure 4.13: Impact of cross-protection on the predicted vector control effectiveness within each of the following time windows: (A) (0y, 5y], (B) (5y, 10y], (C) (10y, 15y], (D) (15y, 20y], (E) (20y, 25y], (F) (25y, 30y]. The parameter l_c referred to the average duration of cross-protection (measured in days).

4.5 Discussion

Our study highlights the contrast between the short- and long-term impacts of sustained dengue vector control in endemic settings, in congruence with the findings from previous studies: Following the suppression of the *Aedes* mosquito population, the equilibrium between the basic reproduction number and level of herd-immunity was disrupted, causing dengue incidence to drastically decline [111,128]. This in turn reduced population immunity over time as new births were added to the susceptible pool and immune individuals were gradually removed from the population due to death [111,128]. Consequently, dengue incidence gradually bounced back, and the existing intervention was no longer effective in the end, requiring the intensity of vector control to be further enhanced to reduce dengue transmission [111]. We have additionally demonstrated that the time point at which vector control stopped having an impact depended mainly on the degree of mosquito suppression, pre-intervention transmission

intensity, and the degree of individual-level heterogeneity in the risk of being bitten by the *Aedes* mosquitoes. It is thus important for future studies to account for the effects of these variables, as well as their interactions, to forecast the time-varying impact of an *Aedes* control program for planning purposes.

For all the levels of vector control intensity considered in this study, a higher pre-intervention transmission intensity was associated with a lower proportion of dengue infections averted within each of the following time windows (0y, 5y], (5y, 10y], and (10y, 15y]. A likely explanation is that an increase in the pre-intervention transmission intensity will generally reduce the average number of years before sustained transmission resumes, thereby causing intervention effectiveness to decrease: Suppose the basic and effective reproduction numbers prior to the intervention are R_0 and 1 respectively ($R_0 > 1$), and hence the fraction of the population being susceptible to dengue at baseline is approximately $1/R_0$. If vector control reduces the basic reproduction number to R_0/k ($R_0 > k > 1$) and if the susceptible fraction of the population increases by c each year during the early phase of the intervention period, the number of years before the effective reproduction number bounces back to one will be $\frac{k-1}{R_0 \cdot c}$, which is inversely proportional to R_0. While the above conclusion may rely on some simplifying assumptions, our individual-based model has produced similar results, which showed that the baseline PE9 had a negative relationship with the critical time point especially at a high level of vector control intensity. It should be noted, however, that the relationship between baseline PE9 and vector control effectiveness became much weaker after 15 years (except when the mosquito population size was suppressed by over 60%), since dengue incidence had already started to rebound by the 15th year in most of the simulation runs, regardless of the baseline PE9 value.

In addition to the important role of pre-intervention transmission intensity in modifying the effectiveness of dengue vector control, our results suggest that the degree of individual-level heterogeneity in the biting risk also needs to be accounted for to forecast the epidemiological impact of a vector control program. Earlier studies have produced strong evidence for the heterogeneous blood feeding patterns of the *Aedes* mosquitoes using DNA profiling of bloodmeals [253,265], and their results have important implications for vector control. For example, maximum impact will not be achieved if a uniform strategy is implemented in settings where a small number of individuals have a disproportionately large contribution to transmission [266]. Besides, heterogeneous biting may increase the success rate of pathogen invasion, thus making disease eradication more difficult [267]. The results obtained in this

study highlight another challenge posed by heterogeneous biting: When the biting risk is more heterogeneous at the individual level, there will be a higher chance for multiple infectious mosquito bites to land on the same susceptible individual, and the resulting infection cannot be averted unless most, if not all, of the aforementioned mosquito bites are prevented. Hence, given a fixed level of reduction in vector density, the percentage decrease in the number of incident dengue infections will be lower in the presence of heterogeneous biting compared with the case where each infectious mosquito chooses to bite a unique individual. Note that to avoid a high degree of collinearity between the baseline PE9 and the variability in biting risk, we set the maximum standard deviation of the biting risk score conditional on the body surface area to be 1, which corresponds to a setting where 55% of the human population receive 90% of all the mosquito bites. In reality, the distribution of mosquito bites may be more skewed in some transmission settings, where the negative impact of heterogeneous biting on the vector control effectiveness can be even larger than what we estimated in this study. More research that quantifies and explains the individual-level heterogeneity in biting risk would be helpful to inform the development of more effective dengue vector control policies that can be implemented to achieve the desired epidemiological impact.

We have observed a weak positive impact of population ageing on the effectiveness of vector control over the 30-year intervention period. However, this should not be interpreted as a minor effect of demographic transition on dengue epidemiology, as previous work has shown that the decline in birth and death rates can cause a substantial decrease in the force of infection, as well as a change in both the age distribution of dengue cases and the periodicity of dengue outbreaks [268]. Instead, our results suggest that population ageing may play a relatively limited role in modifying the effect of a vector control program on dengue incidence. Similarly, we did not find host mobility to have an appreciable impact on vector control effectiveness despite its important influence on the epidemic variability and the risk of serotype extinction [36]. Furthermore, our study has shown that the average duration of cross-immunity has a minimal impact on vector control effectiveness, while the impact of antibody-dependent enhancement was found to be somewhat higher. Sensitivity analysis should remain an important component of future modelling studies to assess the uncertainty in the predicted effectiveness of dengue vector control with respect to the uncertainty in the strengths of immune interactions.

In the absence of vector control tools that are currently available to be rolled out on a large scale to achieve dengue elimination, the results obtained in this study highlight the need to adopt an integrated approach to achieve a sustained reduction in dengue transmission. Special considerations are needed to maximize the cost-effectiveness of combined interventions in each particular context, given that there may be interactions between different intervention strategies. For example, the effectiveness of the CYD-TDV vaccine was predicted to become lower when combined with targeted indoor residual spraying due to the reduced rate of natural infection [129]. Nonetheless, the combination was shown to outperform either alone in high-transmission settings [129], where it may be still beneficial to consider introducing a combined intervention program in view of the relatively low vector control effectiveness based on our model estimates.

There are several limitations in this study. First, we did not explore the impact of spatial heterogeneity in mosquito density on the effectiveness of vector control, and the host movement behaviors have also been simplified in our simulations. More work will be required to study the spatio-temporal variation in vector control effectiveness on heterogeneous landscapes, and how this may be modified by complex host movement patterns. In addition, we also did not consider alternative age structures in the warm-up period of the simulations, and hence the estimated magnitude of interaction between demographic transition and vector control may not generalize to other settings. Besides, our study did not account for the practical deployment of vector control and assumed an instant reduction in the mosquito population size across the entire study area at the start of the intervention period. Scenario planning would also need to be incorporated in future work, to predict the change in the incidence of dengue infections due to unforeseen circumstances such as failure to achieve sustained mosquito suppression. Finally, we may have overestimated the variability in the distribution of vector control effectiveness due to the highly stochastic outputs from the simulation model. Hence, our study has mainly focused on the average effect of each predictor variable on the predicted intervention effectiveness, which we derived separately under different levels of vector control intensity.

4.6 Conclusion

We used a detailed individual-based model to demonstrate the contrast between the short-term and long-term impacts of sustained vector control in dengue endemic settings. The finding that short term gains are mostly reversed as population immunity wanes may lie behind the resurgence of dengue in settings such as Singapore where previously successful vector control created a generation with very little immunity to dengue. Such a reduction in immunity necessitates ever stronger vector control efforts to continue to suppress incidence. Our model has also identified factors that have a substantial impact on the effectiveness of vector control, and how the magnitudes of their impacts may vary over time under different levels of vector control intensity. These insights will be useful to inform future model-based studies that seek to predict the impact of dengue vector control in their local contexts. As more countries start to direct their investment towards implementing novel dengue control measures, timely and detailed surveillance data complemented by mathematical modelling will provide valuable evidence to guide the development of public health policies to achieve sustained reduction in the burden of dengue.

Chapter 5: Discussion and conclusion

Globally, an estimated total of 6.1 billion people will live in areas at risk of dengue transmission in 2080 [98], and the burden of Zika, yellow fever, and chikungunya is also anticipated to grow as the geographical range of the *Aedes* mosquitoes further expands [97]. While huge progress has been made in the recent decades in developing diagnostics, vaccines, and novel vector control technologies to reduce the transmission of *Aedes*-borne diseases, numerous knowledge gaps still remain that pose challenges to the optimized resource allocation for public health surveillance and the formulation of more effective disease control strategies. In this thesis, I used mathematical and statistical modelling to address some of these important research gaps, and the results of my studies are of direct relevance to the making of evidence-based public health policies going forward.

5.1 Main findings and their public health implications

In chapter 2, I developed a novel modelling framework integrating diverse sources of information to unveil locations at a high risk of undetected ZIKV transmission during the 2015—2016 global epidemic. Although more than four years have passed since the drastic decline of Zika cases worldwide, the results obtained in this study still remain useful to guide ZIKV surveillance systems strengthening efforts: In July 2021, the Kerala state in India, which is among the 24 high-risk locations identified in our study, reported its first Zika outbreak in history, with more than 60 cases detected as of mid-August [269]. Notably, phylogenetic analysis of partial sequences suggest that the virus from Kerala belonged to the Asian outbreak strain that was previously circulating during the 2015—2016 Zika pandemic, and retrospective testing of serum samples indicates that ZIKV had already started to spread silently prior to the index case infection [269]. Moreover, the Maharashtra state in India (again one of the 24 high-risk locations) also detected their first ever Zika case in July 2021 [270]. The recent Zika outbreak in India has demonstrated the value of our study results for pinpointing locations that should be prioritized for surveillance system strengthening, which is critical for mitigating the risk of another large-scale Zika epidemic in the future.

In chapter 3, I built regression models using longitudinal data generated from Singapore's Gravitrap surveillance system to identify the environmental and/or anthropogenic variables that best explain the spatio-temporal

dynamics of the *Aedes* population in the country's high-rise public residential zones. I found building age to be the strongest predictor for the abundance of both *Ae. aegypti* and *Ae. albopictus* in Singapore, and this new finding can be used to improve the allocation of resources for vector control inspections and environmental management. Besides, a lot of land-cover variables (e.g., managed vegetation cover) were found to have a statistically significant association with *Aedes* abundance, and the interesting insights generated from this work can inspire future studies that attempt to utilize satellite imagery data to predict the spatial abundance patterns of *Aedes* mosquitoes in their local urban environments. Furthermore, my results showed that weather had a much weaker impact on the *Aedes* abundance in Singapore's public housing estates than previously thought, suggesting that policy makers may need to re-evaluate the use of meteorological data to guide vector control operations.

In chapter 4, I developed an agent-based model to study the time-varying epidemiological impact of sustained *Aedes* control in a wide range of dengue endemic settings. I showed that while a fixed level of reduction in vector density may result in a temporary cessation of dengue transmission, the expected value of intervention effectiveness would gradually diminish towards zero in the long run unless the lowered vector density is able to reduce the basic reproduction number to below one, which can be extremely difficult to achieve given the currently available tools for *Aedes* control. Furthermore, I have identified the main drivers of the difference in vector control effectiveness across different transmission settings. Specifically, the percentage reduction in dengue infections is determined not only by the level of reduction in mosquito density, but also by the transmission intensity prior to intervention roll-out and the individual-level heterogeneity in biting risk. I further showed that the intensity of vector control interacts with the other two factors mentioned earlier, and the interaction magnitude also changes over time. My results underscore the need for policy makers to consider the joint effects of mosquito density reduction, pre-intervention transmission intensity, and heterogeneous biting on dengue incidence during vector control policy planning. Moreover, the contrast between the short-term and long-term impacts of sustained vector control shown in this study highlight the importance of continued investments in developing and testing more effective vector control technologies, as well as the need to adopt an integrated approach whereby vector control is implemented in combination with vaccination to achieve sustained reduction in the burden of dengue.

5.2 Future work

A key contribution of my study in chapter 2 is the identification of locations that should be prioritized for ZIKV surveillance systems enhancement in the future. More testing should also be conducted in these locations to examine whether any past or ongoing ZIKV transmission is present, which will help to further narrow the gaps in understanding the geographical extent of the virus and to strengthen global Zika outbreak preparedness and response capacity. Besides, for countries that were hard-hit by ZIKV during the 2015—2016 pandemic, further research is required to study the long-term antibody kinetics after ZIKV infection and how this may impact protective immunity, which will shed light on the future risk of a new Zika outbreak. Reliable diagnostics also need to be developed to distinguish between the Asian and African ZIKV lineages, the latter of which has been demonstrated to have higher transmissibility and pathogenicity [271] and has recently been isolated from both *Aedes* mosquitoes and non-human primates in Brazil [272].

To date, little consensus has been reached as to where dengue transmission occurs. While my study in chapter 3 has generated interesting insights into the spatio-temporal variation of *Aedes* abundance in Singapore's public housing estates, more research is required to quantify the proportion of dengue cases attributable to infections inside versus outside of home so that the epidemiological impact of vector control can be further enhanced. Future research should also seek to understand how the spatial distribution of *Aedes* mosquitoes, host movement behaviors, and the immune landscape jointly influence the complex transmission dynamics of dengue in Singapore. Filling this knowledge gap will be critical to optimizing resource allocation to reduce dengue transmission.

Finally, my results in chapter 4 highlight the important influence of heterogeneous biting on the effectiveness of dengue vector control. More research is needed to empirically quantify the individual-level heterogeneity in exposure to mosquito bites in their local contexts and to evaluate its potential impact on the effectiveness of dengue control measures, including both vector control and vaccination. Besides, as more countries start to implement intensive *Aedes* control measures, it will be beneficial to develop new approaches to integrate diverse sources of information collected from the field during the early phase of the intervention, such as dengue seroprevalence, proportion of reported cases being imported versus locally acquired, and individual-level dengue case data, to predict how long the impact of the current intervention will last. Recent studies that created new

methods to infer malaria transmission dynamics in near-elimination settings may provide useful insights into how this might be achieved [185,273].

5.3 Final statement

In this thesis, I used mathematical and statistical modelling to expand what is currently known about the epidemiology and ecology of *Aedes*-borne diseases as well as the impact of *Aedes* control on disease transmission. It is hoped that the work produced in this thesis will help to improve the evidence base for the making of key public health policy decisions to reduce the burden of *Aedes*-borne diseases at the city, national, and global scales.

Bibliography

1. Dwivedi VD, Tripathi IP, Tripathi RC, Bharadwaj S, Mishra SK. Genomics, proteomics and evolution of dengue virus. Brief Funct Genomics [Internet]. 2017 Jan 10;elw040. Available from: https://academic.oup.com/bfg/article-lookup/doi/10.1093/bfgp/elw040
2. World Health Organization. Dengue: guidelines for diagnosis, treatment, prevention, and control [Internet]. WHO/HTM/NTD/DEN/2009.1. 2009. Available from: http://whqlibdoc.who.int/publications/2009/9789241547871_eng.pdf
3. Mustafa MS, Rasotgi V, Jain S, Gupta V. Discovery of fifth serotype of dengue virus (DENV-5): A new public health dilemma in dengue control. Med J Armed Forces India [Internet]. 2015 Jan;71(1):67–70. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0377123714001725
4. ten Bosch QA, Clapham HE, Lambrechts L, Duong V, Buchy P, Althouse BM, Lloyd AL, Waller LA, Morrison AC, Kitron U, Vazquez-Prokopec GM, Scott TW, Perkins TA. Contributions from the silent majority dominate dengue virus transmission. Ferguson NM, editor. PLOS Pathog [Internet]. 2018 May 3;14(5):e1006965. Available from: https://dx.plos.org/10.1371/journal.ppat.1006965
5. Kalayanarooj S. Standardized Clinical Management: Evidence of Reduction of Dengue Haemorrhagic Fever Case-Fatality Rate in Thailand [Internet]. 1999. Available from: https://apps.who.int/iris/handle/10665/148661
6. Cattarino L, Rodriguez-Barraquer I, Imai N, Cummings DAT, Ferguson NM. Mapping global variation in dengue transmission intensity. Sci Transl Med [Internet]. 2020 Jan 29;12(528):eaax4144. Available from: https://stm.sciencemag.org/lookup/doi/10.1126/scitranslmed.aax4144
7. Holmes EC. Evolutionary History and Phylogeography of Human Viruses. Annu Rev Microbiol [Internet]. 2008 Oct;62(1):307–28. Available from: http://www.annualreviews.org/doi/10.1146/annurev.micro.62.081307.162912
8. Rabaa MA. The Molecular Evolution and Phylogeography of Dengue Viruses in Asia [Internet]. The Pennsylvania State University; 2012. Available from: https://etda.libraries.psu.edu/files/final_submissions/8205
9. Cafferata ML, Bardach A, Rey-Ares L, Alcaraz A, Cormick G, Gibbons L, Romano M, Cesaroni S, Ruvinsky S. Dengue Epidemiology and Burden of Disease in Latin America and the Caribbean: A Systematic Review of the Literature and Meta-Analysis. Value Heal Reg Issues [Internet]. 2013 Dec;2(3):347–56. Available from: https://linkinghub.elsevier.com/retrieve/pii/S2212109913001295
10. Lazzarini L, Barzon L, Foglia F, Manfrin V, Pacenti M, Pavan G, Rassu M, Capelli G, Montarsi F, Martini S, Zanella F, Padovan MT, Russo F, Gobbi F. First autochthonous dengue outbreak in Italy, August 2020. Eurosurveillance [Internet]. 2020 Sep 10;25(36). Available from: https://www.eurosurveillance.org/content/10.2807/1560-7917.ES.2020.25.36.2001606
11. News Desk. France reports 1st local transmission of dengue in Var department [Internet]. Outbreak News Today. 2021 [cited 2021 Sep 30]. Available from: http://outbreaknewstoday.com/france-reports-1st-local-transmission-of-dengue-in-var-department-70036/
12. Bouri N, Sell TK, Franco C, Adalja AA, Henderson DA, Hynes NA. Return of Epidemic Dengue in the United States: Implications for the Public Health Practitioner. Public Health Rep [Internet]. 2012 May 1;127(3):259–66. Available from: http://journals.sagepub.com/doi/10.1177/003335491212700305
13. Vazquez-Prokopec GM, Kitron U, Montgomery B, Horne P, Ritchie SA. Quantifying the Spatial Dimension of Dengue Virus Epidemic Spread within a Tropical Urban Environment. Gubler DJ, editor. PLoS Negl Trop Dis [Internet]. 2010 Dec 21;4(12):e920. Available from: https://dx.plos.org/10.1371/journal.pntd.0000920
14. Sahak MN. Dengue fever as an emerging disease in Afghanistan: Epidemiology of the first reported cases. Int J Infect Dis [Internet]. 2020 Oct;99:23–7. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1201971220305774
15. Brady OJ, Gething PW, Bhatt S, Messina JP, Brownstein JS, Hoen AG, Moyes CL, Farlow AW, Scott TW, Hay SI. Refining the Global Spatial Limits of Dengue Virus Transmission by Evidence-Based Consensus. Reithinger R, editor. PLoS Negl Trop Dis [Internet]. 2012 Aug 7;6(8):e1760. Available from: https://dx.plos.org/10.1371/journal.pntd.0001760
16. Guzman MG, Harris E. Dengue. Lancet [Internet]. 2015 Jan;385(9966):453–65. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0140673614605729
17. Nishiura H. Duration of short-lived cross-protective immunity against a clinical attack of dengue: A preliminary estimate. Available from: https://apps.who.int/iris/handle/10665/170468
18. Reich NG, Shrestha S, King AA, Rohani P, Lessler J, Kalayanarooj S, Yoon I-K, Gibbons R V., Burke DS, Cummings DAT. Interactions between serotypes of dengue highlight epidemiological impact of cross-immunity. J R Soc Interface [Internet]. 2013 Sep 6;10(86):20130414. Available from: https://royalsocietypublishing.org/doi/10.1098/rsif.2013.0414
19. Forshey BM, Reiner RC, Olkowski S, Morrison AC, Espinoza A, Long KC, Vilcarromero S, Casanova W, Wearing HJ, Halsey ES, Kochel TJ, Scott TW, Stoddard ST. Incomplete Protection against Dengue Virus Type 2 Re-infection in Peru. Messer WB, editor. PLoS Negl Trop Dis [Internet]. 2016 Feb 5;10(2):e0004398. Available from: https://dx.plos.org/10.1371/journal.pntd.0004398
20. Waggoner JJ, Balmaseda A, Gresh L, Sahoo MK, Montoya M, Wang C, Abeynayake J, Kuan G, Pinsky BA, Harris E.

Homotypic Dengue Virus Reinfections in Nicaraguan Children. J Infect Dis [Internet]. 2016 Oct 1;214(7):986–93. Available from: https://academic.oup.com/jid/article-lookup/doi/10.1093/infdis/jiw099
21. Halstead SB, Nimmannitya S, Cohen SN. Observations related to pathogenesis of dengue hemorrhagic fever. IV. Relation of disease severity to antibody response and virus recovered. Yale J Biol Med [Internet]. 1970 Apr;42(5):311–28. Available from: https://pubmed.ncbi.nlm.nih.gov/5419206
22. SANGKAWIBHA N, ROJANASUPHOT S, AHANDRIK S, VIRIYAPONGSE S, JATANASEN S, SALITUL V, PHANTHUMACHINDA B, HALSTEAD SB. RISK FACTORS IN DENGUE SHOCK SYNDROME: A PROSPECTIVE EPIDEMIOLOGIC STUDY IN RAYONG, THAILAND. Am J Epidemiol [Internet]. 1984 Nov;120(5):653–69. Available from: https://academic.oup.com/aje/article-lookup/doi/10.1093/oxfordjournals.aje.a113932
23. Guzmán MG, Kourí G, Valdés L, Bravo J, Vázquez S, Halstead SB. Enhanced severity of secondary dengue-2 infections: death rates in 1981 and 1997 Cuban outbreaks. Rev Panam Salud Pública [Internet]. 2002 Apr;11(4):223–7. Available from: http://www.scielosp.org/scielo.php?script=sci_arttext&pid=S1020-49892002000400003&lng=en&nrm=iso&tlng=en
24. Halstead S, O'Rourke E. Dengue viruses and mononuclear phagocytes. I. Infection enhancement by non-neutralizing antibody. J Exp Med [Internet]. 1977 Jul 1;146(1):201–17. Available from: https://rupress.org/jem/article/146/1/201/48539/Dengue-viruses-and-mononuclear-phagocytes-I
25. Dejnirattisai W, Jumnainsong A, Onsirisakul N, Fitton P, Vasanawathana S, Limpitikul W, Puttikhunt C, Edwards C, Duangchinda T, Supasa S, Chawansuntati K, Malasit P, Mongkolsapaya J, Screaton G. Cross-Reacting Antibodies Enhance Dengue Virus Infection in Humans. Science (80-) [Internet]. 2010 May 7;328(5979):745–8. Available from: https://www.sciencemag.org/lookup/doi/10.1126/science.1185181
26. Jain A, Chaturvedi UC. Dengue in infants: an overview. FEMS Immunol Med Microbiol [Internet]. 2010 Jul;59(2):119–30. Available from: https://academic.oup.com/femspd/article-lookup/doi/10.1111/j.1574-695X.2010.00670.x
27. Mathew A, Rothman AL. Understanding the contribution of cellular immunity to dengue disease pathogenesis. Immunol Rev [Internet]. 2008 Oct;225(1):300–13. Available from: https://onlinelibrary.wiley.com/doi/10.1111/j.1600-065X.2008.00678.x
28. Aguas R, Dorigatti I, Coudeville L, Luxemburger C, Ferguson NM. Cross-serotype interactions and disease outcome prediction of dengue infections in Vietnam. Sci Rep [Internet]. 2019 Dec 28;9(1):9395. Available from: http://www.nature.com/articles/s41598-019-45816-6
29. Bell SM, Katzelnick L, Bedford T. Dengue genetic divergence generates within-serotype antigenic variation, but serotypes dominate evolutionary dynamics. Elife [Internet]. 2019 Aug 6;8. Available from: https://elifesciences.org/articles/42496
30. Olkowski S, Forshey BM, Morrison AC, Rocha C, Vilcarromero S, Halsey ES, Kochel TJ, Scott TW, Stoddard ST. Reduced Risk of Disease During Postsecondary Dengue Virus Infections. J Infect Dis [Internet]. 2013 Sep 15;208(6):1026–33. Available from: https://academic.oup.com/jid/article-lookup/doi/10.1093/infdis/jit273
31. Lourenço J, Tennant W, Faria NR, Walker A, Gupta S, Recker M. Challenges in dengue research: A computational perspective. Evol Appl [Internet]. 2018 Apr 1;11(4):516–33. Available from: https://doi.org/10.1111/eva.12554
32. Wearing HJ, Rohani P. Ecological and immunological determinants of dengue epidemics. Proc Natl Acad Sci [Internet]. 2006 Aug 1;103(31):11802–7. Available from: http://www.pnas.org/cgi/doi/10.1073/pnas.0602960103
33. Cummings DAT, Schwartz IB, Billings L, Shaw LB, Burke DS. Dynamic effects of antibody-dependent enhancement on the fitness of viruses. Proc Natl Acad Sci [Internet]. 2005 Oct 18;102(42):15259–64. Available from: http://www.pnas.org/cgi/doi/10.1073/pnas.0507320102
34. Schwartz I, Shaw L, Cummings D, Billings L, McCrary M, Burke D. Chaotic desynchronization of multistrain diseases. Phys Rev E [Internet]. 2005 Dec 1;72(6):066201. Available from: https://link.aps.org/doi/10.1103/PhysRevE.72.066201
35. Recker M, Blyuss KB, Simmons CP, Hien TT, Wills B, Farrar J, Gupta S. Immunological serotype interactions and their effect on the epidemiological pattern of dengue. Proc R Soc B Biol Sci [Internet]. 2009 Jul 22;276(1667):2541–8. Available from: https://royalsocietypublishing.org/doi/10.1098/rspb.2009.0331
36. Lourenço J, Recker M. Natural, Persistent Oscillations in a Spatial Multi-Strain Disease System with Application to Dengue. Pascual M, editor. PLoS Comput Biol [Internet]. 2013 Oct 24;9(10):e1003308. Available from: https://dx.plos.org/10.1371/journal.pcbi.1003308
37. Lourenço J, Recker M. Dengue serotype immune-interactions and their consequences for vaccine impact predictions. Epidemics [Internet]. 2016 Sep;16:40–8. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1755436516300226
38. DICK GWA, KITCHEN SF, HADDOW AJ. Zika virus. I. Isolations and serological specificity. Trans R Soc Trop Med Hyg [Internet]. 1952 Sep;46(5):509–20. Available from: http://www.ncbi.nlm.nih.gov/pubmed/12995440
39. SMITHBURN KC. Neutralizing antibodies against certain recently isolated viruses in the sera of human beings residing in East Africa. J Immunol [Internet]. 1952 Aug;69(2):223–34. Available from: http://www.ncbi.nlm.nih.gov/pubmed/14946416
40. Kindhauser MK, Allen T, Frank V, Santhana R, Dye C. Zika: the origin and spread of a mosquito-borne virus. Bull World Heal Organ. 2016;94(9):675–686C.

41. Rasmussen SA, Jamieson DJ, Honein MA, Petersen LR. Zika Virus and Birth Defects — Reviewing the Evidence for Causality. N Engl J Med [Internet]. 2016 May 19;374(20):1981–7. Available from: http://www.nejm.org/doi/10.1056/NEJMsr1604338
42. Foy BD, Kobylinski KC, Foy JLC, Blitvich BJ, da Rosa AT, Haddow AD, Lanciotti RS, Tesh RB. Probable Non-Vector-borne Transmission of Zika Virus, Colorado, USA. Emerg Infect Dis. 2011;17(5):880–2.
43. Musso D, Roche C, Robin E, Nhan T, Teissier A, Cao-Lormeau V-M. Potential Sexual Transmission of Zika Virus. Emerg Infect Dis [Internet]. 2015 Feb;21(2):359–61. Available from: http://wwwnc.cdc.gov/eid/article/21/2/14-1363_article.htm
44. Duffy MR, Chen T-H, Hancock WT, Powers AM, Kool JL, Lanciotti RS, Pretrick M, Marfel M, Holzbauer S, Dubray C, Guillaumot L, Griggs A, Bel M, Lambert AJ, Laven J, Kosoy O, Panella A, Biggerstaff BJ, Fischer M, Hayes EB. Zika virus outbreak on Yap Island, Federated States of Micronesia. N Engl J Med [Internet]. 2009 Jun 11;360(24):2536–43. Available from: http://www.ncbi.nlm.nih.gov/pubmed/19516034
45. Musso D, Bossin H, Mallet HP, Besnard M, Broult J, Baudouin L, Levi JE, Sabino EC, Ghawche F, Lanteri MC, Baud D. Zika virus in French Polynesia 2013-14: anatomy of a completed outbreak. Lancet Infect Dis [Internet]. 2018;18(5):e172–82. Available from: http://www.ncbi.nlm.nih.gov/pubmed/29150310
46. Pan American Health Organization. Cases of Zika Virus Disease, by Country or Territory [Internet]. Available from: http://www.paho.org/data/index.php/en/mnu-topics/zika/524-zika-weekly-en.html
47. Pettersson JH-O, Bohlin J, Dupont-Rouzeyrol M, Brynildsrud OB, Alfsnes K, Cao-Lormeau V-M, Gaunt MW, Falconar AK, de Lamballerie X, Eldholm V, Musso D, Gould EA. Re-visiting the evolution, dispersal and epidemiology of Zika virus in Asia. Emerg Microbes Infect [Internet]. 2018 Dec 1;7(1):1–8. Available from: https://www.tandfonline.com/doi/full/10.1038/s41426-018-0082-5
48. de Araújo TVB, Ximenes RA de A, Miranda-Filho D de B, Souza WV, Montarroyos UR, de Melo APL, Valongueiro S, de Albuquerque M de FPM, Braga C, Filho SPB, Cordeiro MT, Vazquez E, Cruz D di CS, Henriques CMP, Bezerra LCA, Castanha PM da S, Dhalia R, Marques-Júnior ETA, Martelli CMT, Rodrigues LC, Dhalia C, Santos M, Cortes F, Kleber de Oliveira W, Evelim Coelho G, Cortez-Escalante JJ, Campelo de Albuquerque de Melo CF, Ramon-Pardo P, Aldighieri S, Mendez-Rico J, Espinal M, Torres L, Nassri Hazin A, Van der Linden A, Coentro M, Santiago Dimech G, Siqueira de Assuncao R, Ismael de Carvalho P, Felix Oliveira V. Association between microcephaly, Zika virus infection, and other risk factors in Brazil: final report of a case-control study. Lancet Infect Dis [Internet]. 2018 Mar;18(3):328–36. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1473309917307272
49. Lopes Moreira ME, Nielsen-Saines K, Brasil P, Kerin T, Damasceno L, Pone M, Carvalho LMA, Pone SM, Vasconcelos Z, Ribeiro IP, Zin AA, Tsui I, Adachi K, Gaw SL, Halai U-A, Salles TS, da Cunha DC, Bonaldo MC, Raja Gabaglia C, Guida L, Malacarne J, Costa RP, Gomes SC, Reis AB, Soares FVM, Hasue RH, Aizawa CYP, Genovesi FF, Aibe M, Einspieler C, Marschik PB, Pereira JP, Portari EA, Janzen C, Cherry JD. Neurodevelopment in Infants Exposed to Zika Virus In Utero. N Engl J Med [Internet]. 2018 Dec 13;379(24):2377–9. Available from: http://www.nejm.org/doi/10.1056/NEJMc1800098
50. Burger-Calderon R, Gonzalez K, Ojeda S, Zambrana JV, Sanchez N, Cerpas Cruz C, Suazo Laguna H, Bustos F, Plazaola M, Lopez Mercado B, Elizondo D, Arguello S, Carey Monterrey J, Nuñez A, Coloma J, Waggoner JJ, Gordon A, Kuan G, Balmaseda A, Harris E. Zika virus infection in Nicaraguan households. Rimoin A, editor. PLoS Negl Trop Dis [Internet]. 2018 May 31;12(5):e0006518. Available from: http://dx.plos.org/10.1371/journal.pntd.0006518
51. Borchering RK, Huang AT, Mier-y-Teran-Romero L, Rojas DP, Rodriguez-Barraquer I, Katzelnick LC, Martinez SD, King GD, Cinkovich SC, Lessler J, Cummings DAT. Impacts of Zika emergence in Latin America on endemic dengue transmission. Nat Commun [Internet]. 2019 Dec 16;10(1):5730. Available from: http://www.nature.com/articles/s41467-019-13628-x
52. Katzelnick LC, Narvaez C, Arguello S, Lopez Mercado B, Collado D, Ampie O, Elizondo D, Miranda T, Bustos Carillo F, Mercado JC, Latta K, Schiller A, Segovia-Chumbez B, Ojeda S, Sanchez N, Plazaola M, Coloma J, Halloran ME, Premkumar L, Gordon A, Narvaez F, de Silva AM, Kuan G, Balmaseda A, Harris E. Zika virus infection enhances future risk of severe dengue disease. Science (80-) [Internet]. 2020 Aug 28;369(6507):1123 LP – 1128. Available from: http://science.sciencemag.org/content/369/6507/1123.abstract
53. Langerak T, Mumtaz N, Tolk VI, van Gorp ECM, Martina BE, Rockx B, Koopmans MPG. The possible role of cross-reactive dengue virus antibodies in Zika virus pathogenesis. Hobman TC, editor. PLOS Pathog [Internet]. 2019 Apr 18;15(4):e1007640. Available from: https://dx.plos.org/10.1371/journal.ppat.1007640
54. Langerak T, Kasbergen LMR, Chandler F, Brinkman T, Faerber Z, Phalai K, Ulbert S, Rockstroh A, Bruin E de, Koopmans MPG, Rockx B, van Gorp ECM, Vreden S. Zika Virus Antibody Titers Three Years after Confirmed Infection. Viruses [Internet]. 2021 Jul 12;13(7):1345. Available from: https://www.mdpi.com/1999-4915/13/7/1345
55. Henderson AD, Aubry M, Kama M, Vanhomwegen J, Teissier A, Mariteragi-Helle T, Paoaafaite T, Teissier Y, Manuguerra J-C, Edmunds J, Whitworth J, Watson CH, Lau CL, Cao-Lormeau V-M, Kucharski AJ. Zika seroprevalence declines and neutralizing antibodies wane in adults following outbreaks in French Polynesia and Fiji. Elife [Internet]. 2020 Jan 28;9. Available from: https://elifesciences.org/articles/48460
56. Moreira-Soto A, de Souza Sampaio G, Pedroso C, Postigo-Hidalgo I, Berneck BS, Ulbert S, Brites C, Netto EM, Drexler JF. Rapid decline of Zika virus NS1 antigen-specific antibody responses, northeastern Brazil. Virus Genes [Internet]. 2020 Oct

15;56(5):632–7. Available from: https://link.springer.com/10.1007/s11262-020-01772-2
57. World Health Organization. Yellow fever [Internet]. 2019 [cited 2021 Oct 11]. Available from: https://www.who.int/news-room/fact-sheets/detail/yellow-fever
58. Suhrbier A, Jaffar-Bandjee M-C, Gasque P. Arthritogenic alphaviruses—an overview. Nat Rev Rheumatol [Internet]. 2012 Jul 8;8(7):420–9. Available from: http://www.nature.com/articles/nrrheum.2012.64
59. Paniz-Mondolfi AE, Rodriguez-Morales AJ, Blohm G, Marquez M, Villamil-Gomez WE. ChikDenMaZika Syndrome: the challenge of diagnosing arboviral infections in the midst of concurrent epidemics. Ann Clin Microbiol Antimicrob [Internet]. 2016 Dec 22;15(1):42. Available from: http://ann-clinmicrob.biomedcentral.com/articles/10.1186/s12941-016-0157-x
60. Santiago FW, Halsey ES, Siles C, Vilcarromero S, Guevara C, Silvas JA, Ramal C, Ampuero JS, Aguilar P V. Long-Term Arthralgia after Mayaro Virus Infection Correlates with Sustained Pro-inflammatory Cytokine Response. Bingham A, editor. PLoS Negl Trop Dis [Internet]. 2015 Oct 23;9(10):e0004104. Available from: https://dx.plos.org/10.1371/journal.pntd.0004104
61. Lataillade L de G de, Vazeille M, Obadia T, Madec Y, Mousson L, Kamgang B, Chen C-H, Failloux A-B, Yen P-S. Risk of yellow fever virus transmission in the Asia-Pacific region. Nat Commun [Internet]. 2020 Dec 16;11(1):5801. Available from: http://www.nature.com/articles/s41467-020-19625-9
62. Puntasecca CJ, King CH, LaBeaud AD. Measuring the global burden of chikungunya and Zika viruses: A systematic review. Mostafa A, editor. PLoS Negl Trop Dis [Internet]. 2021 Mar 4;15(3):e0009055. Available from: https://dx.plos.org/10.1371/journal.pntd.0009055
63. World Health Organization. Mayaro virus disease - French Guiana [Internet]. 2020 [cited 2021 Oct 11]. Available from: https://www.who.int/emergencies/disease-outbreak-news/item/mayaro-virus-disease---french-guiana-france
64. Pereira TN, Carvalho FD, De Mendonça SF, Rocha MN, Moreira LA. Vector competence of Aedes aegypti, Aedes albopictus, and Culex quinquefasciatus mosquitoes for Mayaro virus. Kittayapong P, editor. PLoS Negl Trop Dis [Internet]. 2020 Apr 14;14(4):e0007518. Available from: https://dx.plos.org/10.1371/journal.pntd.0007518
65. Schoof HF. Mating, resting habits and dispersal of Aedes aegypti. Bull World Health Organ [Internet]. 1967;36(4):600–1. Available from: http://www.ncbi.nlm.nih.gov/pubmed/5299459
66. Ponlawat A, Harrington LC. Blood Feeding Patterns of Aedes aegypti and Aedes albopictus in Thailand. J Med Entomol [Internet]. 2005 Sep 1;42(5):844–9. Available from: https://academic.oup.com/jme/article-lookup/doi/10.1093/jmedent/42.5.844
67. Baik LS, Nave C, Au DD, Guda T, Chevez JA, Ray A, Holmes TC. Circadian Regulation of Light-Evoked Attraction and Avoidance Behaviors in Daytime- versus Nighttime-Biting Mosquitoes. Curr Biol [Internet]. 2020 Aug;30(16):3252-3259.e3. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0960982220308265
68. Brady OJ, Hay SI. The Global Expansion of Dengue: How Aedes aegypti Mosquitoes Enabled the First Pandemic Arbovirus. Annu Rev Entomol [Internet]. 2020 Jan 7;65(1):annurev-ento-011019-024918. Available from: https://www.annualreviews.org/doi/10.1146/annurev-ento-011019-024918
69. Verdonschot PFM, Besse-Lototskaya AA. Flight distance of mosquitoes (Culicidae): A metadata analysis to support the management of barrier zones around rewetted and newly constructed wetlands. Limnologica [Internet]. 2014 Mar;45:69–79. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0075951113001011
70. Ritchie SA. Dengue Vector Bionomics: Why Aedes aegypti is Such a Good Vector. In: Dengue and Dengue Hemorrhagic Fever. 2nd ed. CABI; 2014. p. 455–80.
71. Guedes RNC, Beins K, Navarro Costa D, Coelho GE, Bezerra HS da S. Patterns of insecticide resistance in Aedes aegypti : meta-analyses of surveys in Latin America and the Caribbean. Pest Manag Sci [Internet]. 2020 Jun 7;76(6):2144–57. Available from: https://onlinelibrary.wiley.com/doi/10.1002/ps.5752
72. Gratz NG. Critical review of the vector status of Aedes albopictus. Med Vet Entomol [Internet]. 2004 Sep;18(3):215–27. Available from: https://onlinelibrary.wiley.com/doi/10.1111/j.0269-283X.2004.00513.x
73. Vontas J, Kioulos E, Pavlidi N, Morou E, della Torre A, Ranson H. Insecticide resistance in the major dengue vectors Aedes albopictus and Aedes aegypti. Pestic Biochem Physiol [Internet]. 2012 Oct;104(2):126–31. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0048357512000818
74. Reiter P, Sprenger D. The used tire trade: a mechanism for the worldwide dispersal of container breeding mosquitoes. J Am Mosq Control Assoc [Internet]. 1987 Sep;3(3):494–501. Available from: http://www.ncbi.nlm.nih.gov/pubmed/2904963
75. Morin CW, Comrie AC, Ernst K. Climate and Dengue Transmission: Evidence and Implications. Environ Health Perspect. 2013;121:1264–72.
76. Mordecai EA, Cohen JM, Evans M V., Gudapati P, Johnson LR, Lippi CA, Miazgowicz K, Murdock CC, Rohr JR, Ryan SJ, Savage V, Shocket MS, Stewart Ibarra A, Thomas MB, Weikel DP. Detecting the impact of temperature on transmission of Zika, dengue, and chikungunya using mechanistic models. Althouse B, editor. PLoS Negl Trop Dis [Internet]. 2017 Apr 27;11(4):e0005568. Available from: http://dx.plos.org/10.1371/journal.pntd.0005568
77. Tesla B, Demakovsky LR, Mordecai EA, Ryan SJ, Bonds MH, Ngonghala CN, Brindley MA, Murdock CC. Temperature

drives Zika virus transmission: evidence from empirical and mathematical models. Proc R Soc B Biol Sci [Internet]. 2018 Aug 15;285(1884):20180795. Available from: https://royalsocietypublishing.org/doi/10.1098/rspb.2018.0795
78. Biggerstaff BJ, Arana-Vizcarrondo N, Johansson MA, Staples JE. Incubation Periods of Yellow Fever Virus. Am J Trop Med Hyg [Internet]. 2010 Jul 1;83(1):183–8. Available from: https://ajtmh.org/doi/10.4269/ajtmh.2010.09-0782
79. Lambrechts L, Paaijmans KP, Fansiri T, Carrington LB, Kramer LD, Thomas MB, Scott TW. Impact of daily temperature fluctuations on dengue virus transmission by Aedes aegypti. Proc Natl Acad Sci [Internet]. 2011 May 3;108(18):7460–5. Available from: http://www.pnas.org/cgi/doi/10.1073/pnas.1101377108
80. Rueda LM, Patel KJ, Axtell RC, Stinner RE. Temperature-Dependent Development and Survival Rates of Culex quinquefasciatus and Aedes aegypti (Diptera: Culicidae). J Med Entomol [Internet]. 1990 Sep 1;27(5):892–8. Available from: https://academic.oup.com/jme/article-lookup/doi/10.1093/jmedent/27.5.892
81. Brady OJ, Johansson MA, Guerra CA, Bhatt S, Golding N, Pigott DM, Delatte H, Grech MG, Leisnham PT, Maciel-de-Freitas R, Styer LM, Smith DL, Scott TW, Gething PW, Hay SI. Modelling adult Aedes aegypti and Aedes albopictus survival at different temperatures in laboratory and field settings. Parasit Vectors [Internet]. 2013 Dec 12;6(1):351. Available from: https://parasitesandvectors.biomedcentral.com/articles/10.1186/1756-3305-6-351
82. Brady OJ, Golding N, Pigott DM, Kraemer MUG, Messina JP, Reiner Jr RC, Scott TW, Smith DL, Gething PW, Hay SI. Global temperature constraints on Aedes aegypti and Ae. albopictus persistence and competence for dengue virus transmission. Parasit Vectors [Internet]. 2014;7(1):338. Available from: http://parasitesandvectors.biomedcentral.com/articles/10.1186/1756-3305-7-338
83. Benedum CM, Seidahmed OME, Eltahir EAB, Markuzon N. Statistical modeling of the effect of rainfall flushing on dengue transmission in Singapore. Reiner RC, editor. PLoS Negl Trop Dis [Internet]. 2018 Dec 6;12(12):e0006935. Available from: https://dx.plos.org/10.1371/journal.pntd.0006935
84. Lowe R, Lee SA, O'Reilly KM, Brady OJ, Bastos L, Carrasco-Escobar G, de Castro Catão R, Colón-González FJ, Barcellos C, Carvalho MS, Blangiardo M, Rue H, Gasparrini A. Combined effects of hydrometeorological hazards and urbanisation on dengue risk in Brazil: a spatiotemporal modelling study. Lancet Planet Heal [Internet]. 2021 Apr;5(4):e209–19. Available from: https://linkinghub.elsevier.com/retrieve/pii/S2542519620302928
85. Lowe R, Gasparrini A, Van Meerbeeck CJ, Lippi CA, Mahon R, Trotman AR, Rollock L, Hinds AQJ, Ryan SJ, Stewart-Ibarra AM. Nonlinear and delayed impacts of climate on dengue risk in Barbados: A modelling study. Thomson M, editor. PLOS Med [Internet]. 2018 Jul 17;15(7):e1002613. Available from: https://dx.plos.org/10.1371/journal.pmed.1002613
86. Christophers SSR. AEDES AEGYPTI (L.) THE YELLOW FEVER MOSQUITO. ITS LIFE HISTORY, BIONOMICS AND STRUCTURE. CAMBRIDGE UNIVERSITY PRESS; 1960.
87. Lowe R, Cazelles B, Paul R, Rodó X. Quantifying the added value of climate information in a spatio-temporal dengue model. Stoch Environ Res Risk Assess [Internet]. 2016 Dec 20;30(8):2067–78. Available from: http://link.springer.com/10.1007/s00477-015-1053-1
88. Colón-González FJ, Soares Bastos L, Hofmann B, Hopkin A, Harpham Q, Crocker T, Amato R, Ferrario I, Moschini F, James S, Malde S, Ainscoe E, Sinh Nam V, Quang Tan D, Duc Khoa N, Harrison M, Tsarouchi G, Lumbroso D, Brady OJ, Lowe R. Probabilistic seasonal dengue forecasting in Vietnam: A modelling study using superensembles. Cook AR, editor. PLOS Med [Internet]. 2021 Mar 4;18(3):e1003542. Available from: https://dx.plos.org/10.1371/journal.pmed.1003542
89. Lauer SA, Sakrejda K, Ray EL, Keegan LT, Bi Q, Suangtho P, Hinjoy S, Iamsirithaworn S, Suthachana S, Laosiritaworn Y, Cummings DAT, Lessler J, Reich NG. Prospective forecasts of annual dengue hemorrhagic fever incidence in Thailand, 2010–2014. Proc Natl Acad Sci [Internet]. 2018 Mar 6;115(10):E2175–82. Available from: http://www.pnas.org/lookup/doi/10.1073/pnas.1714457115
90. Johansson MA, Apfeldorf KM, Dobson S, Devita J, Buczak AL, Baugher B, Moniz LJ, Bagley T, Babin SM, Guven E, Yamana TK, Shaman J, Moschou T, Lothian N, Lane A, Osborne G, Jiang G, Brooks LC, Farrow DC, Hyun S, Tibshirani RJ, Rosenfeld R, Lessler J, Reich NG, Cummings DAT, Lauer SA, Moore SM, Clapham HE, Lowe R, Bailey TC, García-Díez M, Carvalho MS, Rodó X, Sardar T, Paul R, Ray EL, Sakrejda K, Brown AC, Meng X, Osoba O, Vardavas R, Manheim D, Moore M, Rao DM, Porco TC, Ackley S, Liu F, Worden L, Convertino M, Liu Y, Reddy A, Ortiz E, Rivero J, Brito H, Juarrero A, Johnson LR, Gramacy RB, Cohen JM, Mordecai EA, Murdock CC, Rohr JR, Ryan SJ, Stewart-Ibarra AM, Weikel DP, Jutla A, Khan R, Poultney M, Colwell RR, Rivera-García B, Barker CM, Bell JE, Biggerstaff M, Swerdlow D, Mier-y-Teran-Romero L, Forshey BM, Trtanj J, Asher J, Clay M, Margolis HS, Hebbeler AM, George D, Chretien J-P. An open challenge to advance probabilistic forecasting for dengue epidemics. Proc Natl Acad Sci [Internet]. 2019 Nov 26;116(48):24268–74. Available from: http://www.pnas.org/lookup/doi/10.1073/pnas.1909865116
91. Magori K, Legros M, Puente ME, Focks DA, Scott TW, Lloyd AL, Gould F. Skeeter Buster: A Stochastic, Spatially Explicit Modeling Tool for Studying Aedes aegypti Population Replacement and Population Suppression Strategies. Kittayapong P, editor. PLoS Negl Trop Dis [Internet]. 2009 Sep 1;3(9):e508. Available from: https://dx.plos.org/10.1371/journal.pntd.0000508
92. Otero M, Schweigmann N, Solari HG. A Stochastic Spatial Dynamical Model for Aedes Aegypti. Bull Math Biol [Internet]. 2008 Jul 22;70(5):1297–325. Available from: http://link.springer.com/10.1007/s11538-008-9300-y
93. Legros M, Otero M, Romeo Aznar V, Solari H, Gould F, Lloyd AL. Comparison of two detailed models of Aedes aegypti

population dynamics. Ecosphere [Internet]. 2016 Oct 19;7(10). Available from: https://onlinelibrary.wiley.com/doi/10.1002/ecs2.1515
94. Kraemer MUG, Sinka ME, Duda KA, Mylne A, Shearer FM, Brady OJ, Messina JP, Barker CM, Moore CG, Carvalho RG, Coelho GE, Van Bortel W, Hendrickx G, Schaffner F, Wint GRW, Elyazar IRF, Teng HJ, Hay SI. The global compendium of Aedes aegypti and Ae. albopictus occurrence. Sci Data. 2015;2:1–8.
95. Kraemer MUG, Sinka ME, Duda KA, Mylne AQN, Shearer FM, Barker CM, Moore CG, Carvalho RG, Coelho GE, Van Bortel W, Hendrickx G, Schaffner F, Elyazar IR, Teng HJ, Brady OJ, Messina JP, Pigott DM, Scott TW, Smith DL, William Wint GR, Golding N, Hay SI. The global distribution of the arbovirus vectors Aedes aegypti and Ae. Albopictus. Elife. 2015;4:1–18.
96. Messina JP, Kraemer MUG, Brady OJ, Pigott DM, Shearer FM, Weiss DJ, Golding N, Ruktanonchai CW, Gething PW, Cohn E, Brownstein JS, Khan K, Tatem AJ, Jaenisch T, Murray CJL, Marinho F, Scott TW, Hay SI. Mapping global environmental suitability for Zika virus. Elife. 2016;2007:1–19.
97. Kraemer MUG, Reiner RC, Brady OJ, Messina JP, Gilbert M, Pigott DM, Yi D, Johnson K, Earl L, Marczak LB, Shirude S, Davis Weaver N, Bisanzio D, Perkins TA, Lai S, Lu X, Jones P, Coelho GE, Carvalho RG, Van Bortel W, Marsboom C, Hendrickx G, Schaffner F, Moore CG, Nax HH, Bengtsson L, Wetter E, Tatem AJ, Brownstein JS, Smith DL, Lambrechts L, Cauchemez S, Linard C, Faria NR, Pybus OG, Scott TW, Liu Q, Yu H, Wint GRW, Hay SI, Golding N. Past and future spread of the arbovirus vectors Aedes aegypti and Aedes albopictus. Nat Microbiol [Internet]. 2019 May 4;4(5):854–63. Available from: http://www.nature.com/articles/s41564-019-0376-y
98. Messina JP, Brady OJ, Golding N, Kraemer MUG, Wint GRW, Ray SE, Pigott DM, Shearer FM, Johnson K, Earl L, Marczak LB, Shirude S, Davis Weaver N, Gilbert M, Velayudhan R, Jones P, Jaenisch T, Scott TW, Reiner RC, Hay SI. The current and future global distribution and population at risk of dengue. Nat Microbiol [Internet]. 2019 Sep 10;4(9):1508–15. Available from: http://www.nature.com/articles/s41564-019-0476-8
99. Reiner RC, Stoddard ST, Vazquez-Prokopec GM, Astete H, Perkins TA, Sihuincha M, Stancil JD, Smith DL, Kochel TJ, Halsey ES, Kitron U, Morrison AC, Scott TW. Estimating the impact of city-wide Aedes aegypti population control: An observational study in Iquitos, Peru. Althouse B, editor. PLoS Negl Trop Dis [Internet]. 2019 May 30;13(5):e0007255. Available from: https://dx.plos.org/10.1371/journal.pntd.0007255
100. Achee NL, Gould F, Perkins TA, Reiner RC, Morrison AC, Ritchie SA, Gubler DJ, Teyssou R, Scott TW. A Critical Assessment of Vector Control for Dengue Prevention. Halstead SB, editor. PLoS Negl Trop Dis [Internet]. 2015 May 7;9(5):e0003655. Available from: https://dx.plos.org/10.1371/journal.pntd.0003655
101. Ooi E-E, Goh K-T, Gubler DJ. Dengue Prevention and 35 Years of Vector Control in Singapore. Emerg Infect Dis [Internet]. 2006 Jun;12(6):887–93. Available from: http://wwwnc.cdc.gov/eid/article/12/6/05-1210_article.htm
102. Tan LK, Low SL, Sun H, Shi Y, Liu L, Lam S, Tan HH, Ang LW, Wong WY, Chua R, Teo D, Ng LC, Cook AR. Force of Infection and True Infection Rate of Dengue in Singapore: Implications for Dengue Control and Management. Am J Epidemiol [Internet]. 2019 Aug 1;188(8):1529–38. Available from: https://academic.oup.com/aje/article/188/8/1529/5486392
103. Guzman MG, Kouri G, Vazquez S, Rosario D, Bravo J, Valdes L. DHF Epidemics in Cuba, 1981 and 1997: Some Interesting Observations. Available from: https://apps.who.int/iris/handle/10665/148771
104. Bowman LR, Donegan S, McCall PJ. Is Dengue Vector Control Deficient in Effectiveness or Evidence?: Systematic Review and Meta-analysis. James AA, editor. PLoS Negl Trop Dis [Internet]. 2016 Mar 17;10(3):e0004551. Available from: https://dx.plos.org/10.1371/journal.pntd.0004551
105. Bardach AE, García-Perdomo HA, Alcaraz A, Tapia López E, Gándara RAR, Ruvinsky S, Ciapponi A. Interventions for the control of Aedes aegypti in Latin America and the Caribbean: systematic review and meta-analysis. Trop Med Int Heal [Internet]. 2019 May 11;24(5):530–52. Available from: https://onlinelibrary.wiley.com/doi/10.1111/tmi.13217
106. Andersson N, Nava-Aguilera E, Arosteguí J, Morales-Perez A, Suazo-Laguna H, Legorreta-Soberanis J, Hernandez-Alvarez C, Fernandez-Salas I, Paredes-Solís S, Balmaseda A, Cortés-Guzmán AJ, Serrano de los Santos R, Coloma J, Ledogar RJ, Harris E. Evidence based community mobilization for dengue prevention in Nicaragua and Mexico (Camino Verde, the Green Way): cluster randomized controlled trial. BMJ [Internet]. 2015 Jul 8;h3267. Available from: https://www.bmj.com/lookup/doi/10.1136/bmj.h3267
107. Toledo ME, Vanlerberghe V, Rosales JP, Mirabal M, Cabrera P, Fonseca V, Gómez Padrón T, Pérez Menzies M, Montada D, Van der Stuyft P. The additional benefit of residual spraying and insecticide-treated curtains for dengue control over current best practice in Cuba: Evaluation of disease incidence in a cluster randomized trial in a low burden setting with intensive routine control. Turell MJ, editor. PLoS Negl Trop Dis [Internet]. 2017 Nov 8;11(11):e0006031. Available from: https://dx.plos.org/10.1371/journal.pntd.0006031
108. Ko Y-C, Chen M-J, Yeh S-M. The Predisposing and Protective Factors against Dengue Virus Transmission by Mosquito Vector. Am J Epidemiol [Internet]. 1992 Jul 15;136(2):214–20. Available from: https://academic.oup.com/aje/article/131573/The
109. Wilson AL, Boelaert M, Kleinschmidt I, Pinder M, Scott TW, Tusting LS, Lindsay SW. Evidence-based vector control? Improving the quality of vector control trials. Trends Parasitol [Internet]. 2015 Aug;31(8):380–90. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1471492215000975
110. Hierlihy C, Waddell L, Young I, Greig J, Corrin T, Mascarenhas M. A systematic review of individual and community

mitigation measures for prevention and control of chikungunya virus. Samy AM, editor. PLoS One [Internet]. 2019 Feb 27;14(2):e0212054. Available from: https://dx.plos.org/10.1371/journal.pone.0212054
111. Ferguson NM. Challenges and opportunities in controlling mosquito-borne infections. Nature [Internet]. 2018 Jul 25;559(7715):490–7. Available from: http://www.nature.com/articles/s41586-018-0318-5
112. Utarini A, Indriani C, Ahmad RA, Tantowijoyo W, Arguni E, Ansari MR, Supriyati E, Wardana DS, Meitika Y, Ernesia I, Nurhayati I, Prabowo E, Andari B, Green BR, Hodgson L, Cutcher Z, Rancès E, Ryan PA, O'Neill SL, Dufault SM, Tanamas SK, Jewell NP, Anders KL, Simmons CP. Efficacy of Wolbachia-Infected Mosquito Deployments for the Control of Dengue. N Engl J Med [Internet]. 2021 Jun 10;384(23):2177–86. Available from: http://www.nejm.org/doi/10.1056/NEJMoa2030243
113. World Mosquito Program. Impact [Internet]. 2021 [cited 2021 Sep 15]. Available from: https://worldmosquitoprogram.org/en/work/wolbachia-method/impact
114. Walker T, Johnson PH, Moreira LA, Iturbe-Ormaetxe I, Frentiu FD, McMeniman CJ, Leong YS, Dong Y, Axford J, Kriesner P, Lloyd AL, Ritchie SA, O'Neill SL, Hoffmann AA. The wMel Wolbachia strain blocks dengue and invades caged Aedes aegypti populations. Nature [Internet]. 2011 Aug 24;476(7361):450–3. Available from: http://www.nature.com/articles/nature10355
115. Aliota MT, Peinado SA, Velez ID, Osorio JE. The wMel strain of Wolbachia Reduces Transmission of Zika virus by Aedes aegypti. Sci Rep [Internet]. 2016 Sep 1;6(1):28792. Available from: http://www.nature.com/articles/srep28792
116. Aliota MT, Walker EC, Uribe Yepes A, Dario Velez I, Christensen BM, Osorio JE. The wMel Strain of Wolbachia Reduces Transmission of Chikungunya Virus in Aedes aegypti. Armstrong PM, editor. PLoS Negl Trop Dis [Internet]. 2016 Apr 28;10(4):e0004677. Available from: https://dx.plos.org/10.1371/journal.pntd.0004677
117. Pereira TN, Rocha MN, Sucupira PHF, Carvalho FD, Moreira LA. Wolbachia significantly impacts the vector competence of Aedes aegypti for Mayaro virus. Sci Rep [Internet]. 2018;8(1):6889. Available from: https://doi.org/10.1038/s41598-018-25236-8
118. Rocha MN, Duarte MM, Mansur SB, Silva BDM e, Pereira TN, Adelino TÉR, Giovanetti M, Alcantara LCJ, Santos FM, Costa VR de M, Teixeira MM, Iani FC de M, Costa VV, Moreira LA. Pluripotency of Wolbachia against Arboviruses: the case of yellow fever. Gates Open Res [Internet]. 2019 Apr 16;3:161. Available from: https://gatesopenresearch.org/articles/3-161/v2
119. Reyes JIL, Suzuki Y, Carvajal T, Muñoz MNM, Watanabe K. Intracellular Interactions Between Arboviruses and Wolbachia in Aedes aegypti. Front Cell Infect Microbiol [Internet]. 2021 Jun 23;11. Available from: https://www.frontiersin.org/articles/10.3389/fcimb.2021.690087/full
120. Gabrieli P, Smidler A, Catteruccia F. Engineering the control of mosquito-borne infectious diseases. Genome Biol [Internet]. 2014 Nov 15;15(11):535. Available from: http://genomebiology.biomedcentral.com/articles/10.1186/s13059-014-0535-7
121. Li M, Yang T, Kandul NP, Bui M, Gamez S, Raban R, Bennett J, Sánchez C HM, Lanzaro GC, Schmidt H, Lee Y, Marshall JM, Akbari OS. Development of a confinable gene drive system in the human disease vector Aedes aegypti. Walczak AM, Bellen HJ, editors. Elife [Internet]. 2020;9:e51701. Available from: https://doi.org/10.7554/eLife.51701
122. Buchman A, Gamez S, Li M, Antoshechkin I, Li H-H, Wang H-W, Chen C-H, Klein MJ, Duchemin J-B, Crowe Jr. JE, Paradkar PN, Akbari OS. Broad dengue neutralization in mosquitoes expressing an engineered antibody. PLOS Pathog [Internet]. 2020 Jan 16;16(1):e1008103. Available from: https://doi.org/10.1371/journal.ppat.1008103
123. World Health Organization. Guidance framework for testing genetically modified mosquitoes, second edition [Internet]. 2021. Available from: https://www.who.int/publications/i/item/9789240025233
124. The Project Wolbachia – Singapore Consortium. Wolbachia-mediated sterility suppresses Aedes aegypti populations in the urban tropics. medRxiv [Internet]. 2021 Jan 1; Available from: http://medrxiv.org/content/early/2021/06/17/2021.06.16.21257922.abstract
125. Reiner RC, Achee N, Barrera R, Burkot TR, Chadee DD, Devine GJ, Endy T, Gubler D, Hombach J, Kleinschmidt I, Lenhart A, Lindsay SW, Longini I, Mondy M, Morrison AC, Perkins TA, Vazquez-Prokopec G, Reiter P, Ritchie SA, Smith DL, Strickman D, Scott TW. Quantifying the Epidemiological Impact of Vector Control on Dengue. Harris E, editor. PLoS Negl Trop Dis [Internet]. 2016 May 26;10(5):e0004588. Available from: https://dx.plos.org/10.1371/journal.pntd.0004588
126. Wolbers M, Kleinschmidt I, Simmons CP, Donnelly CA. Considerations in the Design of Clinical Trials to Test Novel Entomological Approaches to Dengue Control. Kittayapong P, editor. PLoS Negl Trop Dis [Internet]. 2012 Nov 29;6(11):e1937. Available from: https://dx.plos.org/10.1371/journal.pntd.0001937
127. Lambrechts L, Ferguson NM, Harris E, Holmes EC, McGraw EA, O'Neill SL, Ooi EE, Ritchie SA, Ryan PA, Scott TW, Simmons CP, Weaver SC. Assessing the epidemiological effect of wolbachia for dengue control. Lancet Infect Dis [Internet]. 2015 Jul;15(7):862–6. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1473309915000912
128. Hladish TJ, Pearson CAB, Patricia Rojas D, Gomez-Dantes H, Halloran ME, Vazquez-Prokopec GM, Longini IM. Forecasting the effectiveness of indoor residual spraying for reducing dengue burden. Vasilakis N, editor. PLoS Negl Trop Dis [Internet]. 2018 Jun 25;12(6):e0006570. Available from: https://dx.plos.org/10.1371/journal.pntd.0006570
129. Hladish TJ, Pearson CAB, Toh K Ben, Rojas DP, Manrique-Saide P, Vazquez-Prokopec GM, Halloran ME, Longini IM.

Designing effective control of dengue with combined interventions. Proc Natl Acad Sci [Internet]. 2020 Feb 11;117(6):3319–25. Available from: http://www.pnas.org/lookup/doi/10.1073/pnas.1903496117
130. Sridhar S, Luedtke A, Langevin E, Zhu M, Bonaparte M, Machabert T, Savarino S, Zambrano B, Moureau A, Khromava A, Moodie Z, Westling T, Mascareñas C, Frago C, Cortés M, Chansinghakul D, Noriega F, Bouckenooghe A, Chen J, Ng S-P, Gilbert PB, Gurunathan S, DiazGranados CA. Effect of Dengue Serostatus on Dengue Vaccine Safety and Efficacy. N Engl J Med [Internet]. 2018 Jul 26;379(4):327–40. Available from: http://www.nejm.org/doi/10.1056/NEJMoa1800820
131. World Health Organization. Vaccines and immunization: Dengue [Internet]. 2018 [cited 2021 Sep 17]. Available from: https://www.who.int/news-room/q-a-detail/dengue-vaccines
132. Lopez AL, Adams C, Ylade M, Jadi R, Daag JV, Molloy CT, Agrupis KA, Kim DR, Silva MW, Yoon I-K, White L, Deen J, de Silva AM. Determining dengue virus serostatus by indirect IgG ELISA compared with focus reduction neutralisation test in children in Cebu, Philippines: a prospective population-based study. Lancet Glob Heal [Internet]. 2021 Jan;9(1):e44–51. Available from: https://linkinghub.elsevier.com/retrieve/pii/S2214109X20303922
133. Ooi EE. Challenges in prevaccination screening for previous dengue infection. Lancet Glob Heal [Internet]. 2021 Jan;9(1):e4–5. Available from: https://linkinghub.elsevier.com/retrieve/pii/S2214109X20305064
134. España G, Yao Y, Anderson KB, Fitzpatrick MC, Smith DL, Morrison AC, Wilder-Smith A, Scott TW, Perkins TA. Model-based assessment of public health impact and cost-effectiveness of dengue vaccination following screening for prior exposure. Morrison A, editor. PLoS Negl Trop Dis [Internet]. 2019 Jul 1;13(7):e0007482. Available from: https://dx.plos.org/10.1371/journal.pntd.0007482
135. Salje H, Alera MT, Chua MN, Hunsawong T, Ellison D, Srikiatkhachorn A, Jarman RG, Gromowski GD, Rodriguez-Barraquer I, Cauchemez S, Cummings DAT, Macareo L, Yoon I-K, Fernandez S, Rothman AL. Evaluation of the extended efficacy of the Dengvaxia vaccine against symptomatic and subclinical dengue infection. Nat Med [Internet]. 2021 Aug 24;27(8):1395–400. Available from: https://www.nature.com/articles/s41591-021-01392-9
136. Perkins TA, Reiner RC, España G, ten Bosch QA, Verma A, Liebman KA, Paz-Soldan VA, Elder JP, Morrison AC, Stoddard ST, Kitron U, Vazquez-Prokopec GM, Scott TW, Smith DL. An agent-based model of dengue virus transmission shows how uncertainty about breakthrough infections influences vaccination impact projections. Alizon S, editor. PLOS Comput Biol [Internet]. 2019 Mar 20;15(3):e1006710. Available from: https://dx.plos.org/10.1371/journal.pcbi.1006710
137. Biswal S, Reynales H, Saez-Llorens X, Lopez P, Borja-Tabora C, Kosalaraksa P, Sirivichayakul C, Watanaveeradej V, Rivera L, Espinoza F, Fernando L, Dietze R, Luz K, Venâncio da Cunha R, Jimeno J, López-Medina E, Borkowski A, Brose M, Rauscher M, LeFevre I, Bizjajeva S, Bravo L, Wallace D. Efficacy of a Tetravalent Dengue Vaccine in Healthy Children and Adolescents. N Engl J Med [Internet]. 2019 Nov 21;381(21):2009–19. Available from: http://www.nejm.org/doi/10.1056/NEJMoa1903869
138. Barrett ADT. Yellow Fever Vaccines. Biologicals [Internet]. 1997 Mar;25(1):17–25. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1045105697900567
139. World Health Organization. Prequalified vaccines [Internet]. [cited 2021 Sep 20]. Available from: https://extranet.who.int/pqweb/vaccines/prequalified-vaccines
140. Pattnaik A, Sahoo BR, Pattnaik AK. Current Status of Zika Virus Vaccines: Successes and Challenges. Vaccines [Internet]. 2020 May 31;8(2):266. Available from: https://www.mdpi.com/2076-393X/8/2/266
141. Gao S, Song S, Zhang L. Recent Progress in Vaccine Development Against Chikungunya Virus. Front Microbiol [Internet]. 2019 Dec 19;10. Available from: https://www.frontiersin.org/article/10.3389/fmicb.2019.02881/full
142. Kuno G, Chang GJ, Tsuchiya KR, Karabatsos N, Cropp CB. Phylogeny of the Genus Flavivirus. J Virol. 1998;72(1):73–83.
143. Dick GWA, Kitchen SF, Haddow AJ. Zika Virus (I). Isolations and Serological Specificity. Trans R Soc Trop Med Hyg. 1952;46(5):509–20.
144. Duffy MR, Chen T-H, Hancock WT, Powers AM, Kool JL, Lanciotti RS, Pretrick M, Marfel M, Holzbauer S, Dubray C, Guillaumot L, Griggs A, Bel M, Lambert AJ, Laven J, Kosoy O, Panella A, Biggerstaff BJ, Fischer M, Hayes EB. Zika Virus Outbreak on Yap Island, Federated States of Micronesia. N Engl J Med [Internet]. 2009 Jun 11;360(24):2536–43. Available from: http://www.nejm.org/doi/abs/10.1056/NEJMoa0805715
145. Ioos S, Mallet H-P, Leparc IG, Gauthier V, Cardoso T, Herida M. Current Zika virus epidemiology and recent epidemics. Med Mal Infect. 2014;44:302–7.
146. World Health Organization. ZIKA EPIDEMIOLOGY UPDATE, JULY 2019 [Internet]. 2019. Available from: https://www.who.int/emergencies/diseases/zika/zika-epidemiology-update-july-2019.pdf?ua=1
147. Giron S, Franke F, Decoppet A, Cadiou B, Travaglini T, Thirion L, Durand G, Jeannin C, L'Ambert G, Grard G, Noël H, Fournet N, Auzet-Caillaud M, Zandotti C, Aboukaïs S, Chaud P, Guedj S, Hamouda L, Naudot X, Ovize A, Lazarus C, de Valk H, Paty M-C, Leparc-Goffart I. Vector-borne transmission of Zika virus in Europe, southern France, August 2019. Eurosurveillance [Internet]. 2019 Nov 7;24(45). Available from: https://www.eurosurveillance.org/content/10.2807/1560-7917.ES.2019.24.45.1900655
148. Brady OJ, Hay SI. The first local cases of Zika virus in Europe. Lancet [Internet]. 2019 Nov;394(10213):1991–2. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0140673619327904
149. Ferguson NM, Cucunubá ZM, Dorigatti I, Gemma L. Nedjati-Gilani, Donnelly CA, Basáñez M-G, Nouvellet P, Lessler J.

Countering the Zika epidemic in Latin America. Science (80-). 2016;353(6297):353–4.
150. Zhang Q, Sun K, Chinazzi M, Pastore A, Dean NE, Patricia D. Spread of Zika virus in the Americas. PNAS. 2017;114:E4334–E4343.
151. Lourenco J, Maia de Lima M, Faria NR, Walker A, Kraemer MUG, Villabona-Arenas CJ, Lambert B, Marques de Cerqueira E, Pybus OG, Alcantara LCJ, Recker M. Epidemiological and ecological determinants of Zika virus transmission in an urban setting. Elife [Internet]. 2017;6:e29820. Available from: http://biorxiv.org/content/early/2017/01/20/101972.abstract
152. Caminade C, Turner J, Metelmann S, Hesson JC, Blagrove MSC, Solomon T, Morse AP, Baylis M. Global risk model for vector-borne transmission of Zika virus reveals the role of El Niño 2015. Proc Natl Acad Sci [Internet]. 2017 Jan 3;114(1):119–24. Available from: http://www.pnas.org/lookup/doi/10.1073/pnas.1614303114
153. O'Reilly KM, Lowe R, Edmunds WJ, Mayaud P, Kucharski A, Eggo RM, Funk S, Bhatia D, Khan K, Kraemer MUG, Wilder-Smith A, Rodrigues LC, Brasil P, Massad E, Jaenisch T, Cauchemez S, Brady OJ, Yakob L. Projecting the end of the Zika virus epidemic in Latin America: a modelling analysis. BMC Med [Internet]. 2018 Dec 3;16(1):180. Available from: https://bmcmedicine.biomedcentral.com/articles/10.1186/s12916-018-1158-8
154. Samy AM, Thomas SM, Abd A, Wahed E, Cohoon KP, Peterson AT. Mapping the global geographic potential of Zika virus spread. Mem Inst Oswaldo Cruz. 2016;111(September):559–60.
155. Bogoch II, Brady OJ, Kraemer MUG, German M, Creatore MI, Brent S, Watts AG, Hay SI, Kulkarni MA, Brownstein JS, Khan K. Potential for Zika virus introduction and transmission in resource-limited countries in Africa and the Asia-Pacific region: a modelling study. Lancet Infect Dis. 2016;16(11):1237–45.
156. Siraj AS, Perkins TA. Assessing the population at risk of Zika virus in Asia – is the emergency really over? BMJ Glob Heal [Internet]. 2017 Aug 25;2(3):e000309. Available from: http://gh.bmj.com/lookup/doi/10.1136/bmjgh-2017-000309
157. Keegan LT, Lessler J, Johansson MA. Quantifying Zika: Advancing the Epidemiology of Zika With Quantitative Models. J Infect Dis [Internet]. 2017 Dec 16;216(suppl_10):S884–90. Available from: https://academic.oup.com/jid/article/216/suppl_10/S884/4753675
158. Grubaugh ND, Saraf S, Gangavarapu K, Watts A, Tan AL, Oidtman RJ, Ladner JT, Oliveira G, Matteson NL, Kraemer MUG, Vogels CBF, Hentoff A, Bhatia D, Stanek D, Scott B, Landis V, Stryker I, Cone MR, Kopp EW, Cannons AC, Heberlein-Larson L, White S, Gillis LD, Ricciardi MJ, Kwal J, Lichtenberger PK, Magnani DM, Watkins DI, Palacios G, Hamer DH, Gardner LM, Perkins TA, Baele G, Khan K, Morrison A, Isern S, Michael SF, Andersen KG. Travel Surveillance and Genomics Uncover a Hidden Zika Outbreak during the Waning Epidemic. Cell [Internet]. 2019 Aug;178(5):1057-1071.e11. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0092867419307834
159. Pan American Health Organization [Internet]. Available from: http://www.paho.org
160. Grubaugh ND, Ladner JT, Kraemer MUG, Dudas G, Tan AL, Gangavarapu K, Wiley MR, White S, Thézé J, Magnani DM, Prieto K, Reyes D, Bingham AM, Paul LM, Robles-Sikisaka R, Oliveira G, Pronty D, Barcellona CM, Metsky HC, Baniecki ML, Barnes KG, Chak B, Freije CA, Gladden-Young A, Gnirke A, Luo C, MacInnis B, Matranga CB, Park DJ, Qu J, Schaffner SF, Tomkins-Tinch C, West KL, Winnicki SM, Wohl S, Yozwiak NL, Quick J, Fauver JR, Khan K, Brent SE, Reiner RC, Lichtenberger PN, Ricciardi MJ, Bailey VK, Watkins DI, Cone MR, Kopp EW, Hogan KN, Cannons AC, Jean R, Monaghan AJ, Garry RF, Loman NJ, Faria NR, Porcelli MC, Vasquez C, Nagle ER, Cummings DAT, Stanek D, Rambaut A, Sanchez-Lockhart M, Sabeti PC, Gillis LD, Michael SF, Bedford T, Pybus OG, Isern S, Palacios G, Andersen KG. Genomic epidemiology reveals multiple introductions of Zika virus into the United States. Nature [Internet]. 2017 Jun 24;546(7658):401–5. Available from: http://www.nature.com/articles/nature22400
161. Ontiveros C. PAHO Zika numbers - Andersen Lab [Internet]. 2017. Available from: https://andersen-lab.com/paho-zika-cases/
162. Lourenço J, Monteiro M, Tomás T, Monteiro Rodrigues J, Pybus O, Rodrigues Faria N. Epidemiology of the Zika Virus Outbreak in the Cabo Verde Islands, West Africa. PLoS Curr [Internet]. 2018; Available from: http://currents.plos.org/outbreaks/article/epidemiology-of-the-zika-virus-outbreak-in-the-cabo-verde-islands-west-africa/
163. Colombian National Institute of Health [Internet]. Available from: http://www.ins.gov.co
164. Siraj AS, Rodriguez-Barraquer I, Barker CM, Tejedor-Garavito N, Harding D, Lorton C, Lukacevic D, Oates G, Espana G, Kraemer MUG, Manore C, Johansson MA, Tatem AJ, Reiner RC, Perkins TA. Spatiotemporal incidence of Zika and associated environmental drivers for the 2015-2016 epidemic in Colombia. Sci Data [Internet]. 2018 Dec 24;5(1):180073. Available from: http://www.nature.com/articles/sdata201873
165. HealthMap [Internet]. Available from: http://www.healthmap.org
166. Reliefweb [Internet]. Available from: https://reliefweb.int/
167. Healy JM, Burgess MC, Chen T-H, Hancock WT, Toews K-AE, Anesi MS, Tulafono RT, Mataia MA, Sili B, Solaita J, Whelen AC, Sciulli R, Gose RB, Uluiviti V, Hennessey M, Utu F, Nua MT, Fischer M. Notes from the Field : Outbreak of Zika Virus Disease — American Samoa, 2016. MMWR Morb Mortal Wkly Rep [Internet]. 2016 Oct 21;65(41):1146–7. Available from: http://www.cdc.gov/mmwr/volumes/65/wr/mm6541a4.htm
168. World Health Organization. Zika virus infection – Papua New Guinea [Internet]. 2016. Available from: https://www.who.int/csr/don/22-april-2016-zika-png/en/

169. Ruchusatsawat K, Wongjaroen P, Posanacharoen A, Rodriguez-Barraquer I, Sangkitporn S, Cummings DAT, Salje H. Long-term circulation of Zika virus in Thailand: an observational study. Lancet Infect Dis [Internet]. 2019 Apr;19(4):439–46. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1473309918307187
170. Rojas DP, Dean NE, Yang Y, Kenah E, Quintero J, Tomasi S, Ramirez EL, Kelly Y, Castro C, Carrasquilla G, Halloran ME, Longini IM. The epidemiology and transmissibility of Zika virus in Girardot and San Andres island, Colombia, September 2015 to January 2016. Eurosurveillance [Internet]. 2016 Jul 14;21(28):30283. Available from: http://www.eurosurveillance.org/ViewArticle.aspx?ArticleId=22529
171. Ho ZJM, Hapuarachchi HC, Barkham T, Chow A, Ng LC, Lee JMV, Leo YS, Prem K, Lim YHG, de Sessions PF, Rabaa MA, Chong CS, Tan CH, Rajarethinam J, Tan J, Anderson DE, Ong X, Cook AR, Chong CY, Hsu LY, Yap G, Lai YL, Chawla T, Pan L, Sim S, Chen I-CM, Thoon KC, Yung CF, Li JH, Ng HLD, Nandar K, Ooi PL, Lin RTP, Aw P, Uehara A, Pratim De P, Soon W, Hibberd ML, Ng HH, Maurer-Stroh S, Sessions OM. Outbreak of Zika virus infection in Singapore: an epidemiological, entomological, virological, and clinical analysis. Lancet Infect Dis [Internet]. 2017 Aug;17(8):813–21. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1473309917302499
172. Centers for Disease Control and Prevention [Internet]. Available from: https://www.cdc.gov/
173. Methodological Notes to the Tourism Statistics Database, 2019 Edition | Notes méthodologiques de la base de données des statistiques du tourisme, édition 2019 | Notas metodológicas de la base de datos de estadísticas de turismo, edición 2019 [Internet]. World Tourism Organization (UNWTO); 2019. Available from: https://www.e-unwto.org/doi/book/10.18111/9789284420476
174. Arino J, Khan K. Using Mathematical Modeling to Integrate Disease Surveillance and Global Air Transportation Data. In: Analyzing and Modeling Spatial and Temporal Dynamics of Infectious Diseases. John Wiley & Sons, Inc.; 2013. p. 1–14.
175. Socioeconomic Data and Applications Center. Gridded Population of the World (GPW), v4 [Internet]. Available from: http://sedac.ciesin.columbia.edu/data/collection/gpw-v4/sets/browse
176. Copernicus Climate Change Service (C3S). ERA5: Fifth generation of ECMWF atmospheric reanalyses of the global climate [Internet]. 2017. Available from: https://cds.climate.copernicus.eu/cdsapp#!/home
177. Dickens BL, Sun H, Jit M, Cook AR, Carrasco LR. Determining environmental and anthropogenic factors which explain the global distribution of Aedes aegypti and Ae. albopictus. BMJ Glob Heal [Internet]. 2018 Sep 3;3(4):e000801. Available from: http://gh.bmj.com/lookup/doi/10.1136/bmjgh-2018-000801
178. Little JDC. A Proof for the Queuing Formula: L = λ W. Oper Res [Internet]. 1961 Jun;9(3):383–7. Available from: http://pubsonline.informs.org/doi/abs/10.1287/opre.9.3.383
179. Armstrong PM, Ehrlich HY, Magalhaes T, Miller MR, Conway PJ, Bransfield A, Misencik MJ, Gloria-Soria A, Warren JL, Andreadis TG, Shepard JJ, Foy BD, Pitzer VE, Brackney DE. Successive blood meals enhance virus dissemination within mosquitoes and increase transmission potential. Nat Microbiol [Internet]. 2020 Feb 9;5(2):239–47. Available from: http://www.nature.com/articles/s41564-019-0619-y
180. Liu-Helmersson J, Stenlund H, Wilder-Smith A, Rocklöv J. Vectorial Capacity of Aedes aegypti: Effects of Temperature and Implications for Global Dengue Epidemic Potential. Moreira LA, editor. PLoS One [Internet]. 2014 Mar 6;9(3):e89783. Available from: https://dx.plos.org/10.1371/journal.pone.0089783
181. Hsieh Y-H. Temporal patterns and geographic heterogeneity of Zika virus (ZIKV) outbreaks in French Polynesia and Central America. PeerJ [Internet]. 2017;5:e3015. Available from: http://www.ncbi.nlm.nih.gov/pubmed/28344900
182. Chowell G, Hincapie-Palacio D, Ospina J, Pell B, Tariq A, Dahal S, Moghadas S, Smirnova A, Simonsen L, Viboud C. Using Phenomenological Models to Characterize Transmissibility and Forecast Patterns and Final Burden of Zika Epidemics. PLoS Curr [Internet]. 2016; Available from: https://currents.plos.org/outbreaks/article/using-phenomenological-models-to-characterize-transmissibility-and-forecast-patterns-and-final-burden-of-zika-epidemics/
183. Perkins TA, Metcalf CJE, Grenfell BT, Tatem AJ. Estimating Drivers of Autochthonous Transmission of Chikungunya Virus in its Invasion of the Americas. PLoS Curr [Internet]. 2015; Available from: http://currents.plos.org/outbreaks/article/estimating-drivers-of-autochthonous-transmission-of-chikungunya-virus-in-its-invasion-of-the-americas/
184. Harris M, Caldwell JM, Mordecai EA. Climate drives spatial variation in Zika epidemics in Latin America. Proc R Soc B Biol Sci [Internet]. 2019 Aug 28;286(1909):20191578. Available from: https://royalsocietypublishing.org/doi/10.1098/rspb.2019.1578
185. Churcher TS, Cohen JM, Novotny J, Ntshalintshali N, Kunene S, Cauchemez S. Measuring the path toward malaria elimination. Science (80-) [Internet]. 2014 Jun 13;344(6189):1230–2. Available from: http://www.sciencemag.org/cgi/doi/10.1126/science.1251449
186. Aubry M, Teissier A, Huart M, Merceron S, Vanhomwegen J, Roche C, Vial A-L, Teururai S, Sicard S, Paulous S, Desprès P, Manuguerra J-C, Mallet H-P, Musso D, Deparis X, Cao-Lormeau V-M. Zika Virus Seroprevalence, French Polynesia, 2014–2015. Emerg Infect Dis [Internet]. 2017 Apr;23(4):669–72. Available from: http://wwwnc.cdc.gov/eid/article/23/4/16-1549_article.htm
187. Kokernot RH, Casaca VMR, Weinbren MP, McIntosh BM. Survey for antibodies against arthropod-borne viruses in the sera of indigenous residents of Angola. Trans R Soc Trop Med Hyg [Internet]. 1965 Sep;59(5):563–70. Available from:

https://academic.oup.com/trstmh/article-lookup/doi/10.1016/0035-9203(65)90159-8
188. Sasmono RT, Dhenni R, Yohan B, Pronyk P, Hadinegoro SR, Soepardi EJ, Ma'roef CN, Satari HI, Menzies H, Hawley WA, Powers AM, Rosenberg R, Myint KSA, Soebandrio A. Zika Virus Seropositivity in 1–4-Year-Old Children, Indonesia, 2014. Emerg Infect Dis [Internet]. 2018 Sep;24(9). Available from: http://wwwnc.cdc.gov/eid/article/24/9/18-0582_article.htm
189. Smithburn KC, Kerr JA, Gatne PB. Neutralizing Antibodies Against Certain Viruses in the Sera of Residents of India. J Immunol [Internet]. 1954 Apr 1;72(4):248 LP – 257. Available from: http://www.jimmunol.org/content/72/4/248.abstract
190. Macnamara FN, Horn DW, Porterfield JS. Yellow fever and other arthropod-borne viruses. Trans R Soc Trop Med Hyg [Internet]. 1959 Mar;53(2):202–12. Available from: https://academic.oup.com/trstmh/article-lookup/doi/10.1016/0035-9203(59)90072-0
191. Herrera BB, Chang CA, Hamel DJ, Mboup S, Ndiaye D, Imade G, Okpokwu J, Agbaji O, Bei AK, Kanki PJ. Continued Transmission of Zika Virus in Humans in West Africa, 1992–2016. J Infect Dis [Internet]. 2017 May 15;215(10):1546–50. Available from: https://academic.oup.com/jid/article/215/10/1546/3200349
192. Gudo ES, Falk KI, Ali S, Muianga AF, Monteiro V, Cliff J. A Historic Report of Zika in Mozambique: Implications for Assessing Current Risk. Carvalho MS, editor. PLoS Negl Trop Dis [Internet]. 2016 Dec 8;10(12):e0005052. Available from: https://dx.plos.org/10.1371/journal.pntd.0005052
193. Nah K, Mizumoto K, Miyamatsu Y, Yasuda Y, Kinoshita R, Nishiura H. Estimating risks of importation and local transmission of Zika virus infection. PeerJ [Internet]. 2016 Apr 5;4:e1904. Available from: https://peerj.com/articles/1904
194. Hill SC, Vasconcelos J, Neto Z, Jandondo D, Zé-Zé L, Aguiar RS, Xavier J, Thézé J, Mirandela M, Micolo Cândido AL, Vaz F, Sebastião C dos S, Wu C-H, Kraemer MUG, Melo A, Schamber-Reis BLF, de Azevedo GS, Tanuri A, Higa LM, Clemente C, da Silva SP, da Silva Candido D, Claro IM, Quibuco D, Domingos C, Pocongo B, Watts AG, Khan K, Alcantara LCJ, Sabino EC, Lackritz E, Pybus OG, Alves M-J, Afonso J, Faria NR. Emergence of the Asian lineage of Zika virus in Angola: an outbreak investigation. Lancet Infect Dis [Internet]. 2019 Oct;19(10):1138–47. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1473309919302932
195. Brock J. Angola reports first two cases of Zika virus [Internet]. Reuters. 2017. Available from: https://www.reuters.com/article/us-health-zika-angola/angola-reports-first-two-cases-of-zika-virus-idUSKBN14V1PQ
196. THE HINDU. Tamil Nadu reports first case of Zika virus [Internet]. 2017. Available from: https://www.thehindu.com/news/national/tamil-nadu/tamil-nadu-reports-first-case-of-zika-virus/article19254690.ece
197. Chien Y-W, Ho T-C, Huang P-W, Ko N-Y, Ko W-C, Perng GC. Low seroprevalence of Zika virus infection among adults in Southern Taiwan. BMC Infect Dis [Internet]. 2019 Dec 23;19(1):884. Available from: https://bmcinfectdis.biomedcentral.com/articles/10.1186/s12879-019-4491-4
198. Outbreak News Today. Dengue outbreak in the Ivory Coast [Internet]. 2019. Available from: http://outbreaknewstoday.com/dengue-outbreak-in-the-ivory-coast-28815/
199. Fofana D, Beugré JMV, Yao-Acapovi GL, Lendzele SS. Risk of Dengue Transmission in Cocody (Abidjan, Ivory Coast). J Parasitol Res [Internet]. 2019 Jan 14;2019:1–7. Available from: https://www.hindawi.com/journals/jpr/2019/4914137/
200. Proesmans S, Katshongo F, Milambu J, Fungula B, Muhindo Mavoko H, Ahuka-Mundeke S, Inocêncio da Luz R, Van Esbroeck M, Ariën KK, Cnops L, De Smet B, Lutumba P, Van geertruyden J-P, Vanlerberghe V. Dengue and chikungunya among outpatients with acute undifferentiated fever in Kinshasa, Democratic Republic of Congo: A cross-sectional study. Blacksell SD, editor. PLoS Negl Trop Dis [Internet]. 2019 Sep 5;13(9):e0007047. Available from: http://dx.plos.org/10.1371/journal.pntd.0007047
201. Cousins S. Three Zika cases are found in India after random tests. BMJ [Internet]. 2017 May 31;j2654. Available from: http://www.bmj.com/lookup/doi/10.1136/bmj.j2654
202. Bhardwaj S, Gokhale MD, Mourya DT. Zika virus: Current concerns in India. Indian J Med Res [Internet]. 2017 Nov;146(5):572–5. Available from: http://www.ncbi.nlm.nih.gov/pubmed/29512599
203. Yadav PD, Malhotra B, Sapkal G, Nyayanit DA, Deshpande G, Gupta N, Padinjaremattathil UT, Sharma H, Sahay RR, Sharma P, Mourya DT. Zika virus outbreak in Rajasthan, India in 2018 was caused by a virus endemic to Asia. Infect Genet Evol [Internet]. 2019 Apr;69:199–202. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1567134819300048
204. Ganeshkumar P, Murhekar M V., Poornima V, Saravanakumar V, Sukumaran K, Anandaselvasankar A, John D, Mehendale SM. Dengue infection in India: A systematic review and meta-analysis. Rodriguez-Barraquer I, editor. PLoS Negl Trop Dis [Internet]. 2018 Jul 16;12(7):e0006618. Available from: https://dx.plos.org/10.1371/journal.pntd.0006618
205. Olson JG, Ksiazek TG, Suhandiman, Triwibowo. Zika virus, a cause of fever in Central Java, Indonesia. Trans R Soc Trop Med Hyg [Internet]. 1981 Jan;75(3):389–93. Available from: https://academic.oup.com/trstmh/article-lookup/doi/10.1016/0035-9203(81)90100-0
206. Olson JG, Ksiazek TG, Gubler DJ, Lubis SI, Simanjuntak G, Lee VH, Nalim S, Juslis K, See R. A survey for arboviral antibodies in sera of humans and animals in Lombok, Republic of Indonesia. Ann Trop Med Parasitol [Internet]. 1983 Apr 15;77(2):131–7. Available from: https://www.tandfonline.com/doi/full/10.1080/00034983.1983.11811687

207. Xiao J-P, He J-F, Deng A-P, Lin H-L, Song T, Peng Z-Q, Wu X-C, Liu T, Li Z-H, Rutherford S, Zeng W-L, Li X, Ma W-J, Zhang Y-H. Characterizing a large outbreak of dengue fever in Guangdong Province, China. Infect Dis Poverty [Internet]. 2016 Dec 3;5(1):44. Available from: http://idpjournal.biomedcentral.com/articles/10.1186/s40249-016-0131-z
208. Tseng S-P, Yen C-H, Chang K, Chen Y-H, Wu D-C, Wang S-F, Chen Y-MA, Wang W-H. Severe Dengue Fever Outbreak in Taiwan. Am J Trop Med Hyg [Internet]. 2016 Jan 6;94(1):193–7. Available from: http://www.ajtmh.org/content/journals/10.4269/ajtmh.15-0422
209. The Government of the Hong Kong SAR. CHP notified of imported case of Zika Virus Infection in Guangdong [Internet]. 2020 [cited 2020 Sep 16]. Available from: https://www.info.gov.hk/gia/general/202001/09/P2020010900783.htm
210. Focus Taiwan. Taiwan reports first imported case of Zika infection for 2020 [Internet]. 2020 [cited 2020 Sep 16]. Available from: https://focustaiwan.tw/society/202002250018
211. Zhou C, Liu J, Qi R, Fang L, Qin X, Han H, Mo R, Yu H, Jiao Y, Lin J, Yu X. Emergence of Zika virus infection in China. Samy AM, editor. PLoS Negl Trop Dis [Internet]. 2020 May 19;14(5):e0008300. Available from: https://dx.plos.org/10.1371/journal.pntd.0008300
212. Alwafi O, McNabb S, Memish Z, Assiri A, Alzahrani S, Asiri S, Alturkstani A, Albar A, Turkistani A. Dengue Fever in Makkah, Kingdom of Saudi Arabia, 2008-2012. Am J Res Commun. 2013 Oct 1;
213. Ahmed QA, Memish ZA. Zika in Singapore: implications for Saudi Arabia. East Mediterr Heal J. 2017;23(4).
214. Alayed MS, Qureshi MA, Ahmed S, Alqahtani AS, Alqahtani AM, Alshaybari K, Alshahrani M, Asaad AM. Seroprevalence of Zika virus among asymptomatic pregnant mothers and their newborns in the Najran region of southwest Saudi Arabia. Ann Saudi Med [Internet]. 2018 Nov;38(6):408–12. Available from: http://www.annsaudimed.net/doi/10.5144/0256-4947.2018.408
215. World Health Organization. Dengue Fever – Uruguay [Internet]. Disease Outbreak News. 2016 [cited 2020 Sep 16]. Available from: https://www.who.int/csr/don/10-march-2016-dengue-uruguay/en/
216. Lew R, Tsai W-Y, Wang W-K. Dengue outbreaks in Hawai'i After WWII – A Review of Public Health Response and Scientific Literature. HAWAI'I J Med PUBLIC Heal. 2018;77(12):315–318.
217. Akter R, Naish S, Gatton M, Bambrick H, Hu W, Tong S. Spatial and temporal analysis of dengue infections in Queensland, Australia: Recent trend and perspectives. Moreira LA, editor. PLoS One [Internet]. 2019 Jul 22;14(7):e0220134. Available from: https://dx.plos.org/10.1371/journal.pone.0220134
218. Faria NR, Quick J, Claro IM, Thézé J, de Jesus JG, Giovanetti M, Kraemer MUG, Hill SC, Black A, da Costa AC, Franco LC, Silva SP, Wu C-H, Raghwani J, Cauchemez S, du Plessis L, Verotti MP, de Oliveira WK, Carmo EH, Coelho GE, Santelli ACFS, Vinhal LC, Henriques CM, Simpson JT, Loose M, Andersen KG, Grubaugh ND, Somasekar S, Chiu CY, Muñoz-Medina JE, Gonzalez-Bonilla CR, Arias CF, Lewis-Ximenez LL, Baylis SA, Chieppe AO, Aguiar SF, Fernandes CA, Lemos PS, Nascimento BLS, Monteiro HAO, Siqueira IC, de Queiroz MG, de Souza TR, Bezerra JF, Lemos MR, Pereira GF, Loudal D, Moura LC, Dhalia R, França RF, Magalhães T, Marques ET, Jaenisch T, Wallau GL, de Lima MC, Nascimento V, de Cerqueira EM, de Lima MM, Mascarenhas DL, Neto JPM, Levin AS, Tozetto-Mendoza TR, Fonseca SN, Mendes-Correa MC, Milagres FP, Segurado A, Holmes EC, Rambaut A, Bedford T, Nunes MRT, Sabino EC, Alcantara LCJ, Loman NJ, Pybus OG. Establishment and cryptic transmission of Zika virus in Brazil and the Americas. Nature [Internet]. 2017 Jun 24;546(7658):406–10. Available from: http://www.nature.com/articles/nature22401
219. Thangamani S, Huang J, Hart CE, Guzman H, Tesh RB. Vertical Transmission of Zika Virus in Aedes aegypti Mosquitoes. Am J Trop Med Hyg [Internet]. 2016 Nov 2;95(5):1169–73. Available from: http://www.ajtmh.org/content/journals/10.4269/ajtmh.16-0448
220. Rodriguez-Barraquer I, Costa F, Nascimento EJM, Nery N, Castanha PMS, Sacramento GA, Cruz J, Carvalho M, De Olivera D, Hagan JE, Adhikarla H, Wunder EA, Coêlho DF, Azar SR, Rossi SL, Vasilakis N, Weaver SC, Ribeiro GS, Balmaseda A, Harris E, Nogueira ML, Reis MG, Marques ETA, Cummings DAT, Ko AI. Impact of preexisting dengue immunity on Zika virus emergence in a dengue endemic region. Science (80-) [Internet]. 2019 Feb 8;363(6427):607–10. Available from: https://www.sciencemag.org/lookup/doi/10.1126/science.aav6618
221. Calvez E, O'Connor O, Pol M, Rousset D, Faye O, Richard V, Tarantola A, Dupont-Rouzeyrol M. Differential transmission of Asian and African Zika virus lineages by Aedes aegypti from New Caledonia. Emerg Microbes Infect [Internet]. 2018 Sep 26;7(1):159. Available from: http://www.ncbi.nlm.nih.gov/pubmed/30254274
222. Balcan D, Colizza V, Goncalves B, Hu H, Ramasco JJ, Vespignani A. Multiscale mobility networks and the spatial spreading of infectious diseases. Proc Natl Acad Sci [Internet]. 2009 Dec 22;106(51):21484–9. Available from: http://www.pnas.org/cgi/doi/10.1073/pnas.0906910106
223. World Health Organization. Questions and Answers on Dengue Vaccines [Internet]. Immunization, Vaccines and Biologicals. 2018 [cited 2020 Jun 30]. Available from: https://www.who.int/immunization/research/development/dengue_q_and_a/en/
224. Fitzpatrick C, Haines A, Bangert M, Farlow A, Hemingway J, Velayudhan R. An economic evaluation of vector control in the age of a dengue vaccine. PLoS Negl Trop Dis. 2017;11(8):1–27.
225. Liew C, Curtis CF, Agency NE, Liew C, Curtis CF. Horizontal and vertical dispersal of dengue vector mosquitoes, Aedes aegypti and Aedes albopictus, in Singapore. Med Vet Entomol [Internet]. 2004;18(4):351–60. Available from:

http://www.ncbi.nlm.nih.gov/pubmed/15642001
226. Vanwambeke SO, Somboon P, Harbach RE, Isenstadt M, Lambin EF, Walton C, Butlin RK. Landscape and Land Cover Factors Influence the Presence of Aedes and Anopheles Larvae. J Med Entomol [Internet]. 2007 Jan 1;44(1):133–44. Available from: https://academic.oup.com/jme/article-lookup/doi/10.1093/jmedent/41.5.133
227. Lee KS, Lo S, Tan SSY, Chua R, Tan LK, Xu H, Ng LC. Dengue virus surveillance in Singapore reveals high viral diversity through multiple introductions and in situ evolution. Infect Genet Evol [Internet]. 2012;12(1):77–85. Available from: http://dx.doi.org/10.1016/j.meegid.2011.10.012
228. Lee C, Vythilingam I, Chong CS, Abdul Razak MA, Tan CH, Liew C, Pok KY, Ng LC. Gravitraps for management of dengue clusters in Singapore. Am J Trop Med Hyg. 2013;88(5):888–92.
229. Sim S, Ng LC, Lindsay SW, Wilson AL. A greener vision for vector control: The example of the Singapore dengue control programme. Kittayapong P, editor. PLoS Negl Trop Dis [Internet]. 2020 Aug 27;14(8):e0008428. Available from: https://dx.plos.org/10.1371/journal.pntd.0008428
230. National Envronmental Agency. NEA Urges Continued Vigilance In Fight Against Dengue [Internet]. 2019 [cited 2019 Oct 14]. Available from: https://www.nea.gov.sg/media/news/news/index/nea-urges-continued-vigilance-in-fight-against-dengue-2019
231. Ong J, Chong C-S, Yap G, Lee C, Abdul Razak MA, Chiang S, Ng L-C. Gravitrap Deployment for Adult Aedes aegypti Surveillance and its Impact on Dengue Cases. Aldridge RL, editor. PLoS Negl Trop Dis [Internet]. 2020 Aug 7;14(8):e0008528. Available from: https://dx.plos.org/10.1371/journal.pntd.0008528
232. Govtech. Data.gov.sg [Internet]. 2020 [cited 2020 Jun 30]. Available from: https://data.gov.sg/
233. Ong J, Liu X, Rajarethinam J, Yap G, Ho D, Ng LC. A novel entomological index, Aedes aegypti Breeding Percentage, reveals the geographical spread of the dengue vector in Singapore and serves as a spatial risk indicator for dengue. Parasit Vectors [Internet]. 2019 Dec 8;12(1):17. Available from: https://parasitesandvectors.biomedcentral.com/articles/10.1186/s13071-018-3281-y
234. Richards DR, Tunçer B. Using image recognition to automate assessment of cultural ecosystem services from social media photographs. Ecosyst Serv [Internet]. 2017 Sep 18; Available from: https://www.sciencedirect.com/science/article/pii/S2212041617301559
235. OpenStreetMap [Internet]. Available from: https://www.openstreetmap.org
236. Housing & Development Board. Resale Flat Prices [Internet]. [cited 2017 Sep 11]. Available from: https://services2.hdb.gov.sg/webapp/BB33RTIS/BB33PReslTrans.jsp
237. Rue H, Martino S, Chopin N. Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations. J R Stat Soc Ser B. 2009;71:319–392.
238. The R-INLA project [Internet]. Available from: www.r-inla.org
239. World Health Organization. Dengue control - The mosquito [Internet]. [cited 2020 Jun 30]. Available from: https://www.who.int/denguecontrol/mosquito/en/
240. Seidahmed OME, Lu D, Chong CS, Ng LC, Eltahir EAB. Patterns of Urban Housing Shape Dengue Distribution in Singapore at Neighborhood and Country Scales. GeoHealth [Internet]. 2018 Jan;2(1):54–67. Available from: http://doi.wiley.com/10.1002/2017GH000080
241. National Environment Agency. Know The Potential Aedes Breeding Sites [Internet]. 2019 [cited 2020 Jun 30]. Available from: https://www.nea.gov.sg/dengue-zika/prevent-aedes-mosquito-breeding/know-the-potential-aedes-breeding-sites
242. Eisen L, Lozano-Fuentes S. Use of Mapping and Spatial and Space-Time Modeling Approaches in Operational Control of Aedes aegypti and Dengue. Tatem AJ, editor. PLoS Negl Trop Dis [Internet]. 2009 Apr 28;3(4):e411. Available from: http://dx.plos.org/10.1371/journal.pntd.0000411
243. Seidahmed OME, Eltahir EAB. A Sequence of Flushing and Drying of Breeding Habitats of Aedes aegypti (L.) Prior to the Low Dengue Season in Singapore. Kittayapong P, editor. PLoS Negl Trop Dis [Internet]. 2016 Jul 26;10(7):e0004842. Available from: http://dx.plos.org/10.1371/journal.pntd.0004842
244. National Environment Agency. NEA Vox [Internet]. 2020 [cited 2020 Jun 30]. Available from: https://www.nea.gov.sg/media/nea-vox/index/why-is-nea-placing-mosquito-traps-outside-my-house-will-this-increase-my-chances-of-being-bitten-by-mosquitoes#.XPTELvi-F3o
245. Ferguson NM, Rodríguez-Barraquer I, Dorigatti I, Mier-y-Teran-Romero L, Laydon DJ, Cummings DAT. Benefits and risks of the Sanofi-Pasteur dengue vaccine: Modeling optimal deployment. Science (80-) [Internet]. 2016 Sep 2;353(6303):1033–6. Available from: https://www.sciencemag.org/lookup/doi/10.1126/science.aaf9590
246. Zheng X, Zhang D, Li Y, Yang C, Wu Y, Liang X, Liang Y, Pan X, Hu L, Sun Q, Wang X, Wei Y, Zhu J, Qian W, Yan Z, Parker AG, Gilles JRL, Bourtzis K, Bouyer J, Tang M, Zheng B, Yu J, Liu J, Zhuang J, Hu Z, Zhang M, Gong J-T, Hong X-Y, Zhang Z, Lin L, Liu Q, Hu Z, Wu Z, Baton LA, Hoffmann AA, Xi Z. Incompatible and sterile insect techniques combined eliminate mosquitoes. Nature [Internet]. 2019 Aug 1;572(7767):56–61. Available from: http://www.nature.com/articles/s41586-019-1407-9
247. Kittayapong P, Ninphanomchai S, Limohpasmanee W, Chansang C, Chansang U, Mongkalangoon P. Combined sterile insect technique and incompatible insect technique: The first proof-of-concept to suppress Aedes aegypti vector

populations in semi-rural settings in Thailand. Wang W-K, editor. PLoS Negl Trop Dis [Internet]. 2019 Oct 28;13(10):e0007771. Available from: https://dx.plos.org/10.1371/journal.pntd.0007771
248. Sippy R, Rivera GE, Sanchez V, Heras F, Morejón B, Beltrán E, Hikida RS, López-Latorre MA, Aguirre A, Stewart-Ibarra AM, Larsen DA, Neira M. Ingested insecticide to control Aedes aegypti: developing a novel dried attractive toxic sugar bait device for intra-domiciliary control. Parasit Vectors [Internet]. 2020 Dec 17;13(1):78. Available from: https://parasitesandvectors.biomedcentral.com/articles/10.1186/s13071-020-3930-9
249. Revay EE, Müller GC, Qualls WA, Kline DL, Naranjo DP, Arheart KL, Kravchenko VD, Yefremova Z, Hausmann A, Beier JC, Schlein Y, Xue R-D. Control of Aedes albopictus with attractive toxic sugar baits (ATSB) and potential impact on non-target organisms in St. Augustine, Florida. Parasitol Res [Internet]. 2014 Jan 12;113(1):73–9. Available from: http://link.springer.com/10.1007/s00436-013-3628-4
250. Carrington LB, Tran BCN, Le NTH, Luong TTH, Nguyen TT, Nguyen PT, Nguyen CVV, Nguyen HTC, Vu TT, Vo LT, Le DT, Vu NT, Nguyen GT, Luu HQ, Dang AD, Hurst TP, O'Neill SL, Tran VT, Kien DTH, Nguyen NM, Wolbers M, Wills B, Simmons CP. Field- and clinically derived estimates of Wolbachia -mediated blocking of dengue virus transmission potential in Aedes aegypti mosquitoes. Proc Natl Acad Sci [Internet]. 2018 Jan 9;115(2):361–6. Available from: http://www.pnas.org/lookup/doi/10.1073/pnas.1715788115
251. Casas-Martínez M, Tamayo-Domínguez R, Bond-Compeán JG, Rojas JC, Weber M, Ulloa-García A. Oogenic development and gonotrophic cycle of Aedes aegypti and Aedes albopictus in laboratory. Salud Publica Mex. 2020 Jun;62(4, jul-ago):372.
252. Hladish TJ, Pearson CAB, Chao DL, Rojas DP, Recchia GL, Gómez-Dantés H, Halloran ME, Pulliam JRC, Longini IM. Projected Impact of Dengue Vaccination in Yucatán, Mexico. Carvalho MS, editor. PLoS Negl Trop Dis [Internet]. 2016 May 26;10(5):e0004661. Available from: https://dx.plos.org/10.1371/journal.pntd.0004661
253. Liebman KA, Stoddard ST, Reiner RC, Perkins TA, Astete H, Sihuincha M, Halsey ES, Kochel TJ, Morrison AC, Scott TW. Determinants of Heterogeneous Blood Feeding Patterns by Aedes aegypti in Iquitos, Peru. Barrera R, editor. PLoS Negl Trop Dis [Internet]. 2014 Feb 13;8(2):e2702. Available from: https://dx.plos.org/10.1371/journal.pntd.0002702
254. United Nations Department of Economic and Social Affairs. United Nations Model Life Tables [Internet]. [cited 2021 May 6]. Available from: https://www.un.org/en/development/desa/population/publications/pdf/manuals/model/lifetables/annex1.pdf
255. Department of Statistics Singapore. SingStat Table Builder [Internet]. [cited 2021 May 6]. Available from: https://www.tablebuilder.singstat.gov.sg/
256. Schoolfield RM, Sharpe PJH, Magnuson CE. Non-linear regression of biological temperature-dependent rate models based on absolute reaction-rate theory. J Theor Biol [Internet]. 1981 Feb;88(4):719–31. Available from: https://linkinghub.elsevier.com/retrieve/pii/0022519381902460
257. Otero M, Solari HG, Schweigmann N. A Stochastic Population Dynamics Model for Aedes Aegypti: Formulation and Application to a City with Temperate Climate. Bull Math Biol. 2006 Nov;68(8):1945–74.
258. Beachkofski B, Grandhi R. Improved Distributed Hypercube Sampling. In: 43rd AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference [Internet]. Reston, Virigina: American Institute of Aeronautics and Astronautics; 2002. Available from: https://arc.aiaa.org/doi/10.2514/6.2002-1274
259. Otero M, Solari HG, Schweigmann N. A Stochastic Population Dynamics Model for Aedes Aegypti: Formulation and Application to a City with Temperate Climate. Bull Math Biol [Internet]. 2006 Nov 3;68(8):1945–74. Available from: http://link.springer.com/10.1007/s11538-006-9067-y
260. Casas-Martínez M, Tamayo-Domínguez R, Bond-Compeán JG, Rojas JC, Weber M, Ulloa-García A. Oogenic development and gonotrophic cycle of Aedes aegypti and Aedes albopictus in laboratory. Salud Publica Mex [Internet]. 2020 Jun 12;62(4, jul-ago):372. Available from: http://saludpublica.mx/index.php/spm/article/view/10164
261. Chan M, Johansson MA. The Incubation Periods of Dengue Viruses. Vasilakis N, editor. PLoS One [Internet]. 2012 Nov 30;7(11):e50972. Available from: https://dx.plos.org/10.1371/journal.pone.0050972
262. Gubler DJ, Suharyono W, Tan R, Abidin M, Sie A. Viraemia in patients with naturally acquired dengue infection. Bull World Health Organ [Internet]. 1981;59(4):623–30. Available from: http://www.ncbi.nlm.nih.gov/pubmed/6976230
263. Chen T, Guestrin C. XGBoost: A Scalable Tree Boosting System. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining [Internet]. New York, NY, USA: ACM; 2016. p. 785–94. Available from: https://dl.acm.org/doi/10.1145/2939672.2939785
264. Lundberg S, Lee S-I. A Unified Approach to Interpreting Model Predictions. 2017 May 22; Available from: http://arxiv.org/abs/1705.07874
265. Harrington LC, Fleisher A, Ruiz-Moreno D, Vermeylen F, Wa C V., Poulson RL, Edman JD, Clark JM, Jones JW, Kitthawee S, Scott TW. Heterogeneous Feeding Patterns of the Dengue Vector, Aedes aegypti, on Individual Human Hosts in Rural Thailand. Bingham A, editor. PLoS Negl Trop Dis [Internet]. 2014 Aug 7;8(8):e3048. Available from: https://dx.plos.org/10.1371/journal.pntd.0003048
266. Irvine MA, Kazura JW, Hollingsworth TD, Reimer LJ. Understanding heterogeneities in mosquito-bite exposure and infection distributions for the elimination of lymphatic filariasis. Proc R Soc B Biol Sci [Internet]. 2018 Jan

31;285(1871):20172253. Available from: https://royalsocietypublishing.org/doi/10.1098/rspb.2017.2253
267. Dye C, Hasibeder G. Population dynamics of mosquito-borne disease: effects of flies which bite some people more frequently than others. Trans R Soc Trop Med Hyg [Internet]. 1986 Jan;80(1):69–77. Available from: https://academic.oup.com/trstmh/article-lookup/doi/10.1016/0035-9203(86)90199-9
268. Cummings DAT, Iamsirithaworn S, Lessler JT, McDermott A, Prasanthong R, Nisalak A, Jarman RG, Burke DS, Gibbons R V. The Impact of the Demographic Transition on Dengue in Thailand: Insights from a Statistical Analysis and Mathematical Modeling. Farrar J, editor. PLoS Med [Internet]. 2009 Sep 1;6(9):e1000139. Available from: https://dx.plos.org/10.1371/journal.pmed.1000139
269. KUMAR NP, SAINI P, SAMUEL P, AJITHLAL PM, SURESH A, MATHEW J, SONIA T, AMJU KP, Veerapathiran A, Selvam A, Balaji T, Raju KHK, Gupta N, KUMAR A. ZIKV outbreak in Thiruvananthapuram, Kerala, India, 2021 – a primary report [Internet]. Research Square; 2021. Available from: http://europepmc.org/abstract/PPR/PPR394444
270. India Today. Maharashtra reports first case of Zika virus in Pune [Internet]. 2021 [cited 2021 Oct 14]. Available from: https://www.indiatoday.in/cities/pune/story/maharashtra-reports-first-case-of-zika-virus-in-pune-1835228-2021-07-31
271. Aubry F, Jacobs S, Darmuzey M, Lequime S, Delang L, Fontaine A, Jupatanakul N, Miot EF, Dabo S, Manet C, Montagutelli X, Baidaliuk A, Gámbaro F, Simon-Lorière E, Gilsoul M, Romero-Vivas CM, Cao-Lormeau V-M, Jarman RG, Diagne CT, Faye O, Faye O, Sall AA, Neyts J, Nguyen L, Kaptein SJF, Lambrechts L. Recent African strains of Zika virus display higher transmissibility and fetal pathogenicity than Asian strains. Nat Commun [Internet]. 2021 Dec 10;12(1):916. Available from: http://www.nature.com/articles/s41467-021-21199-z
272. Kasprzykowski JI, Fukutani KF, Fabio H, Fukutani ER, Costa LC, Andrade BB, Queiroz ATL. A recursive sub-typing screening surveillance system detects the appearance of the ZIKV African lineage in Brazil: Is there a risk of a new epidemic? Int J Infect Dis [Internet]. 2020 Jul;96:579–81. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1201971220303970
273. Routledge I, Lai S, Battle KE, Ghani AC, Gomez-Rodriguez M, Gustafson KB, Mishra S, Unwin J, Proctor JL, Tatem AJ, Li Z, Bhatt S. Tracking progress towards malaria elimination in China: Individual-level estimates of transmission and its spatiotemporal variation using a diffusion network approach. Pascual M, editor. PLOS Comput Biol [Internet]. 2020 Mar 23;16(3):e1007707. Available from: https://dx.plos.org/10.1371/journal.pcbi.1007707

Appendix A: Mapping the cryptic spread of the 2015–2016 global Zika virus epidemic

A1. Additional information on the collection and pre-processing of data

A1.1 Reported autochthonous Zika case data:

[Group 1]: American Samoa, Papua New Guinea, Thailand.

Cases were reported monthly at a country level. We assumed the daily case count to be uniform within each month to calculate the weekly incidence. If the monthly number of reported cases was below four, we randomly assigned each case to a Eweek within that month.

[Group 2]: Anguilla, Antigua and Barbuda, Argentina, Aruba, Bahamas, Barbados, Belize, Bolivia, British Virgin Islands, Cayman Islands, Costa Rica, Cuba, Curacao, Dominica, Dominican Republic, Ecuador, El Salvador, French Guiana, Grenada, Guadeloupe, Guatemala, Guyana, Haiti, Honduras, Jamaica, Martinique, Mexico, Nicaragua, Panama, Paraguay, Peru, Puerto Rico, Saint Barthelemy, Saint Kitts and Nevis, Saint Vincent and the Grenadines, Suriname, Trinidad and Tobago, Turks and Caicos Islands, United States, US Virgin Islands, Venezuela.

Weekly autochthonous case counts were published by the Pan American Health Organization (PAHO) as bar charts, which were digitized by the Anderson Lab and made publicly available. Except the United States, where data were available for each state (Florida and Texas), the rest of the countries in Group 2 had incidence data published at a country level only.

[Group 3]: Bonaire, Sint Eustatius, and Saba; Cabo Verde; Micronesia; Montserrat; Saint Martin; Samoa; Sint Maarten; Tonga.

We digitized the weekly country-level autochthonous case counts from published bar charts, except for Tonga, for which data were directly obtained from the table published by the ReliefWeb.

[Group 4]: Brazil

Weekly number of autochthonous cases reported by Brazil during 2016 were digitized by the Anderson Lab based on bar charts published by the PAHO. We manually digitized the weekly number of notified Zika cases in Brazil during 2015 published by Lourenço et al. (2018), since these data were not released by the PAHO.

Ideally, we would want to obtain case data reported at a subdivision level, given that Brazil reported the largest number of autochthonous Zika cases during 2015–2016 and these cases were not evenly distributed across space. We were unable to obtain the subdivision-level data however and therefore ran the analysis at a country level.

[Group 5]: Colombia

For each first-level country subdivision except San Andrés, weekly cumulative case counts between Eweek 37 and Eweek 52 of 2016 were obtained from the Colombian National Institute of Health's website, and each time series was then differenced to derive the weekly case counts. The weekly ZIKV incidence data reported prior to Eweek 38 of 2016 were collected and made publicly available by Siraj et al. (2018).

For San Andrés, weekly incidence data were not published by Siraj et al. (2018). Hence, we obtained daily incidence data prior to Eweek 5 of 2016 from Rojas et al. (2016), which were subsequently aggregated within each Eweek. As before, weekly incidence data from Eweek 5 of 2016 onwards were derived from the weekly cumulative case counts published by the Colombian National Institute of Health.

[Group 6] Fiji, Maldives, Philippines, Solomon Islands, Vanuatu, Vietnam.

These countries were excluded from our analyses, because either the autochthonous case counts were not publicly available, or the time of each reported case could not even be narrowed down to a specific month. Overall, the number of autochthonous Zika cases reported by these countries only accounted for less than 0.09% of the total case count reported worldwide during 2015–2016.

[Group 7]: Guinea-Bissau, Indonesia, Laos, Malaysia, Myanmar, New Caledonia, Palau.

The total number of autochthonous cases reported by each of these countries during 2015–2016 was relatively small (less than ten per country). The Eweek during which each case was reported can be found using HealthMap. We located each autochthonous case reported by Indonesia, Malaysia, and Myanmar to a first-level country subdivision, and data for the rest of the countries were publicly available at a country level only.

[Group 8]: Marshall Islands, Singapore.

We digitized the daily country-level autochthonous case counts from published bar charts, which were subsequently aggregated within each Eweek.

[Group 9] Saint Lucia

The PAHO published the total number of autochthonous Zika cases reported at a country level by the end of 2016, and the time when the first autochthonous case was identified. Since the total number of cases was reasonably large (872), we did not exclude Saint Lucia from our analysis, and assumed the shape of the epidemiological curve (post-standardization) was equal to that of Saint Vincent and the Grenadines, which is situated closed to Saint Lucia.

A1.2 Merging of subdivisions

For each first-level country subdivision s with no incoming air passengers based on the 2015–2016 OAG data, we performed an estimation of the most likely airport a^* (within the same country) that the population of subdivision s would rely on when they returned home. In the equation below, the distance between the population-weighted centroid coordinates of subdivision s and an airport a (denoted by $d(a,s)$) was adjusted by the total number of incoming air passengers during 2015–2016 (v_a) using Holling's type 2 function, so that major airports would not be overemphasized. We used the notation "a within c_s" to restrict ourselves to all airports a within the country c_s, to which subdivision s belonged (similarly for "a' within c_s"):

$$a^* = \mathrm{argmin}_{a \text{ within } c_s} \frac{d(a,s)}{\sigma(a)}$$

$$\sigma(a) = \max_{a' \text{within } c_s}(v_{a'}) \cdot \frac{v_a}{v_a + \mathrm{median}_{a' \text{within } c_s}(v_{a'})}$$

Consequently, subdivision $\boldsymbol{s}$ was merged with the one where airport $\boldsymbol{a}^*$ was located, and they were modelled as a single unit from then on. However, if the reported autochthonous cases cannot be located to the subdivision level, the entire country was treated as a single unit of analysis. Hence, the spatial unit of analysis in our study can be a single first-level country subdivision, a combination of subdivisions, or an entire country (hereinafter referred to as "spatial unit").

A1.3 Countries or territories with zero incoming air passenger during 2015–2016, and hence were excluded from the analyses.

Andorra, Bouvet Island, British Indian Ocean Territory, Clipperton, French Southern Territories, Heard Island and McDonald Islands, Liechtenstein, Palestinian Territory, Pitcairn, Saint Helena, San Marino, South Georgia, Tokelau, Vatican City.

A2. Additional information on the global risk model validation

We took the list of R_0 estimates that were obtained by Armstrong et al. (2020) via a systematic search within Web of Science and PubMed, and applied the following selection criteria:

1) The R_0 value was derived by estimating the early growth rate of the epidemic based on the reported case data (i.e., using a simple exponential growth model, or Richards' growth model).

In other words, we did not include estimates derived from mechanistic models, which could vary substantially in terms of the functional forms as well as the input parameter values used. In fact, even for the same location and time window, there can be very different literature estimates of R_0 obtained by these models, where the differences in model specifications played an important role. This is not to say that these model results were all invalid, but the complex nature of the models has made it challenging to standardize the results, as well as to examine the validity of the methods given the large number of studies included in the list. Hence, we only included estimates that were directly driven by the reported case data as described in the beginning.

2) The time period of the reported case data that were used to estimate R_0 was clearly specified.

This would allow us to average our R_0 estimates across the corresponding time window to be compared with the literature estimate.

3) The spatial resolution of the location of the reported case data that were used to estimate R_0 should not be higher than the spatial unit of analysis in our study.

4) The width of the uncertainty interval of the R_0 estimate should not exceed 10.

A3. Data and tables

Data S2.1: Weekly number of reported autochthonous Zika cases by each country or subdivision during 2015–2016.

Available at https://static-content.springer.com/esm/art%3A10.1186%2Fs12916-020-01845-x/MediaObjects/12916_2020_1845_MOESM2_ESM.xlsx

Table S2.1: Reported / simulated total number of Zika cases / ZIKV infections imported in each susceptible spatial unit during 2015–2016, together with the estimated reporting index.

Available at https://static-content.springer.com/esm/art%3A10.1186%2Fs12916-020-01845-x/MediaObjects/12916_2020_1845_MOESM6_ESM.xlsx

Table S2.2: Estimated probability that no onward spread of ZIKV occurred following importation during 2015–2016 for each susceptible spatial unit.

Available at https://static-content.springer.com/esm/art%3A10.1186%2Fs12916-020-01845-x/MediaObjects/12916_2020_1845_MOESM7_ESM.xlsx

Table S2.3: List of first-level country subdivisions belonging to each susceptible spatial unit.

Available at https://static-content.springer.com/esm/art%3A10.1186%2Fs12916-020-01845-x/MediaObjects/12916_2020_1845_MOESM8_ESM.xlsx

Figure S3.1: Weekly mean catch per trap for *Aedes aegypti* (red) and *Ae. albopictus* (blue) at Bukit Batok Street 52, Jurong West Avenue 1, Tampines Street 21, and Yishun Avenue 9. Week 1 refers to the first epidemiological week of 2017, and week 104 the last epidemiological week of 2018. Due to the large number of sites (n=552) in our study, we picked these four sites to visualize *Aedes* abundance over time for demonstration purposes.

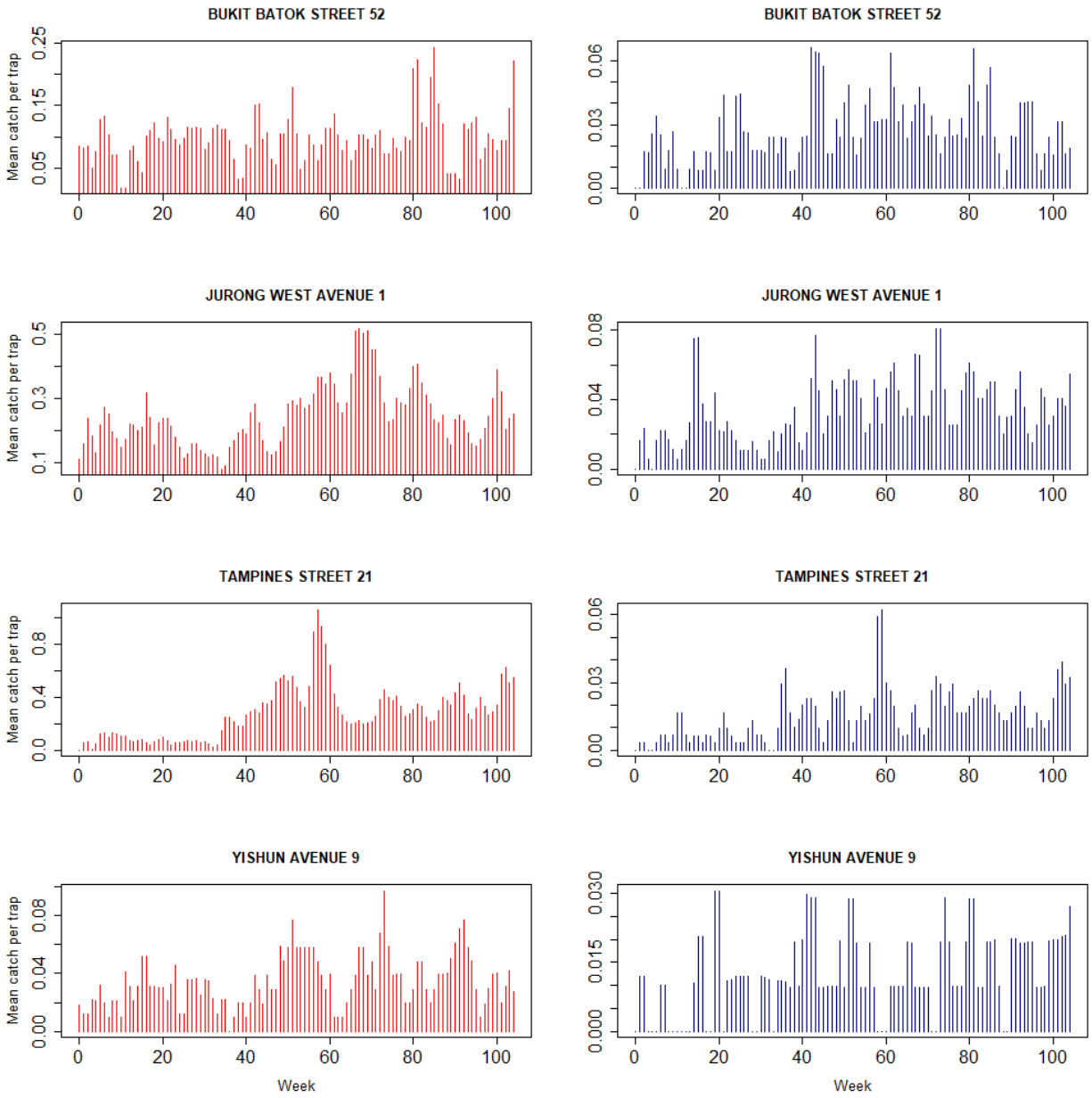


Figure S3.2: Visualizations of spatial variables: (a) distance to the nearest waterway (m), (b) Drain line density (km^{-1}), and (c) distance to the nearest water area (m) for each site, as well as (d) the land classification map produced by Richards & Tunçer.

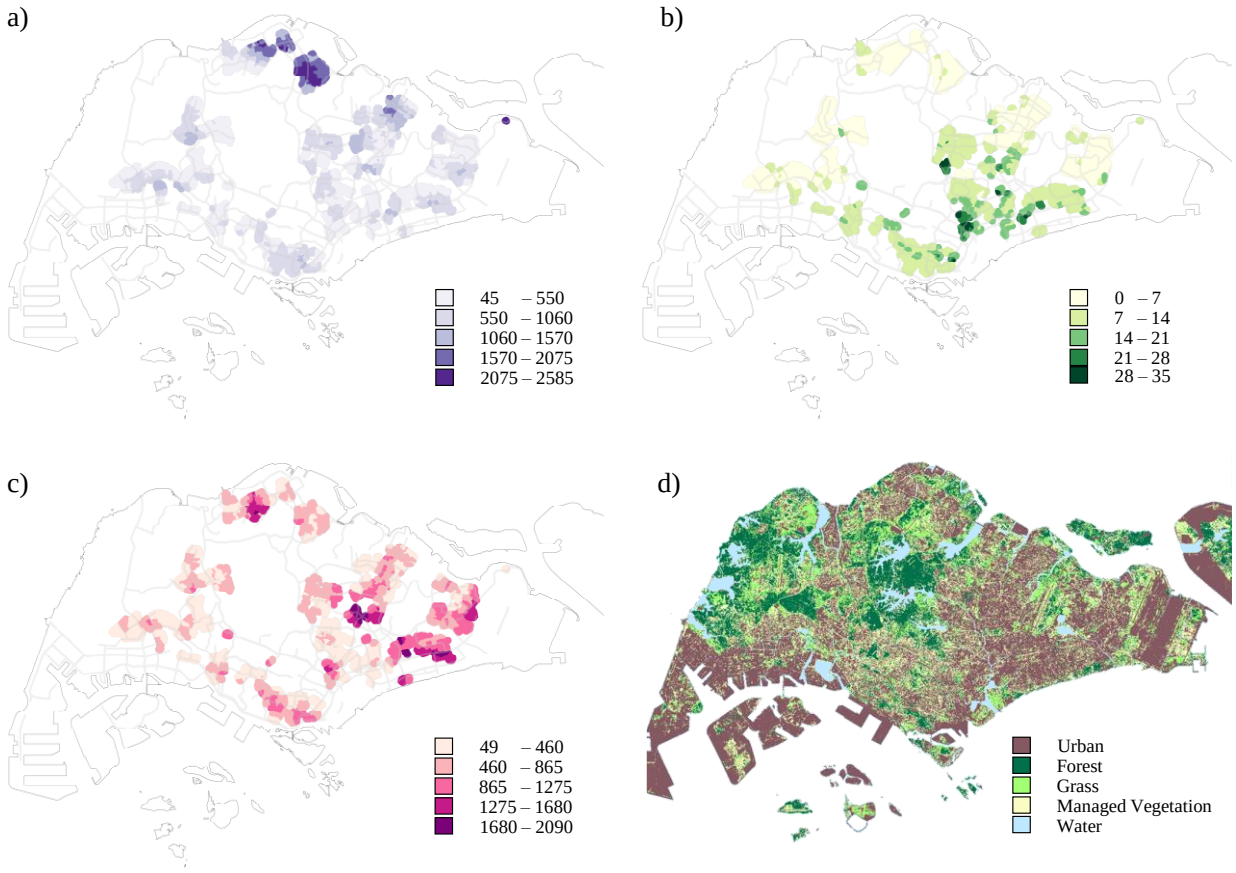


Figure S3.3: Visualizations of weather variables. Each sub-plot shows a weather variable whose weekly value was averaged across all the sites in the study.

